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Class II Cast-Metal Restorations

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The cast-metal restoration is versatile and is especially applicable to Class II onlay preparations. The process has many steps, involves numerous dental materials, and requires meticulous attention to detail. Typically a dental laboratory is involved, and the dentist and the laboratory technician must be devoted to perfection. The high degree of satisfaction and service derived from a properly made cast-metal restoration is a reward for the painstaking application required.[1] The Class II *inlay* involves the occlusal surface and one or more proximal surfaces of a posterior tooth. When cusp tips are restored, the term *onlay* is used. The procedure requires two appointments: the first for preparing the tooth and making an impression, and the second for delivering the restoration to the patient. The fabrication process is referred to as an *indirect procedure* because the casting is made on a replica of the prepared tooth in a dental laboratory.

Material Qualities

Cast-metal restorations can be made from a variety of casting alloys. Although the physical properties of these alloys vary, their major advantages are their high compressive and tensile strengths. These high strengths are especially valuable in restorations that rebuild most or all of the occlusal surface.

The American Dental Association (ADA) Specification No. 5 for Dental Casting Gold Alloys requires a minimum total gold-plus-platinum-metals content of 75 weight percent (wt%). Such traditional high-gold alloys are unreactive in the oral environment and are some of the most biocompatible materials available to the restorative dentist.[2] At present, four distinct groups of alloys are in use for cast restorations: (1) traditional high-gold alloys, (2) low-gold alloys, (3) palladium–silver alloys, and (4) base metal alloys. Each of the alternatives to high-gold alloys has required some modification of technique or acceptance of reduced performance, most commonly related to decreased tarnish resistance and decreased burnishability.[3] Also they have been associated with higher incidences of postrestorative allergy, most often exhibited by irritated soft tissue adjacent to the restoration.[2]

Indications

Large Restorations

The cast-metal inlay is an alternative to amalgam or composite when the higher strength of a casting alloy is needed or when the superior control of contours and contacts that the indirect procedure provides is desired. The cast-metal onlay is often an excellent alternative to a crown for teeth that have been greatly weakened by caries or by large, failing restorations but where the facial and lingual tooth surfaces are relatively unaffected by disease or injury. For such weakened teeth, the superior physical properties of a casting alloy are desirable to withstand the occlusal loads placed on the restoration; also the onlay can be designed to distribute occlusal loads over the tooth in a manner that decreases the chance of tooth fracture in the future. Preserving intact facial and lingual surfaces is conducive to maintaining the health of the pulp and the contiguous soft tissue (gingiva). When margins are supragingival they are easier to inspect for adaptation and seal. When proximal surface caries is extensive, favorable consideration should be given to the cast inlay or onlay. The indirect procedure used to develop the cast restoration allows more control of contours and contacts (proximal and occlusal).

Endodontically Treated Teeth

A molar or premolar with treatment root canal filling can be restored with a cast-metal onlay, provided that the onlay has been thoughtfully designed to distribute occlusal loads in such a manner as to reduce the chance of tooth fracture.

Teeth at Risk for Fracture

Fracture lines in enamel and dentin, especially in teeth having extensive restorations, should be recognized as cleavage planes for possible future fracture of the tooth. Restoring these teeth with a restoration that braces the tooth against fracture injury may be warranted sometimes. Such restorations are cast onlays and crowns.

Dental Rehabilitation With Cast-Metal Alloys

When cast-metal restorations have been used to restore adjacent or opposing teeth, the continued use of the same material may be considered to eliminate electrical and corrosive activity that sometimes occurs between dissimilar metals in the mouth, particularly when they come in contact with each other (galvanism).

Diastema Closure and Occlusal Plane Correction

Often the cast inlay or onlay is indicated when extension of the mesiodistal dimension of the tooth is necessary to form a contact with an adjacent tooth. Cast onlays also can be used to correct the occlusal plane of a slightly tilted tooth.

Removable Prosthodontic Abutment

Teeth that are to serve as abutments for a removable partial denture can be restored with cast-metal restorations. The advantages of cast restorations are as follows: (1) The superior physical properties of cast-metal alloys allow restorations to better withstand forces imparted by a partial denture, and (2) rest seats, guiding planes, and other aspects of contour relating to partial dentures are better controlled when the indirect technique is used.

Contraindications

High Caries Rate

Facial and lingual (especially lingual) smooth-surface caries indicates a high caries activity that should be brought under control before expensive cast-metal restorations are used.

Young Patients

With younger patients, direct restorative materials (e.g., composite or amalgam) are indicated unless the tooth is severely broken or endodontically treated. An indirect procedure requires longer and more numerous appointments, access is more difficult, the clinical crowns are shorter, and younger patients may neglect oral hygiene, resulting in additional caries.

Esthetics

The dentist must consider the esthetic impact (display of metal) of the cast-metal restoration. This factor usually limits the use of cast-metal restorations to tooth surfaces that are hidden at a conversational distance. Composite and porcelain restorations are alternatives in esthetically sensitive areas.

Small Restorations

Because of the success of amalgam and composite, few cast-metal inlays are done in small Class I and II restorations.

Advantages

Strength

The inherent strength of dental casting alloys allows them to restore large damaged or missing areas and be used in ways that protect the tooth from future fracture injury. Such restorations include onlays and crowns.

Biocompatibility

As previously mentioned, high-gold dental casting alloys are unreactive in the oral environment. This biocompatibility can be helpful for many patients who have allergies or sensitivities to other restorative materials.

Low Wear

Although individual casting alloys vary in their wear resistance, castings are able to withstand occlusal loads with minimal changes. This is especially important in large restorations that restore a large percentage of occlusal contacts.

Control of Contours and Contacts

Through the use of the indirect technique, the dentist has great control over contours and contacts. This control becomes especially important when the restoration is larger and more complex.

Disadvantages

Number of Appointments and Higher Chair Time

The cast inlay or onlay requires at least two appointments and much more time than a direct restoration, such as amalgam or composite.

Temporary Restorations

Patients must have temporary restorations between the preparation and delivery appointments. Temporaries occasionally loosen or break, requiring additional visits.

Cost

In some instances, cost to the patient becomes a major consideration in the decision to restore teeth with cast-metal restorations. The cost of materials, laboratory bills, and the time involved make indirect cast restorations more expensive than direct restorations.

Technique Sensitivity

Every step of the indirect procedure requires diligence and attention to detail. Errors at any part of the long, multistep process tend to be compounded, resulting in less than ideal fits.

Splitting Forces

Small inlays may produce a wedging effect on facial or lingual tooth structure and increase the potential for splitting the tooth. Onlays do not have this disadvantage.

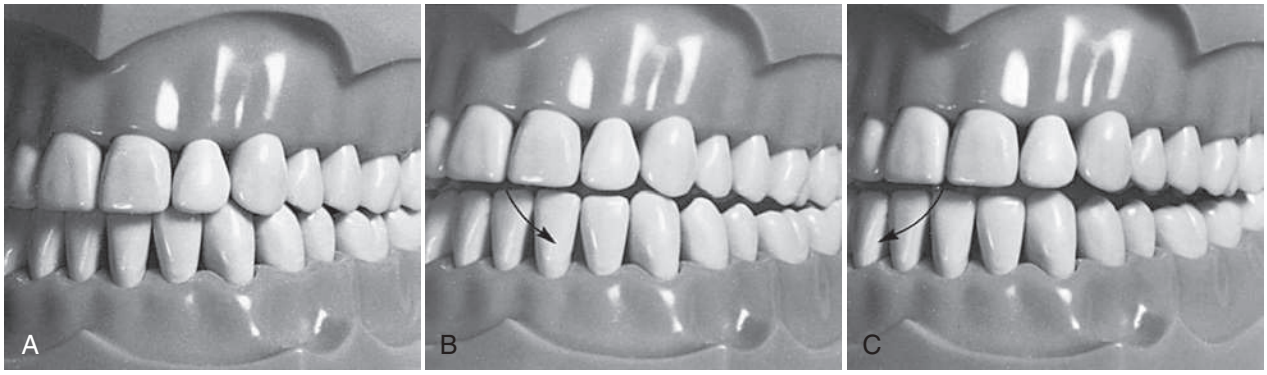
Initial Procedures

Occlusion

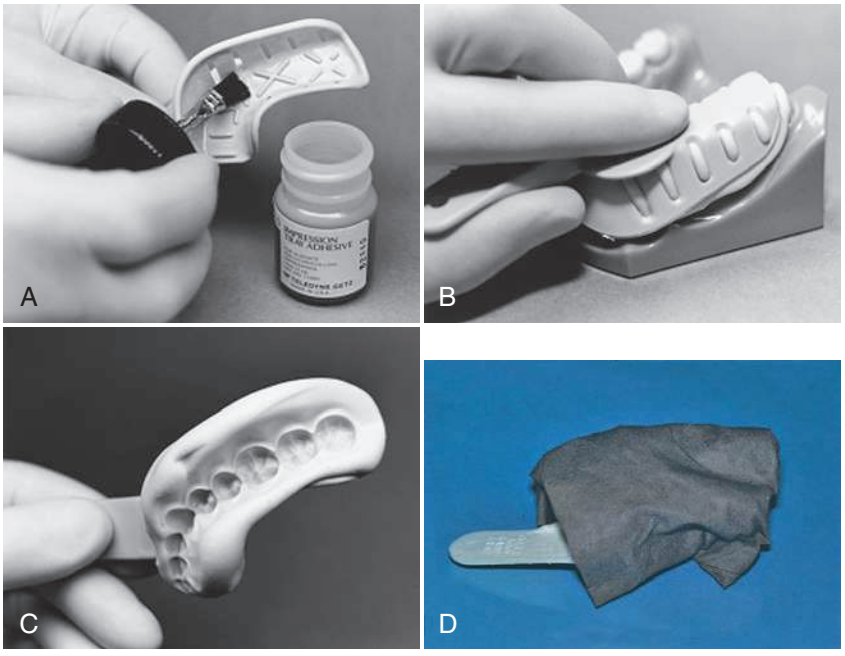
Before the anesthetic is administered and before preparation of any tooth, the occlusal contacts of teeth should be evaluated. As part of this evaluation, the dentist must decide if the existing occlusal relationships can be improved with the cast-metal restoration. An evaluation should include (1) the occlusal contacts in maximum intercuspation where teeth are brought into full interdigitation and (2) the occlusal contacts that occur during mandibular movements (Fig. 18.1). The pattern of occlusal contacts influences the preparation design, selection of interocclusal records, and type of articulator or cast development needed.

Anesthesia

Local anesthesia of the tooth to be operated on and of adjacent soft tissue usually is recommended. Anesthesia in these areas eliminates pain and reduces salivation, resulting in a more pleasant procedure for the patient and the operator.



• **Fig. 18.1** A–C, Evaluate occlusal relationships in maximum intercuspation (A) and during mandibular movements (B and C). Be alert for problems with tooth alignment and contact position. Note the amount of posterior separation provided by the guidance of anterior teeth (working side) and articular eminence (nonworking side).



• **Fig. 18.2** A, Applying tray adhesive to stock quadrant tray. B, Making preoperative impression. C, Inspecting preoperative impression for completeness. D, When using alginate, wrap the impression with wet paper towels to serve as a humidor.

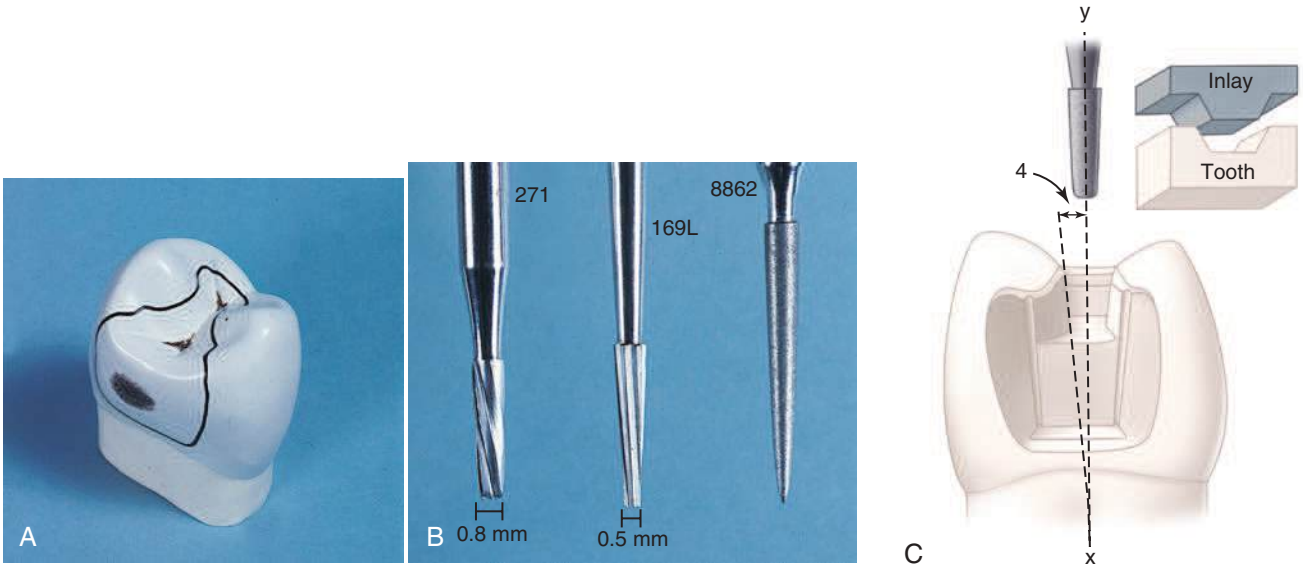
Considerations for Temporary Restorations

Before preparation of the tooth, consideration must be given to the method that will be used to fabricate the temporary restoration. Most temporary restoration techniques require the use of a preoperative impression to reproduce the occlusal, facial, and lingual surfaces of the temporary restoration to the preoperative contours.

The technique involves making a preoperative impression with an elastic impression material. Alginate impression materials may be used and are relatively inexpensive. The preoperative impression may be made with a polyvinyl siloxane (PVS) impression material if additional accuracy, stability, and durability are required. If the tooth to be restored has large defects such as a missing cusp, an instrument can be used to carve away impression material in the area of the missing cusp or tooth structure, to simulate the desired form for the temporary restoration. Alternatively, a material such as rope wax can be added to the tooth before the impression in the area of the defect. An impression is typically made using a sectional tray (Fig. 18.2A). The tray filled with impression material is seated (see Fig. 18.2B). After the impression has set, the impression is removed and examined for completeness (see Fig. 18.2C). Alginate impressions can distort quickly if they are allowed to gain or lose moisture, so the impression is wrapped in wet paper towels to serve as a humidor (see Fig. 18.2D). Preoperative PVS impressions do not need to be wrapped. The preoperative impression is placed aside for later use in forming the temporary restoration.

Tooth Preparations for Class II Cast-Metal Restorations

A small, distal, cavitated caries lesion in the maxillary right first premolar is used to illustrate the classic two-surface preparation



• **Fig. 18.3** A, Proposed outline form for distoocclusal preparation. B, Dimensions and configuration of No. 271, No. 169L, and No. 8862 instruments. C, Conventional 4-degree divergence from line of draw *(xy)*.

for an inlay (Fig. 18.3A). Treatment principles for other defects are presented later. As indicated previously, few small one-surface or two-surface inlays are done. Because the description of a small tooth preparation presents the basic concepts, it is used to illustrate the technique. More extensive tooth preparations are presented later.

Tooth Preparation for Class II Cast-Metal Inlays

Initial Preparation

Carbide burs used to develop the vertical internal walls of the preparation for cast-metal inlays and onlays are plane cut, tapered fissure burs. These burs are plane cut so that the vertical walls are smooth. The side and end surfaces of the bur should be straight to aid in the development of uniformly tapered walls and smooth pulpal and gingival walls. Recommended dimensions and configurations of the burs to be used are shown in Fig. 18.3B. Suggested burs are the No. 271 and the No. 169L burs (Brasseler USA, Inc., Savannah, GA). Before using unfamiliar burs, the operator is cautioned to verify measurements to judge the depth into the tooth during preparation. The sides and end surface of the No. 271 bur meet in a slightly rounded manner so that sharp, stress-inducing internal angles are not formed in the preparation.[4] The marginal bevels are placed with a slender, fine-grit, flame-shaped diamond instrument such as the No. 8862 bur (Brasseler USA, Inc.).

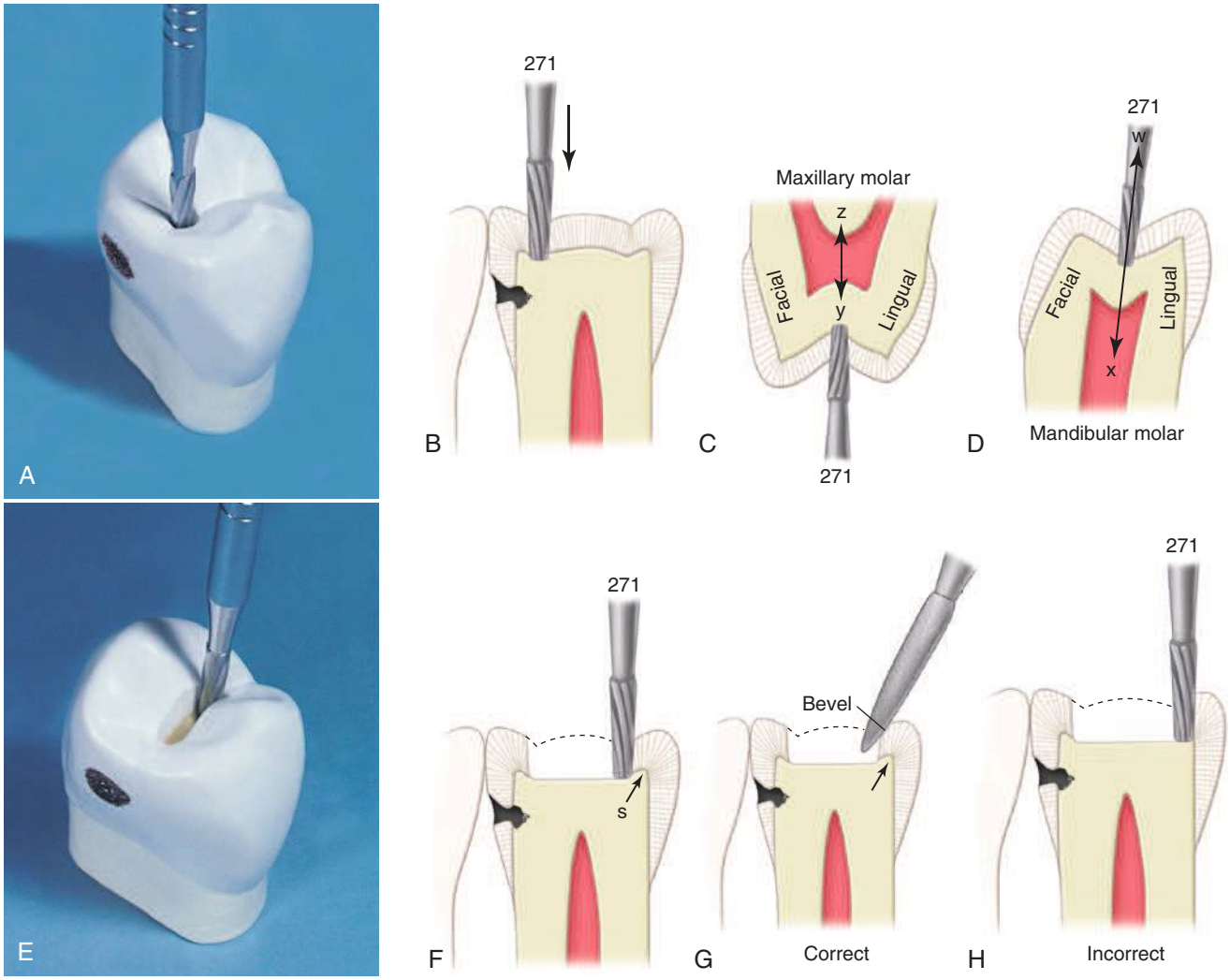
Throughout the preparation for a cast inlay, the cutting instruments used to develop the vertical walls are oriented to a single "draw" path, usually the long axis of the tooth crown, so that the completed preparation has draft (no undercuts) (see Fig. 18.3C). The gingival-to-occlusal divergence of these preparation walls may range from 2 to 5 degrees per wall from the line of draw. If the vertical walls are unusually short, a maximum of 2 degrees occlusal divergence is desirable to increase retention potential. As the occlusogingival height increases, the occlusal divergence should increase because lengthy preparations with minimal divergence (more parallel) may present difficulties during the seating and withdrawal of the restoration.

Occlusal Step

With the No. 271 carbide bur held parallel to the long axis of the tooth crown, the dentist enters the fossa or pit closest to the involved marginal ridge, using a punch cut to a depth of 1.5 mm to establish the depth of the pulpal wall (Fig. 18.4A and B). In the initial preparation, this specified depth should not be exceeded, regardless of whether the bur end is in dentin, caries, old restorative material, or air. The bur should be rotating at high speed (with air-water spray) before application to the tooth and should not stop rotating until it is removed; this minimizes perceptible vibration and prevents breakage or chipping of the bur blades. A general rule is to maintain the long axis of the bur parallel to the long axis of the tooth crown at all times (see Fig. 18.4B and C). For mandibular molars and second premolars whose crowns tilt slightly lingually, this rule dictates that the bur should also be tilted slightly (5–10 degrees) lingually to conserve the strength of the lingual cusps (see Fig. 18.4D). When the operator is cutting at high speeds, a properly directed air-water spray is used to provide the necessary cooling and cleansing effects.[5]

Maintaining the 1.5-mm initial depth and the same bur orientation, the dentist extends the preparation outline mesially along the central groove or fissure to include the mesial fossa or pit (see Fig. 18.4E and F). Ideally, the faciolingual dimension of this cut should be minimal. The dentist takes care to keep the mesial marginal ridge strong by not removing the dentin support of the ridge (see Fig. 18.4F and H). The use of light intermittent pressure minimizes heat production on the tooth surface and reduces the incidence of enamel crazing ahead of the bur. Occasionally a fissure extends onto the mesial marginal ridge. This defect, if shallow, may be treated with enameloplasty, or it may be included in the outline form with the cavosurface bevel, which is applied in a later step in the tooth preparation (see Fig. 18.4G).

Enameloplasty, as presented in earlier chapters, occasionally reduces extension along the fissures, conserving the tooth structure vital for pulp protection and the strength of the remaining tooth crown. The extent to which enameloplasty can be used usually cannot be determined until the operator is in the process of

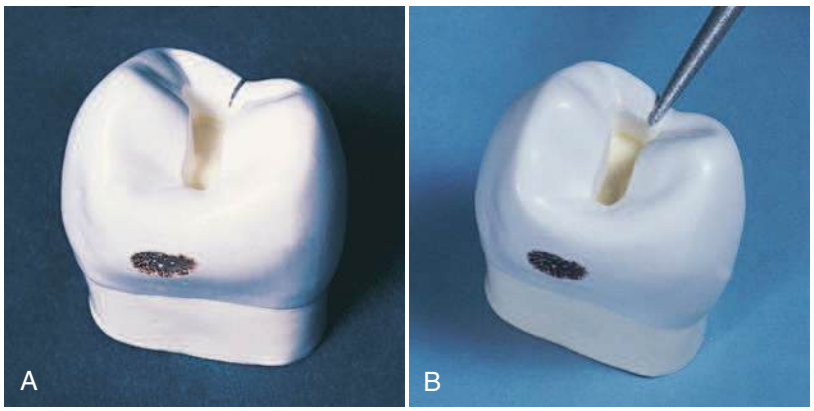


• **Fig. 18.4** A and B, Bur after punch cut to a depth of 1.5 mm. C, For maxillary posterior teeth, the long axis of the bur should parallel the long axis of the tooth crown *(yz)*. D, For molar and second premolar teeth of mandibular dentition, the long axis of the bur should tilt slightly lingually to parallel the long axis of the tooth crown *(wx)*. E and F, Extending the mesial wall, taking care to conserve dentin that supports marginal ridge *(s)*. G, The marginal bevel can provide additional extension. H, Improper extension that has weakened the marginal ridge.

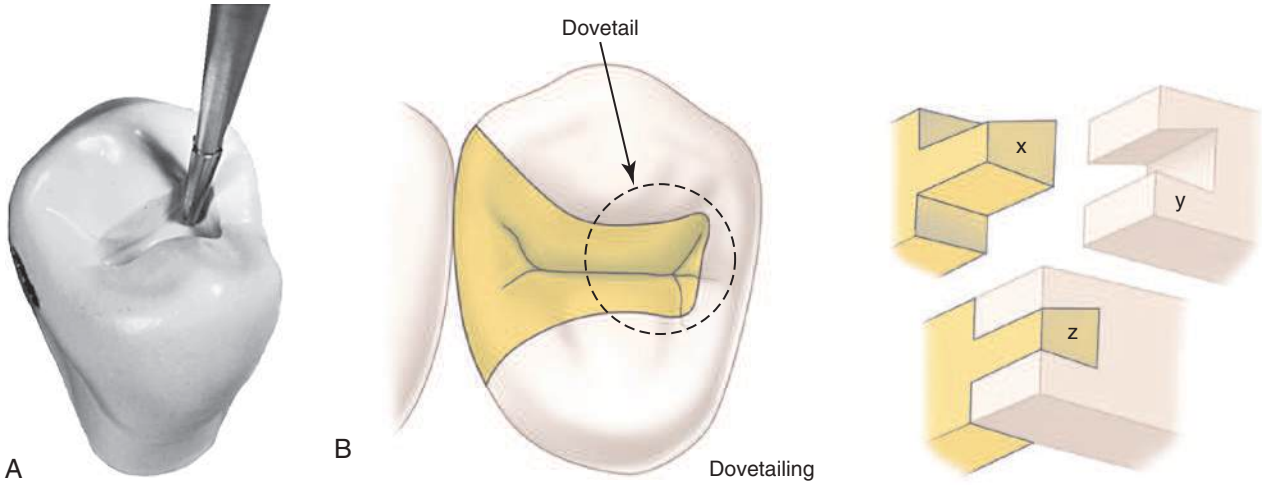
extending the preparation wall, when the depth of the fissure in the enamel wall can be observed (Fig. 18.5). When enameloplasty shows a fissure in a marginal ridge to be deeper than one third the thickness of enamel, the procedures described in the later section should be used.

Extend to include faulty facial and lingual fissures radiating from the mesial pit. During this extension cutting, the operator is cautioned again not to remove the dentin support of the proximal marginal ridge. To conserve the tooth structure and the strength of the remaining tooth, the final extension up these fissures can be accomplished with the slender No. 169L carbide bur (Fig. 18.6A). The tooth structure and strength can be conserved further by using (1) enameloplasty of the fissure ends, when possible, and (2) the marginal bevel of the final preparation to include (eliminate) the terminal ends of these fissures in the outline form. The facial and lingual extensions in the mesial pit region should provide the desired dovetail retention form, which resists distal displacement of the inlay (see Fig. 18.6B). When these facial and lingual grooves are not faulty, sufficient facial extension in the mesial pit region should be made to provide this dovetail retention form against distal displacement. Minor extension in the transverse ridge area to include any remaining facial or lingual caries may necessitate additional facial or lingual extension in the mesial pit to provide this dovetail feature. (During such facial or lingual extensions to sound tooth structure, the bur depth is maintained at 1.5 mm.) If major facial or lingual extension is required to remove undermined occlusal enamel, capping the weak remaining cuspal structure and additional features in the preparation to provide adequate retention and resistance forms may be indicated. These considerations are discussed in subsequent sections.

Continuing at the initial depth, the occlusal step is extended distally into the distal marginal ridge sufficiently to expose the junction of the proximal enamel and dentin (Fig. 18.7A and B). While extending distally, the dentist progressively widens the preparation to the desired faciolingual width in anticipation of the proximal box preparation. The increased faciolingual width



• **Fig. 18.5** A, Shallow enamel fault that is no deeper than one third the thickness of enamel. B, Using fine-grit diamond instrument to remove enamel that contains shallow fault.

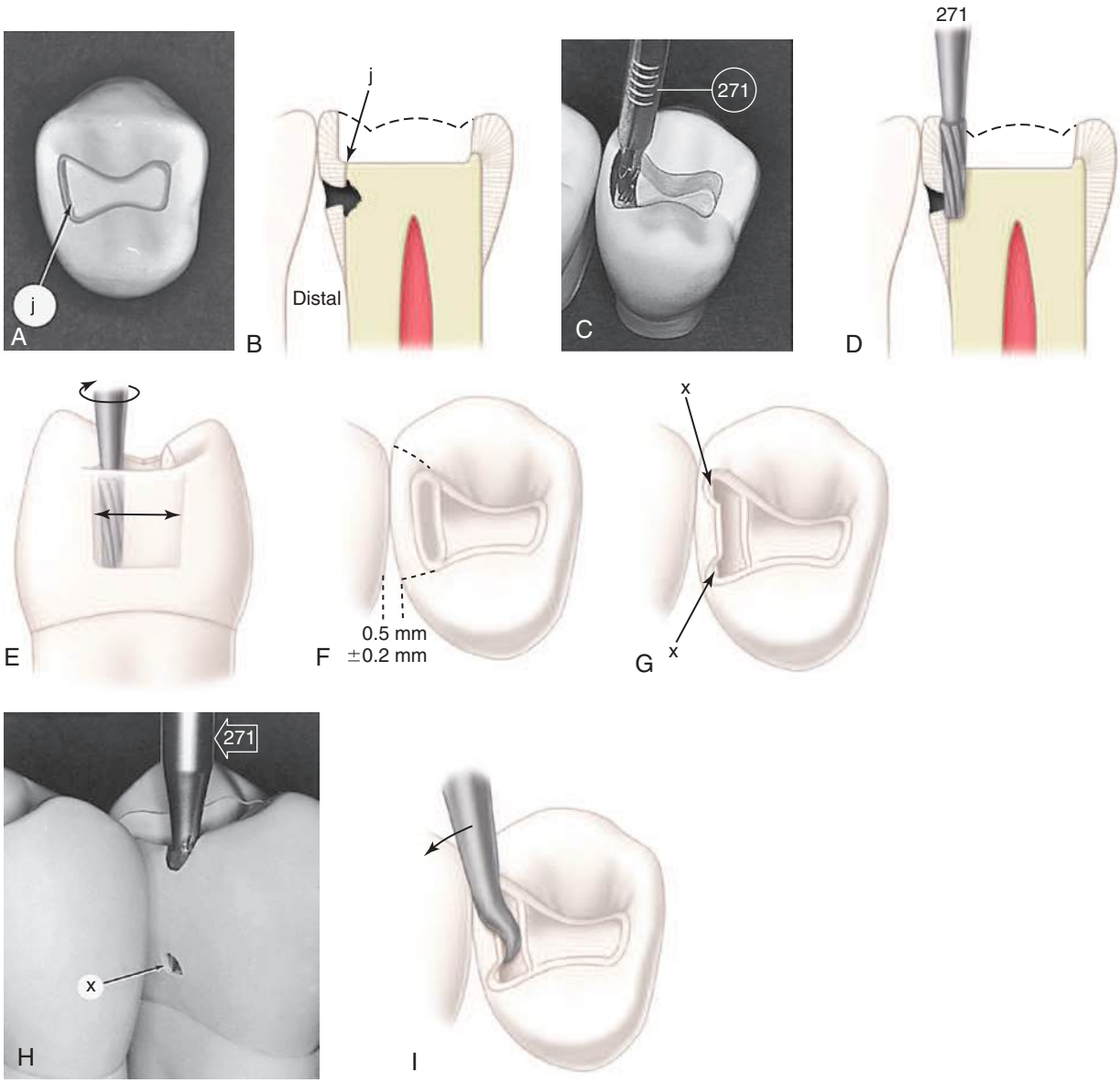


• **Fig. 18.6** A, Extending up the mesiofacial triangular groove using the slender No. 169L bur. B, Dovetail retention form is created by extension shown in *A*. As *x* fits into *y* only in one direction resulting in *z*, similarly dovetail portion of inlay fits into the dovetail portion of the preparation only in an occlusal-to-gingival direction.

enables the facial and lingual walls of the box to project (visually) perpendicularly to the proximal surface at positions that clear the adjacent tooth by 0.2 to 0.5 mm (see Fig. 18.7F). The facial and lingual walls of the occlusal step should go around the cusps in graceful curves, and the prepared isthmus in the transverse ridge ideally should be only slightly wider than the bur, thus conserving the dentinal protection for the pulp and maintaining the strength of the cusps. If the occlusal step has been prepared correctly, any caries on the pulpal floor should be uncovered by facial and lingual extensions to sound enamel (supported by dentin).

Proximal Box

Continuing with the No. 271 carbide bur, the distal enamel is isolated by cutting a proximal ditch (see Fig. 18.7C–F). The harder enamel should guide the bur. Slight pressure toward enamel is necessary to prevent the bur from cutting only dentin. If the bur is allowed to cut only dentin, the resulting axial wall would be too deep. The mesiodistal width of the ditch should be 0.8 mm (the tip diameter of the bur) and prepared approximately two thirds (0.5 mm) at the expense of dentin and one third (0.3 mm) at the expense of enamel. The gingival extension of this cut may be checked with the length of the bur by first measuring the depth from the height of the marginal ridge and then removing the bur and holding it beside the tooth. A periodontal probe also may be used for this measurement. While penetrating gingivally, the dentist extends the proximal ditch facially and lingually beyond the caries to the desired position of the facioaxial and linguoaxial line angles. If the caries lesion is minimal, the ideal extension facially and lingually is performed as previously described (see Fig. 18.7F). Ideal gingival extension of a minimal, cavitated lesion eliminates caries on the gingival floor and provides a 0.5-mm clearance of the unbeveled gingival margin with the adjacent tooth. Moderate to extensive caries on the proximal surface dictates continued extension of the proximal ditch to the extent of the caries at the dentinoenamel junction (DEJ), but not pulpally (see Fig. 18.11D later in the chapter). When preparing the proximal portion of the preparation, the dentist maintains the side of the bur at the specified axial wall depth regardless of whether it is in dentin, caries, old restorative material, or air. The operator should guard against overcutting the facial, lingual, and gingival walls, which would not conserve the tooth structure and could result in (1) overextension of the margins in the completed preparation, (2) a weakened tooth,



• **Fig. 18.7** A, After exposing the junction *(j)* of proximal enamel and dentin. B, Sectional drawing of A. C, Cutting the proximal ditch. D, Sectional drawing of C. E, Proximal view of D. F, Occlusal view of the proximal ditch with proposed ideal clearance with the adjacent tooth. G and H, Proximal ditch extended distally. *x*, penetration of enamel by side of bur at its gingival end. I, Breaking away isolated enamel.

and (3) possible injury of soft tissue. Because the proximal enamel diminishes in thickness from the occlusal to gingival level, the end of the bur is closer to the external tooth surface as the cutting progresses gingivally. The axial wall should follow the contour of the tooth faciolingually. Any carious dentin on the axial wall should not be removed at this stage of the preparation.

With the No. 271 carbide bur, the dentist makes two cuts, one at the facial limit of the proximal ditch and the other at the lingual limit, extending from the ditch perpendicularly toward the enamel surface (in the direction of the enamel rods) (see Fig. 18.7G). These cuts are extended until the bur is nearly through the marginal ridge enamel (the side of the bur may emerge slightly through the surface at the level of the gingival floor) as shown in Fig. 18.7H. This weakens the enamel by which the remaining isolated portion is held. Also the level of the gingival floor is verified by observing where the end of the bur emerged through the proximal surface. If indicated, additional gingival extension can be accomplished while the remaining enamel still serves to guide the bur and to prevent it from marring the proximal surface of the adjacent tooth. At this time, however, the remaining wall of enamel often breaks away during cutting, especially when high speeds are employed. If the isolated wall of enamel is still present, it can be fractured out with a spoon excavator (see Fig. 18.7I). At this stage, the ragged enamel edges left from breaking away the proximal surface may be touching the adjacent tooth.

Planing the distofacial, distolingual, and gingival walls by hand instruments to remove all undermined enamel may be indicated if minimal extension is needed to fulfill an esthetic objective.

Depending on access, the operator can use a No. 15 (width) straight chisel, bin-angle chisel (Fig. 18.8), or enamel hatchet. For a right-handed operator, the distal beveled bin-angle chisel is used on the distofacial wall of a distoocclusal preparation for the maxillary right premolar. The dentist planes the wall by holding the instrument in the modified palm-and-thumb grasp and uses a chisel-like motion in an occlusal-to-gingival direction (see Fig. 18.8A and B). The dentist planes the gingival wall by using the same instrument as a hoe, scraping in a lingual-to-facial direction (see Fig. 18.8C). In this latter action, the axial wall may be planed with the side edge (secondary edge) of the blade. The distolingual wall is planed smooth by using the bin-angle chisel with the mesial bevel (see Fig. 18.8D). When proximal caries is minimal, ideal facial and lingual extensions at this step in the preparation result in margins that clear the adjacent tooth by 0.2 to 0.5 mm.

The experienced operator usually does not use chisel hand instruments during the preparation for inlays, considering that the narrow, flame-shaped, fine-grit diamond instrument, when artfully used, removes ragged, weak enamel during application of the cavosurface bevel and flares and causes the patient to be less apprehensive (see Figs. 18.12 and 18.13 later in the chapter). If the diamond instrument is to be used exclusively in finishing the enamel walls and margins, this procedure is postponed until after the removal of soft dentin, old restorative material, or both and the application of any necessary base. Waiting prevents any hemorrhage (which occasionally follows the beveling of the gingival margin) from hindering (1) the suitable removal of remaining soft dentin and old restorative material and (2) the proper application of a necessary base. Hand instruments are more useful on the mesiofacial surfaces of maxillary premolars and first molars, where minimal extension is desired to prevent an unsightly display of metal.

Shallow (0.3-mm deep) retention grooves may be cut in the facioaxial and linguoaxial line angles with the No. 169L carbide bur (see Fig. 18.8E–I). These grooves are indicated especially when the prepared tooth is short. When properly positioned, the grooves are in sound dentin, close to but not contacting the DEJ. The long axis of the bur must be held parallel to the line of draw. Preparing these grooves may be postponed until after any required bases are applied during the final preparation.

Final Preparation

Removal of Soft Dentin and Pulp Protection

After the initial preparation has been completed, the dentist evaluates the internal walls of the preparation visually and tactilely (with an explorer) for indications of any remaining soft dentin. If soft dentin remains and if it is judged to be shallow or moderate (≥1 mm of remaining dentin between the caries and the pulp), satisfactory isolation for the removal of such caries and the application of any necessary base may be attained by reducing salivation through anesthesia and the use of cotton rolls, a saliva ejector, and gingival retraction cord. The retraction cord also serves to widen the gingival sulcus and slightly retract the gingiva in preparation for beveling and flaring the proximal margins (Fig. 18.9; see also Fig. 18.12A and B). For insertion of the cord, see the sections on preparation of bevels and flares and tissue retraction. The removal of the remaining caries and placement of a necessary base can be accomplished during the time required for the full effect of the inserted cord. A slowly revolving round bur (No. 2 or No. 4) or spoon excavator is used to remove soft dentin (see Fig. 18.9F and G). If a bur is used, visibility can be improved by using air alone. This excavation is done just above stall-out speed with light, intermittent cutting. The operator should avoid unnecessarily desiccating the exposed dentin during this procedure.

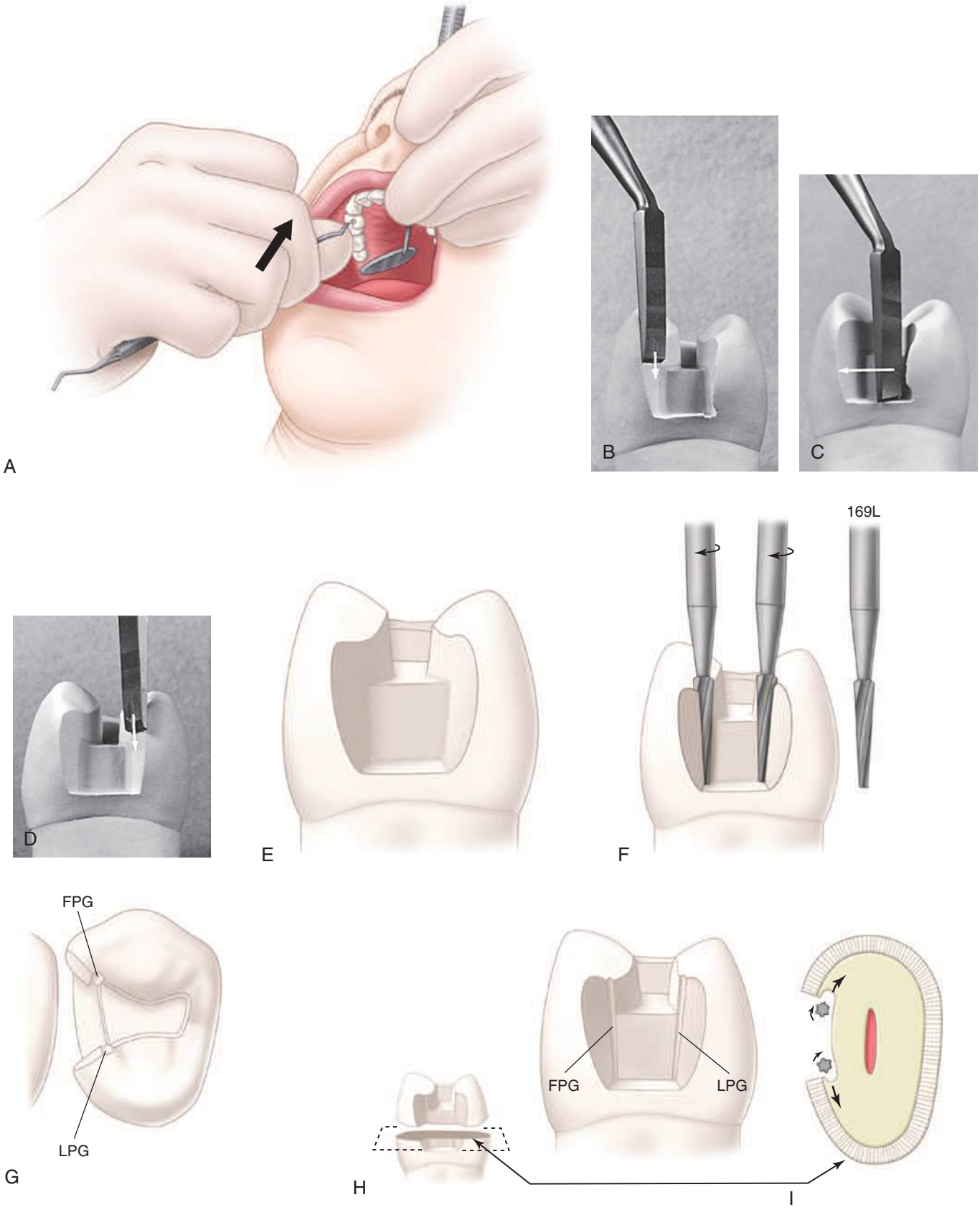
Light-cured glass ionomer cement may be applied with a suitable applicator to these shallow (or moderately deep) excavated regions to the depth and form of the ideally prepared surface. Placing the material takes little time and should be considered because it results in preparation walls with no undercuts and "ideal" position and contour. This facilitates making the temporary restoration, and also applying the light-cured glass ionomer at this time minimizes additional irritation of the pulp during subsequent procedures necessary for the completion of the restoration. The material is applied by conveying small portions on the end of a periodontal probe and is light-cured when the correct form has been achieved (see Fig. 18.9H and I). Any excess cement can be trimmed back to the ideal form with the No. 271 carbide bur after the cement has hardened.

If the caries lesion is judged to approach the pulp closely, a rubber dam should be applied and the soft dentin removed as indicated (see Chapter 2). Rubber dam provides the optimal environment for successfully treating a pulp exposure should it occur. If removal of soft dentin leads directly to a pulpal exposure (carious pulpal exposure), root canal treatment should be accomplished before completing the cast-metal restoration.

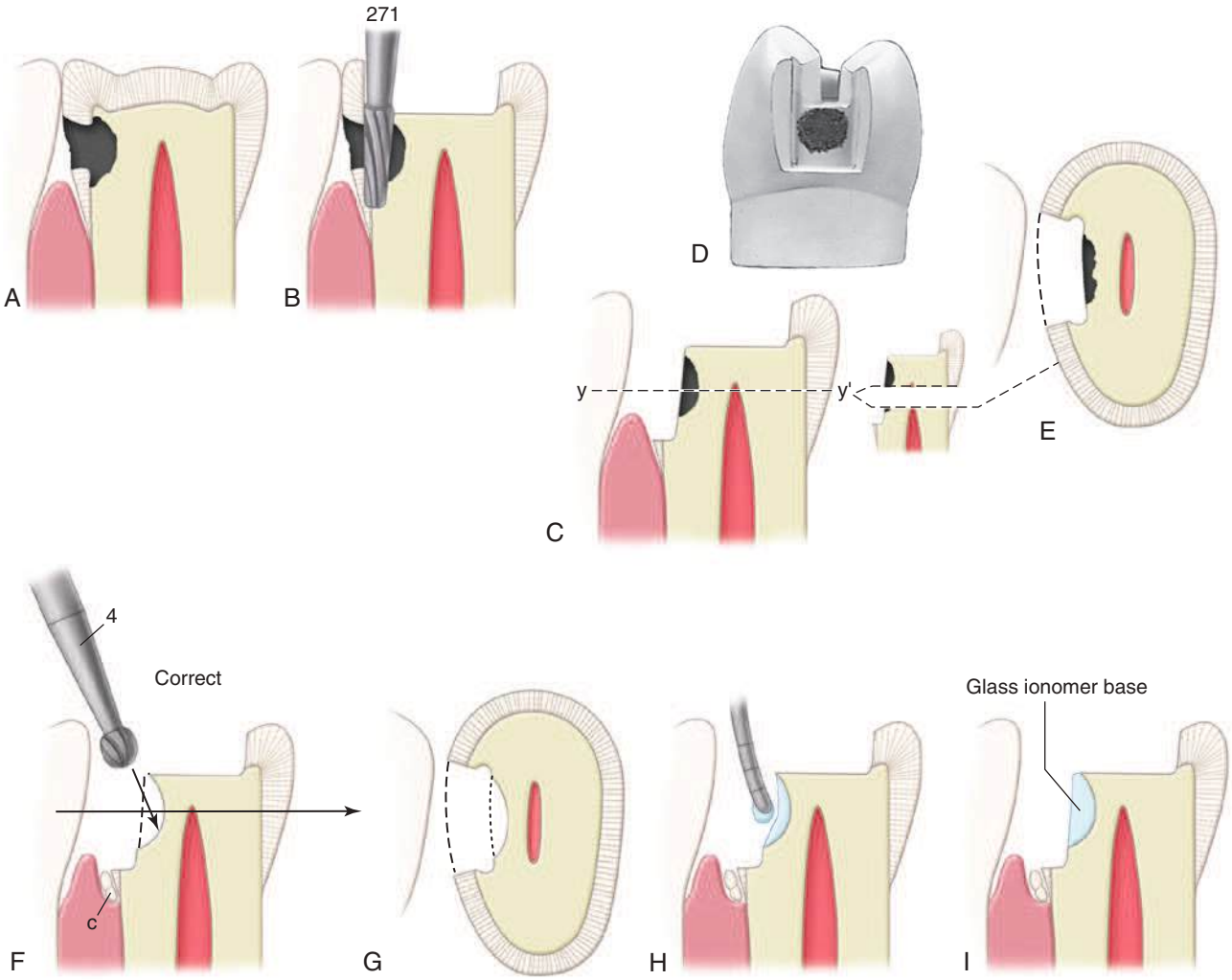
If the pulp is inadvertently exposed as a result of operator error or misjudgment (mechanical pulpal exposure), the operator must decide whether to proceed with the root canal treatment or to attempt a direct pulp capping procedure. A clinical evaluation should be made to determine the health of the pulp. A favorable prognosis for the pulp after direct pulp capping may be expected if the following criteria are met:

- The exposure is small (<0.5 mm in diameter).
- The tooth has been asymptomatic, showing no signs of pulpitis.
- Any hemorrhage from the exposure site is easily controlled.
- The invasion of the pulp chamber was relatively atraumatic with little physical irritation to the pulp tissue.
- A clean, uncontaminated operating field is maintained (i.e., by using a rubber dam).

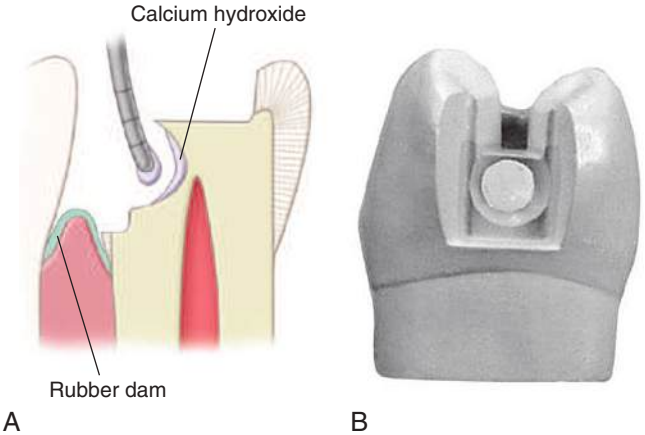
If the excavation closely approaches the pulp or if a direct pulp cap is indicated, the dentist should first apply a lining of calcium hydroxide using a flow technique (without pressure). This calcium hydroxide liner should cover and protect any possible near or actual exposure and extend over a major portion of the excavated dentinal surface (Fig. 18.10). Although undetected, an exposed recessional tract of a pulp horn may exist in any deep excavation. Calcium hydroxide treatment of an exposed, healthy pulp promotes the formation of a dentin bridge, which would close the exposure.[3] A resin-modified glass ionomer base should be applied and should completely cover the calcium hydroxide lining as well as some peripheral dentin for good adhesion (Fig. 18.11). The base should be sufficiently thick in dimension to protect the thin underlying dentin and the calcium hydroxide liner from subsequent stresses. Usually good resistance form dictates that the pulpal wall should not be formed entirely by a base; rather, in at least two regions, one diametrically across the excavation from the other, the pulpal wall should be in normal position, flat, and formed by sound dentin (see region *S* in Fig. 18.11E, which depicts basing in a mandibular molar). The dentist should consider the addition of other retention features such as proximal grooves if a major portion of a proximal axial wall is composed mostly of base because this base should not be relied on for contributing to retention of the cast restoration (see Fig. 18.8F).



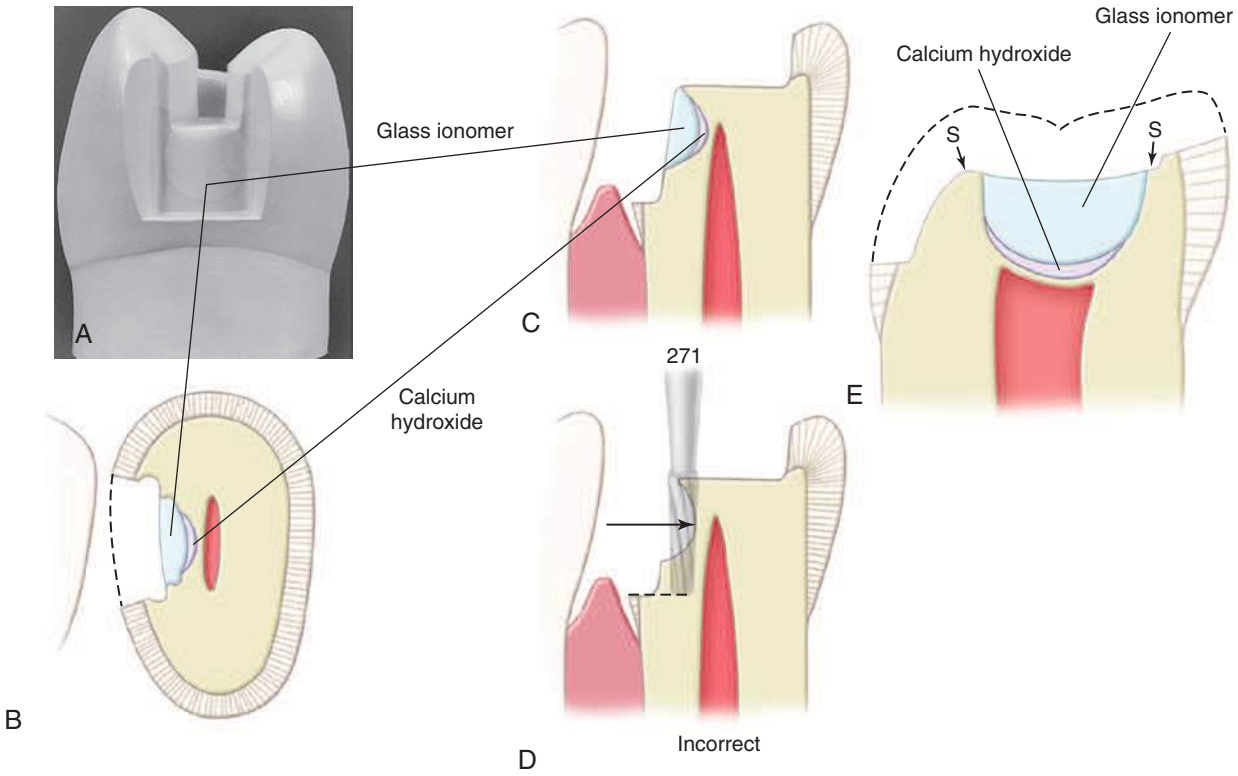
• **Fig. 18.8** A–D, Using modified palm-and-thumb grasp (A) to plane distofacial and distolingual walls (B and D) and to scrape gingival wall (C). E, Before cutting retention grooves. F, Cutting retention grooves. G and H, Facial proximal groove *(FPG)* and lingual proximal groove *(LPG)*. I, Section in plane *x*. Large arrows depict the direction of translation of the rotating bur.



• **Fig. 18.9** Moderately deep caries lesion. A–C, Extending the proximal ditch gingivally (B) to a sound floor free from caries (C). D, Remaining caries on the axial wall. E, Section of C in plane *yy*. F, Removing the remaining soft dentin. *c*, Inserted retraction cord. G, Section of F. H, Inserting glass ionomer base with periodontal probe. I, Completed base.



• **Fig. 18.10** A and B, Deep caries excavations near the pulp are lined with calcium hydroxide. Note the rubber dam.



• **Fig. 18.11** A–C, Completed base for the treatment of deep caries. D, *Never* deepen entire axial wall with the side of a fissure bur to remove a caries lesion because the pulp would be greatly irritated from the resulting closeness of the gingivoaxial region of the preparation. E, Base placed deep in the excavation on the mandibular molar. Note the flat seats in sound dentin *(S)* that are desirable for adequate resistance form.

Any remaining old restorative material on the internal walls should be removed if any of the following conditions are present: (1) The old material is judged to be thin, nonretentive, or both; (2) radiographic evidence of a caries lesion under the old material is present; (3) the pulp was symptomatic preoperatively; or (4) the periphery of the remaining restorative material is not intact (i.e., some breach exists in the junction of the material with the adjacent tooth structure that may indicate a caries lesion under the material). If none of these conditions is present, the operator may elect to leave the remaining restorative material to serve as a base, rather than risk unnecessary removal of sound dentin or irritation or exposure of the pulp. The same isolation conditions described previously for the removal of soft dentin also apply for the removal of old restorative material.

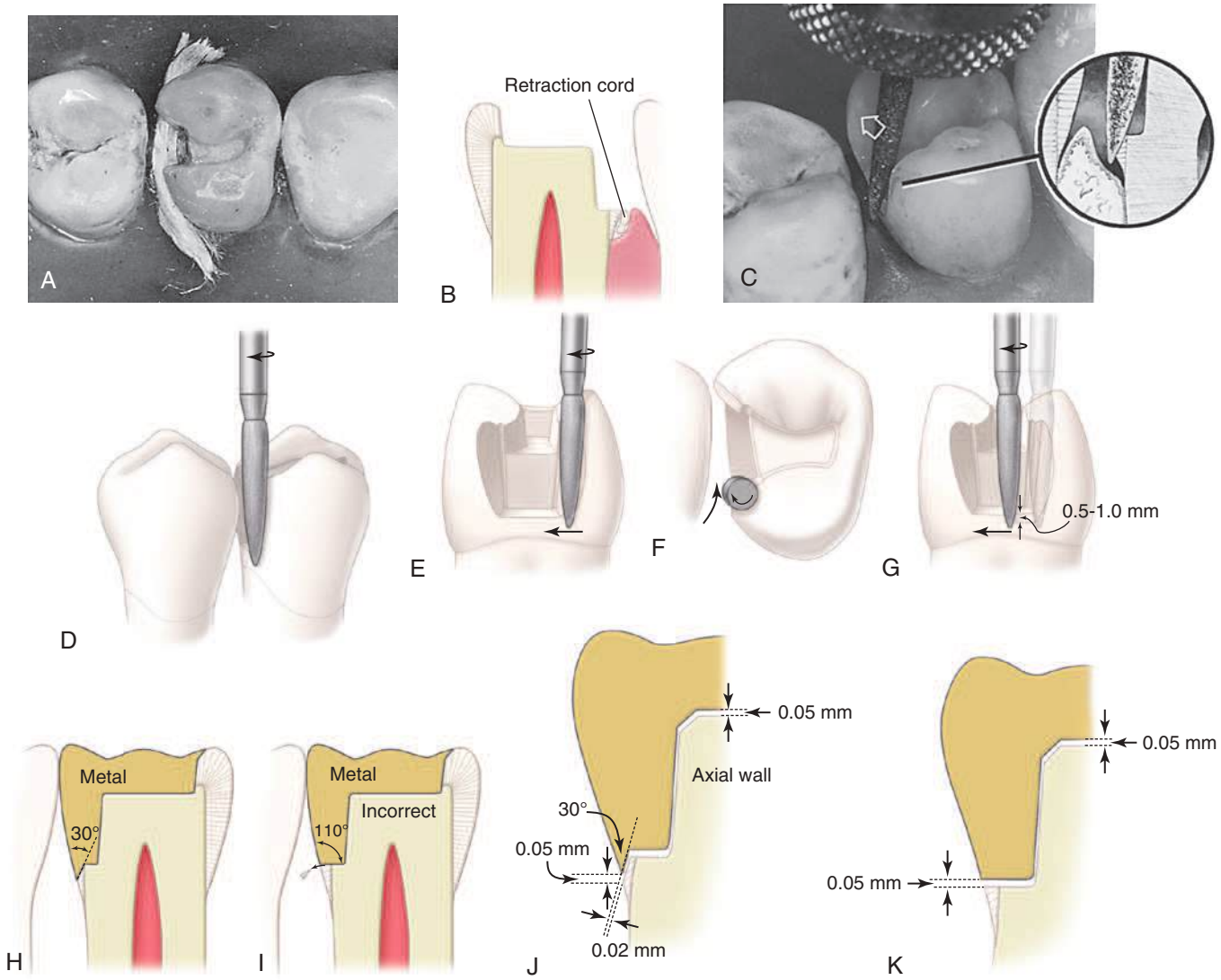
Future root canal therapy is a possibility for any tooth treated for deep caries that approximates or exposes the pulp. When treating a tooth that has had such extensive caries, the following should be considered: (1) reducing all cusps to cover the occlusal surface with metal, for better distribution of occlusal loads, and (2) adding skirts to the preparation to augment the resistance form because teeth are more prone to fracture after root canal therapy.

Preparation of Bevels and Flares

After the base (where indicated) is completed, the slender, flame-shaped, fine-grit diamond instrument is used to bevel the occlusal and gingival margins and to apply the secondary flare on the distolingual and distofacial walls. This should result in 30- to 40-degree marginal metal on the inlay (refer to upcoming Figs. 18.12H, 18.13J, and 18.14B). This cavosurface design helps seal and protect the margins and results in a strong enamel margin with an angle of 140 to 150 degrees. A cavosurface enamel angle of more than 150 degrees is incorrect because it results in a less-defined enamel margin (finish line), and the marginal cast-metal alloy is too thin and weak if its angle is less than 30 degrees. Conversely, if the enamel margin is 140 degrees or less, the metal is too bulky and difficult to burnish when its angle is greater than 40 degrees (see Fig. 18.14F later in the chapter).

Usually it is helpful to insert a gingival retraction cord of suitable diameter into the gingival sulcus adjacent to the gingival margin and leave it in place for several minutes just before the use of the flame-shaped diamond instrument on the proximal margins (Fig. 18.12A–C). The cord should be small enough in diameter to permit relatively easy insertion and to preclude excessive pressure against the gingival tissue, and yet it should be large enough to widen the sulcus to about 0.5 mm. Immediately before the flame-shaped diamond instrument is used, the cord may be removed to create an open sulcus that improves visibility for beveling the gingival margin and helps prevent injury and subsequent hemorrhage of gingival tissue. Some operators prefer to leave the cord in the sulcus while placing the gingival bevel.

Using the flame-shaped diamond instrument that is rotating at high speed, the dentist prepares the lingual secondary flare (Fig. 18.13A; also see Fig. 18.12D–F). The dentist approaches from the lingual embrasure (see Fig. 18.12F), moving the instrument mesiofacially. The direction of the distolingual wall and the position of the distolingual margin are compared before and after this extension (see Figs. 18.8G and 18.13A). The distolingual wall extends from the linguoaxial line angle into the lingual embrasure

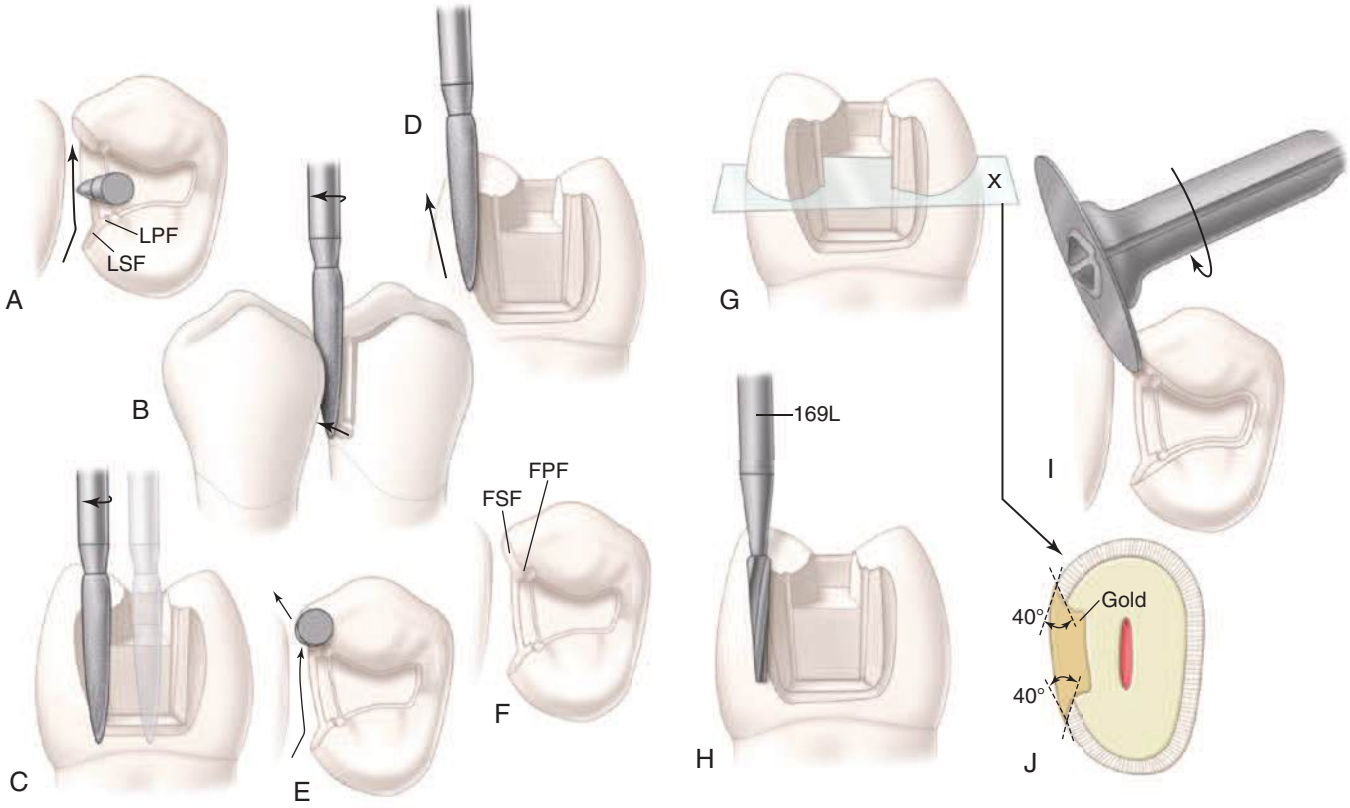


• **Fig. 18.12** A and B, The retraction cord is inserted in the gingival sulcus and left for several minutes. C, An open gingival sulcus after the cord shown in *A* is removed facilitates beveling the gingival margin with a diamond instrument. D–F, Diamond instrument preparing lingual secondary flare. Large arrow in F indicates the direction of the translation. G, Beveling the gingival margin. Note in *C* the mesial tilting of diamond instrument to produce a bevel that is properly directed to result in 30-degree marginal metal as shown in H. H, Properly directed gingival bevel resulting in 30-degree marginal metal. I, Failure to bevel the gingival margin results in a weak margin formed by undermined rods (note the easily displaced wedge of enamel) and 110-degree marginal metal, an angular design unsuitable for burnishing. J, Lap, sliding fit of prescribed bevel metal decreases the 50-μm error of seating to 20 μm. K, A 50-μm error of seating produces an equal cement line of 50 μm along the unbeveled gingival margin.

in two planes (see Fig. 18.13A). The first is termed *lingual primary flare*; the second is termed *lingual secondary flare*. During this (secondary) flaring operation, the long axis of the instrument is held nearly parallel to the line of draw, with only a slight tilting mesially and lingually for assurance of draft (see Fig. 18.12D and E), and the direction of translation of the instrument is that which results in a marginal metal angle of 40 degrees (see Figs. 18.12F and 18.13J).

The dentist bevels the gingival margin by moving the instrument facially along the gingival margin (see Figs. 18.12G and 18.13A). While cutting the gingival bevel, the rotational speed should be reduced to increase the sense of touch; otherwise, overbeveling may result. The instrument should be tilted slightly mesially to produce a gingival bevel with the correct steepness to result in 30-degree marginal metal (see Fig. 18.12C, H, and J). If the instrument is not tilted in this manner, the bevel is too steep, resulting in gingival bevel metal that is too thin (<30-degree metal) and too weak. Although the instrument is tilted mesially, its long axis must not tilt facially or lingually (see Fig. 18.12G). The gingival bevel should be 0.5 to 1 mm wide and should blend with the lingual secondary flare.

The operator completes the gingival bevel and prepares the facial secondary flare (see Fig. 18.13A–F). The long axis of the instrument during this secondary flare is again returned nearly to



• **Fig. 18.13** A, Occlusal view of Fig. 18.12G. *LSF,* Lingual secondary flare; *LPF,* lingual primary flare. B–E, Preparing the facial secondary flare. Large arrows in *B*, *D*, and *E* indicate the direction of the translation. F, Completed facial secondary flare. *FSF,* Facial secondary flare; *FPF*, facial primary flare. G, Distal view of *F*. *x*, Plane of cross section shown in J. H and I, Preparing the secondary flare with the No. 169L carbide bur (H) or with paper disk (I). J, The secondary flares are directed to result in 40-degree marginal metal and 140-degree marginal enamel.

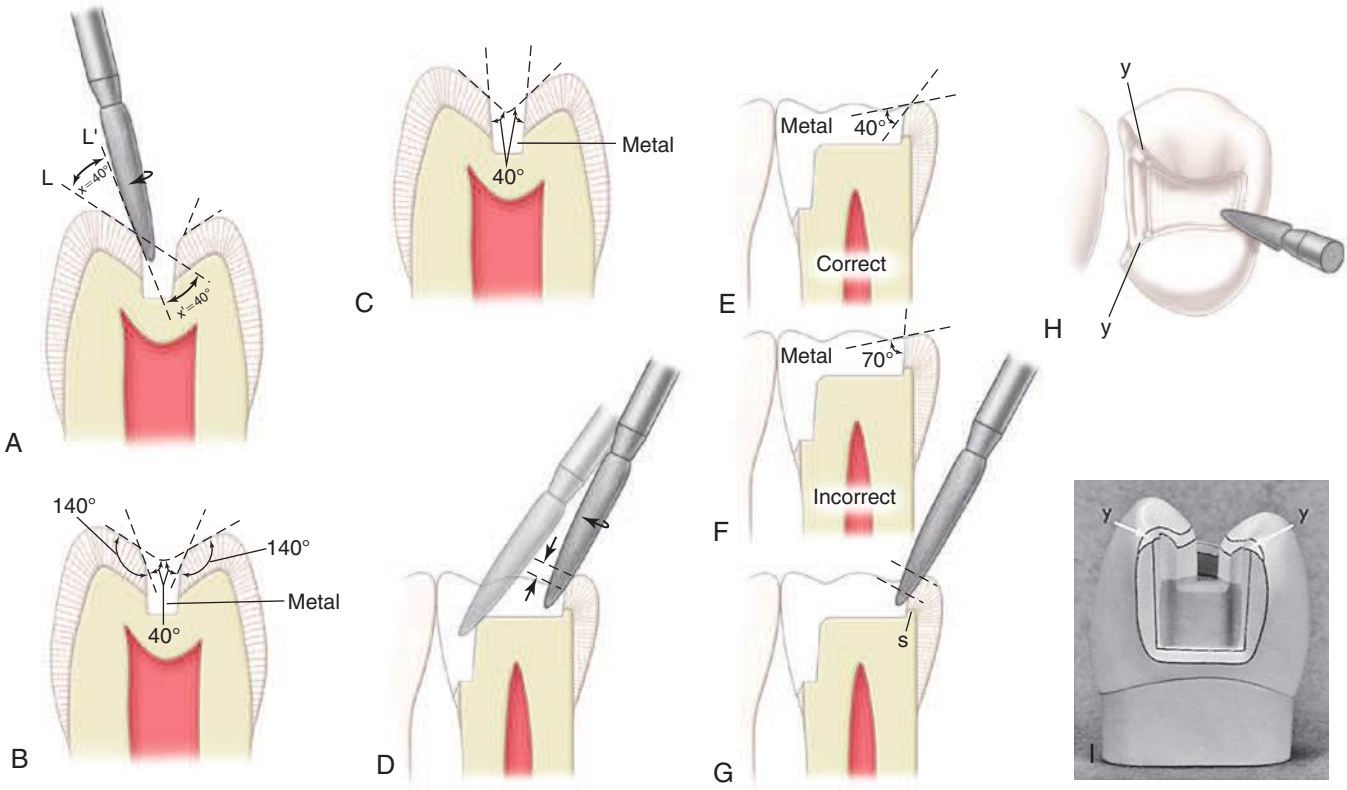
the line of draw, with only a small tilting mesially and facially, and the direction of translation of the instrument is that which results in 40-degree marginal metal (see Fig. 18.13E and J). When the adjacent proximal surface (mesial of the second premolar) is not being prepared, care must be exercised to avoid abrading the adjacent tooth and overextending the distofacial margin. To prevent such abrasion or overextension, the instrument may be raised occlusally (using the narrower portion at its tip end) to complete the most facial portion of the wall and margin (see Fig. 18.13D). Also the more slender No. 169L carbide bur may be used, rather than the flame-shaped diamond instrument (see Fig. 18.13H). The No. 169L bur produces an extremely smooth surface to the secondary flare and a smooth, straight distofacial margin. When access permits, a fine-grit sandpaper disk may be used on the facial and lingual walls and on the margins of the proximal preparation, especially when minimal extension of the facial margin is desired (see Fig. 18.13I). This produces smooth walls and helps create respective margins that are straight (not ragged) and sound.

In the flaring and beveling of the proximal margins, as described in the previous paragraphs, the procedure began at the lingual surface and proceeded to the facial surface. The direction may be reversed, however, starting at the facial surface and moving toward the lingual surface. On the mesiofacial surface of maxillary premolars and first molars where extension of the facial margin should be minimal, it is usually desirable to use the lingual-to-facial direction.

The gingival bevel serves the following purposes:

- Weak enamel is removed. If the gingival margin is in the enamel, it would be weak if not beveled because of the gingival declination of the enamel rods (see Fig. 18.12I).
- The bevel results in 30-degree metal that is burnishable (on the die) because of its angular design (see Fig. 18.12H). Bulky 110-degree metal along an unbeveled margin is not burnishable (see Fig. 18.12I).
- A lap, sliding fit is produced at the gingival margin (see Fig. 18.12J). This helps improve the fit of the casting in this region. With the prescribed gingival bevel, if the inlay fails to seat by 50 µm, the void between the bevel metal and the gingival bevel on the tooth may be 20 µm; however, failure to apply such a bevel would result in a void (and a cement line) as great as in the failure to seat (see Fig. 18.12K).

Uninterrupted blending of the gingival bevel into the secondary flares of the distolingual and distofacial walls results in the distolingual and distofacial margins joining the gingival margin in a desirable arc of a small circle; also the gingivofacial and gingivolingual line angles no longer extend to the marginal outline. If such line angles are allowed to extend to the preparation outline, early failure may follow because of an "open" margin, dissolution



• **Fig. 18.14** A, The diamond instrument beveling the occlusal margin when it is indicated to result in 40-degree marginal metal, as shown in *B*. Angles *x* and *x′* are equal because the opposite angles are equal when two lines (*L* and *L′*) intersect. The diamond instrument is always directed such that an angle of 40 degrees is made by the side of the instrument and the external enamel surface. B, Occlusal marginal metal is approximately 40 degrees in cross section, making the enamel angle 140 degrees. C, When the cuspal inclines are steep, no beveling is indicated considering that 40-degree metal would result without beveling. D, Beveling the mesial margin and the axiopulpal line angle. E, The mesial bevel is directed correctly to result in 40-degree marginal metal. F, An unbeveled mesial margin is incorrect because it results in a weak enamel margin and unburnishable marginal metal. G, To conserve dentinal support *(s)*, occlusal defects on the marginal ridge are included in the outline form by applying a cavosurface bevel, which may be wider than usual, when necessary. H, Occlusal view of G. Preparing a 140-degree cavosurface enamel angle at regions labeled *y* usually dictates that the occlusal bevel be extended over the marginal ridges into the secondary flares. I, Distal view of *H*.

of exposed cement, and eventual leakage, all potentially resulting in caries.

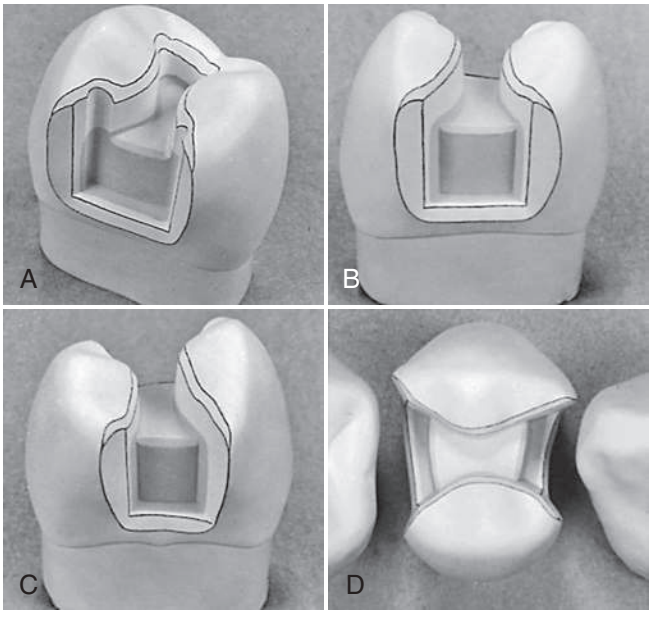
The secondary flare is necessary for several reasons: (1) The secondary flaring of the proximal walls extends the margins into the embrasures, making these margins more self-cleaning and more accessible to finishing procedures during the inlay insertion appointment, and does so with conservation of dentin. (2) The direction of the flare results in 40-degree marginal metal (see Fig. 18.13J). Metal with this angular design is burnishable; however, metal shaped at a larger angle is unsatisfactory for burnishing; metal with an angle less than 30 degrees is too thin and weak, with a corresponding enamel margin that is too indefinite and ragged. (3) A more blunted and stronger enamel margin is produced because of the secondary flare.

In a later section, the secondary flare is omitted for esthetic reasons on the mesiofacial proximal wall of preparations on premolars and first molars of the maxillary dentition. In this location, the wall is completed with minimal extension by using either hand instruments (straight or bin-angle chisel) followed by a fine-grit sandpaper disk or very thin rotary instruments.

The flame-shaped, fine-grit diamond instrument also is used for occlusal bevels. The width of the cavosurface bevel on the occlusal margin should be approximately one fourth the depth of the respective wall (Fig. 18.14A and B). The exception to the rule is when a wider bevel is desired to include an enamel defect (see Fig. 18.14G and H). The resulting occlusal marginal metal of the inlay should be 40-degree metal; the occlusal marginal enamel is 140-degree enamel (see Fig. 18.14B and E). Beveling the occlusal margins in this manner increases the strength of the marginal enamel and helps seal and protect the margins. While beveling the occlusal margins, a guide to diamond positioning is to maintain an approximate 40-degree angle between the side of the instrument and the external enamel surface; this also indicates when an occlusal bevel is necessary (see Fig. 18.14A). If the cusp inclines are so steep that the diamond instrument, when positioned at a 40-degree angle to the external enamel surface, is parallel with the enamel preparation wall, no bevel is indicated (see Fig. 18.14C). By using this technique, it can be seen that margins on the proximal marginal ridges always require a cavosurface bevel (see Fig. 18.14D and I).

Failure to apply a bevel in these regions leaves the enamel margin weak and subject to injury by fracture before the inlay insertion appointment and during the try-in of the inlay when burnishing the marginal metal. Also failure to bevel the margins on the marginal ridges results in metal alloy that is difficult to burnish because it is too bulky (see Fig. 18.14F). Similarly, the importance of extending the occlusal bevel to include the portions of the occlusal margin that cross over the marginal ridge cannot be overemphasized (see Fig. 18.14H and I). These margins are beveled to result in 40-degree marginal metal. Otherwise fracture of the enamel margin in such stress-vulnerable regions may occur in the interim between the preparation and the cementation appointment.

The diamond instrument also is used to bevel the axiopulpal line angle lightly (see Fig. 18.14D). Such a bevel provides a thicker and stronger wax pattern at this critical region. The desirable metal angle at the margins of inlays is 40 degrees except at the gingival margins, where the metal angle should be 30 degrees. The completed preparation is illustrated in Fig. 18.15A.



• **Fig. 18.15** A, Completed distoocclusal preparation for the inlay. B, Mesioocclusodistal preparation for the inlay on the maxillary right first premolar, distoocclusal view. C, Same preparation as in B, mesioocclusal view. D, Same preparation as in B, occlusal view. Note the absence, for esthetic reasons, of secondary flare on the mesiofacial aspect and minimal extension of the mesiofacial margin.

Modifications in Inlay Tooth Preparations

Because the indications for small inlays are rare, the following sections provide procedural information that may promote better understanding of their applications in more complex and larger inlay or onlay restorations.

Mesioocclusodistal Preparation

If a marginal ridge is severely weakened because of excessive extension, the preparation outline often should be altered to include the proximal surface. The distoocclusal preparation illustrated in the previous section would be extended to a mesioocclusodistal preparation (Fig. 18.16; see also Fig. 18.15B–D). The decision to extend the preparation in this manner calls for clinical judgment as to whether the remaining marginal ridge would withstand occlusal forces without fracture. A fortunate factor in favor of not extending the preparation is that such ridge enamel usually is composed of gnarled enamel and is stronger than it appears. Caries lesions present on both proximal surfaces would result in a mesioocclusodistal preparation and restoration. The only difference in technique as described previously is the inclusion of the other proximal surface.

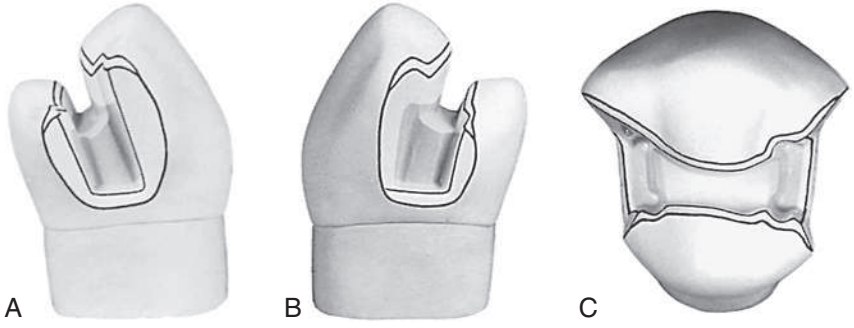
Modifications of Class II Preparation for Esthetics

For esthetic reasons, minimal flare is desired for the mesiofacial proximal wall in maxillary premolars and first molars in Class II cast-metal preparations (see Fig. 18.15D). The mesiofacial margin is minimally extended facially of the contact to such a position that the margin is barely visible from a facial viewing position. To accomplish this, the secondary flare is omitted, and the wall and margin are developed with (1) a chisel or enamel hatchet and final smoothing with a fine-grit paper disk or (2) a narrow diamond or bur when access permits.

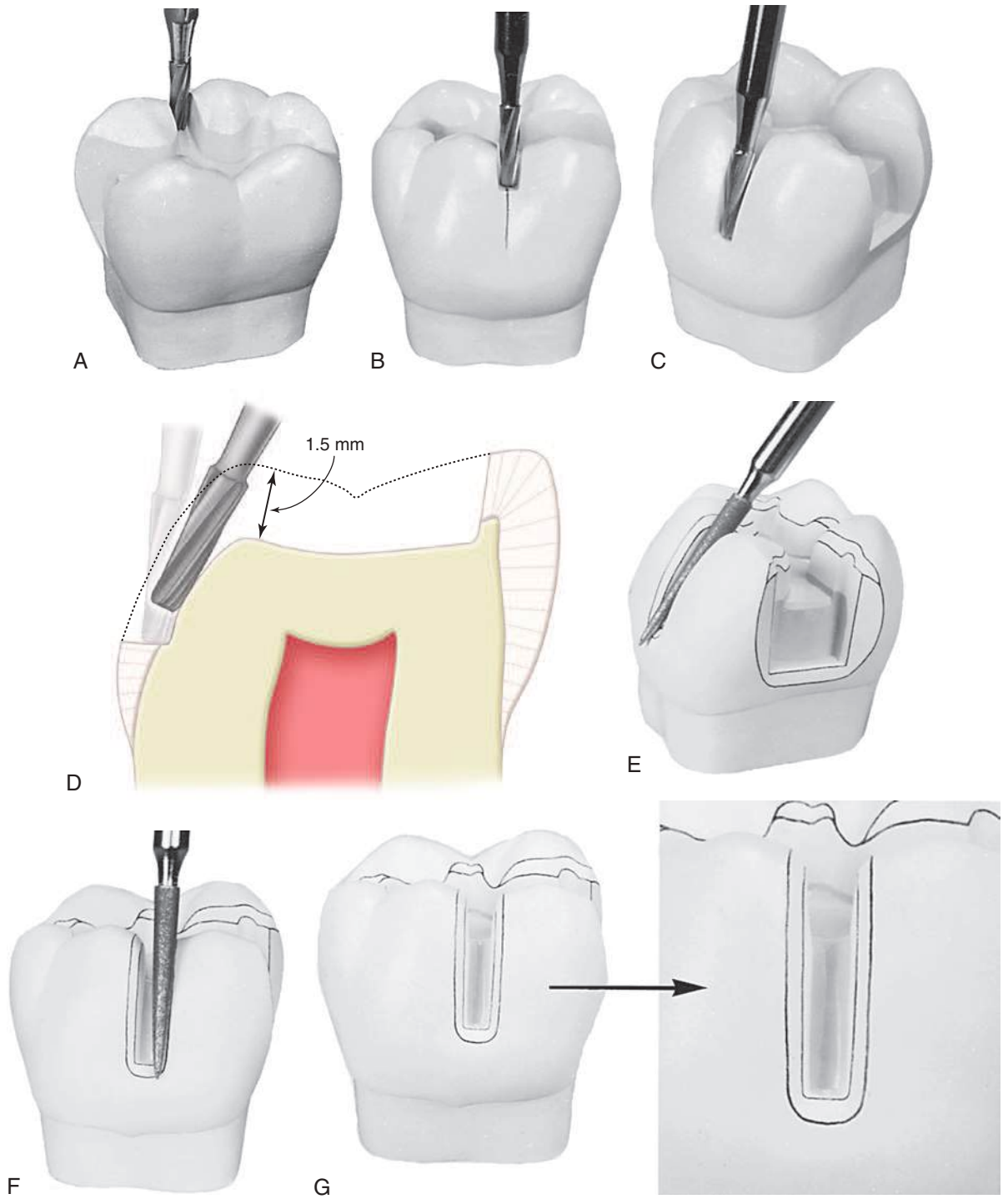
Facial or Lingual Surface Groove Extension

Sometimes, a faulty facial groove (fissure) on the occlusal surface is continuous with a faulty facial surface groove (mandibular molars), or a faulty distal oblique groove on the occlusal surface is continuous with a faulty lingual surface groove (maxillary molar). This situation requires extension of the preparation outline to include the fissure to its termination (Fig. 18.17; see also Fig. 18.19C). Occasionally the operator may extend further gingivally than the fissure length to improve retention form. Such groove extensions, when sufficiently long, are effective for increasing retention. Likewise, this extension may be indicated to provide sufficient retention form even though the facial or lingual surface grooves are not fissured.

For extension onto the facial surface, the dentist uses the No. 271 carbide bur held parallel to the line of draw and extends



• **Fig. 18.16** Mandibular first premolar prepared for the mesioocclusodistal inlay. Distal view (A), mesial view (B), and occlusal view (C).



• **Fig. 18.17** A–C, Extending to include the occlusal fissure that is continuous with the facial fissure on the facial surface. D, Section of C. E and F, Beveling the gingival margin (E) and the mesial and distal margins (F) of fissure extension. G, Beveling completed.

through the facial ridge (see Fig. 18.17A and B). The depth of the cut should be 1.5 mm. The floor (pulpal wall) should be continuous with the pulpal wall of the occlusal portion of the preparation (see Fig. 18.17D).

With the bur still aligned with the path of draw, the dentist uses the side of the bur to cut the facial surface portion of this extension (see Fig. 18.17C). The diameter of the bur serves as a depth gauge for the axial wall, which is in dentin. The blade portion of the No. 271 bur is 0.8 mm in diameter at its tip end and 1 mm at the neck; the axial wall depth should approximate 1 mm or slightly more. The bur should be tilted lingually as it is drawn occlusally, to develop the uniform depth of the axial wall (see Fig. 18.17D). The same principles apply for the extension of a lingual surface groove.

When a facial or lingual groove is included, it also must be beveled. With the flame-shaped, fine-grit diamond instrument, the operator bevels the gingival margin (using no more than one third the depth of the gingival floor) to provide for 30-degree marginal metal (see Fig. 18.17D). The operator applies a light bevel on the mesial and distal margins that is continuous with the occlusal and gingival bevels and results in 40-degree metal at these margins (see Fig. 18.17F and G). The bevel width around the extended groove is approximately 0.5 mm.

Class II Preparation for Abutment Teeth and Extension Gingivally to Include Root-Surface Lesions

Extending the facial, lingual, and gingival margins may be indicated on the proximal surfaces of abutments for removable partial dentures to increase the surface area for the development of guiding planes. In addition, the occlusal outline form must be wide enough faciolingually to accommodate any contemplated rest preparation without involving the margins of the restoration. These extensions may be accomplished by simply increasing the width of the bevels.

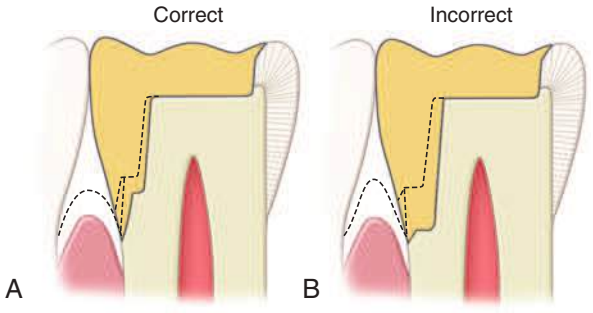
The following modified preparation is recommended when further gingival extension is indicated to include a root lesion on the proximal surface. The gingival extension should be accomplished primarily by lengthening the gingival bevel, especially when preparing a tooth that has a longer clinical crown than normal as a result of gingival recession. It is necessary to extend (gingivally) the gingival floor only slightly, and although the axial wall consequently must be moved pulpally, this should be minimal. If additional extension of the gingival floor is necessary, it should not be as wide pulpally as when the floor level is at a normal position (Fig. 18.18A). These considerations are necessary because of the draft requirement and because the tooth is smaller apically. Extending the preparation gingivally without these modifications would result in a dangerous encroachment of the axial wall on the pulp (see Fig. 18.18B).

Maxillary First Molar With Unaffected, Strong Oblique Ridge

When a maxillary first molar is to be restored, consideration should be given to preserving the oblique ridge if it is strong and unaffected, especially if only one proximal surface is carious. A mesioocclusal preparation for an inlay is illustrated in Fig. 18.19A and B. If a distal surface lesion appears subsequent to the insertion of a mesioocclusal restoration, the tooth may be prepared for a distoocclusolingual inlay (see Fig. 18.19H and I). The distoocclusolingual restoration that caps the distolingual cusp is preferable to the distoocclusal restoration because it protects the miniature distolingual cusp from subsequent fracture. The distoocclusolingual preparation requires diligent application to develop satisfactory retention and resistance forms. Retention form is attained by (1) creating a maximum of 2-degree occlusal divergence of the vertical walls, (2) accentuating some line angles, and (3) extending the lingual surface groove to create an axial wall height in this extension of at least 2.5 mm occlusogingivally. The proper resistance form dictates (1) routine capping of the distolingual cusp and (2) maintaining sound tooth structure between the lingual surface groove extension and the distolingual wall of the proximal boxing.

To prepare the distoocclusolingual preparation, the operator first reduces the distolingual cusp with the side of the No. 271 carbide bur. The cusp should be reduced a uniform 1.5 mm. Next the operator completes the remaining occlusal step of the preparation with the No. 271 carbide bur. The operator prepares the proximal box portion of the preparation. The lingual groove extension is prepared only after the position of the distolingual wall of the proximal boxing is established. This permits the operator to judge the best position of the lingual surface groove extension to maintain a minimum of 3 mm of sound tooth structure between this extension and the distolingual wall; if this is not possible because of an extensive caries lesion, a more extensive type of preparation may be indicated (one that crosses the oblique ridge). One can use the side of the No. 271 carbide bur to produce the lingual surface groove extension (see Fig. 18.19C). The diameter of the bur is the gauge for the depth (pulpally) of the axial wall in this extension, and the occlusogingival dimension of this axial wall is a minimum of 2.5 mm. With the end of this bur, the operator also establishes a 2-mm depth to the portion of the pulpal floor that connects the proximal boxing to the lingual surface groove extension. This additional depth to the pulpal floor helps strengthen the wax pattern and casting in later steps of fabrication. This should create a definite 0.5-mm step from the reduced distolingual cusp to the pulpal floor. Using the No. 169L carbide bur, the operator increases retention form in the distoocclusolingual preparation by (1) creating mesioaxial and distoaxial grooves in the lingual surface groove extension (see Fig. 18.19D) and (2) preparing facial and lingual retention grooves in the distal boxing (see Fig. 18.19E).

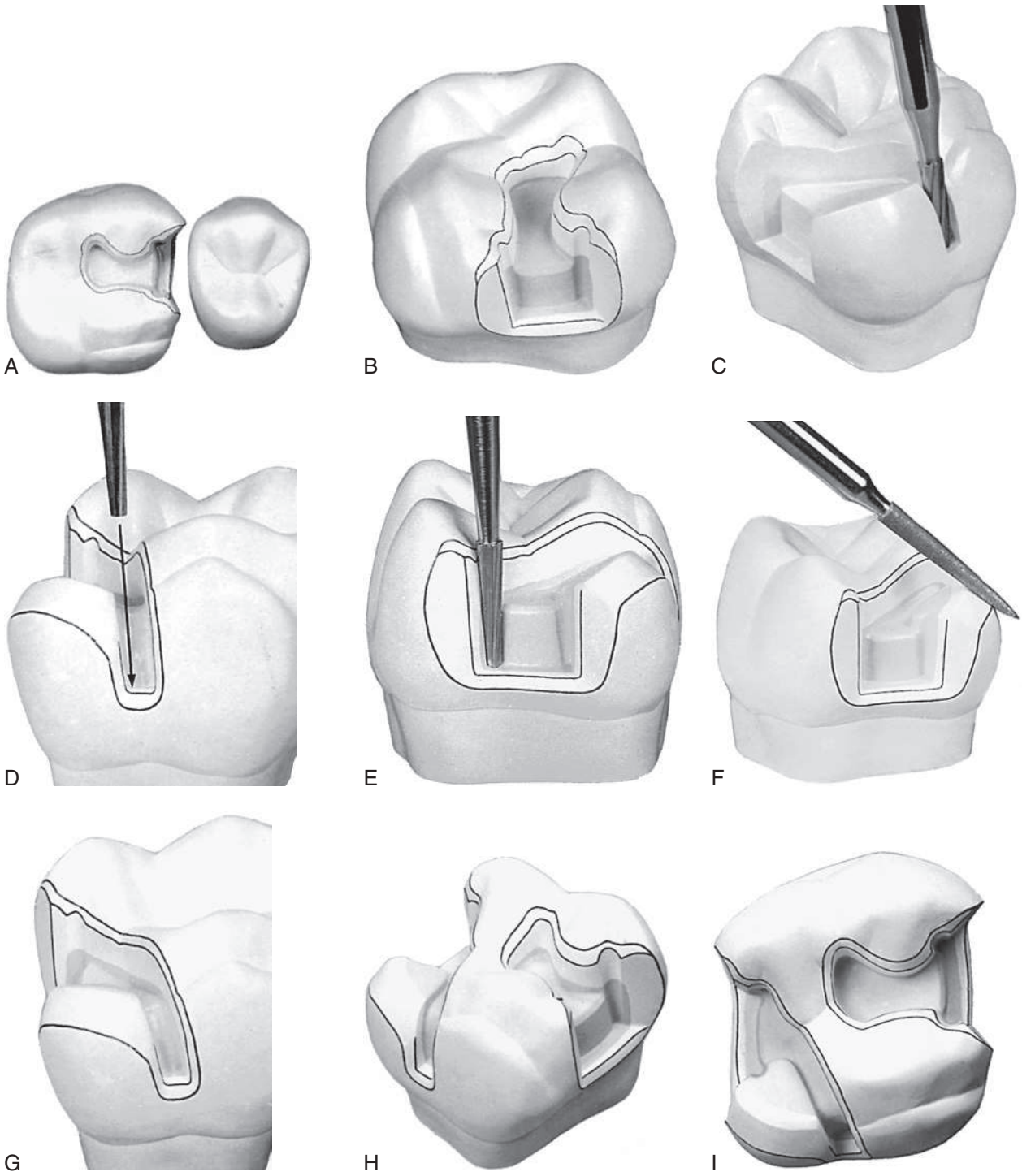
The dentist uses the flame-shaped, fine-grit diamond instrument to bevel the proximal gingival margin and to prepare the secondary flares on the proximal enamel walls and to bevel the lingual margins. A lingual counterbevel is prepared on the distolingual cusp that is generous in width and results in 30-degree metal at the margin (see Fig. 18.19F). Occlusion should be checked at this point because the counterbevel should be sufficiently wide to extend beyond any occlusal contacts, either in maximum intercuspation or during mandibular movements. The bevel on the gingival margin of the lingual extension should be 0.5 mm wide and should provide for a 30-degree metal angle. The bevels on the mesial and distal margins of the lingual extension also are approximately 0.5 mm wide and result in 40-degree marginal metal.



• **Fig. 18.18** Modifications of the preparation when extending to include the proximal root-surface lesions after moderate gingival recession. A, Correct. B, Incorrect. Note the decreased dentinal protection of the pulp compared with the management depicted in *A*.

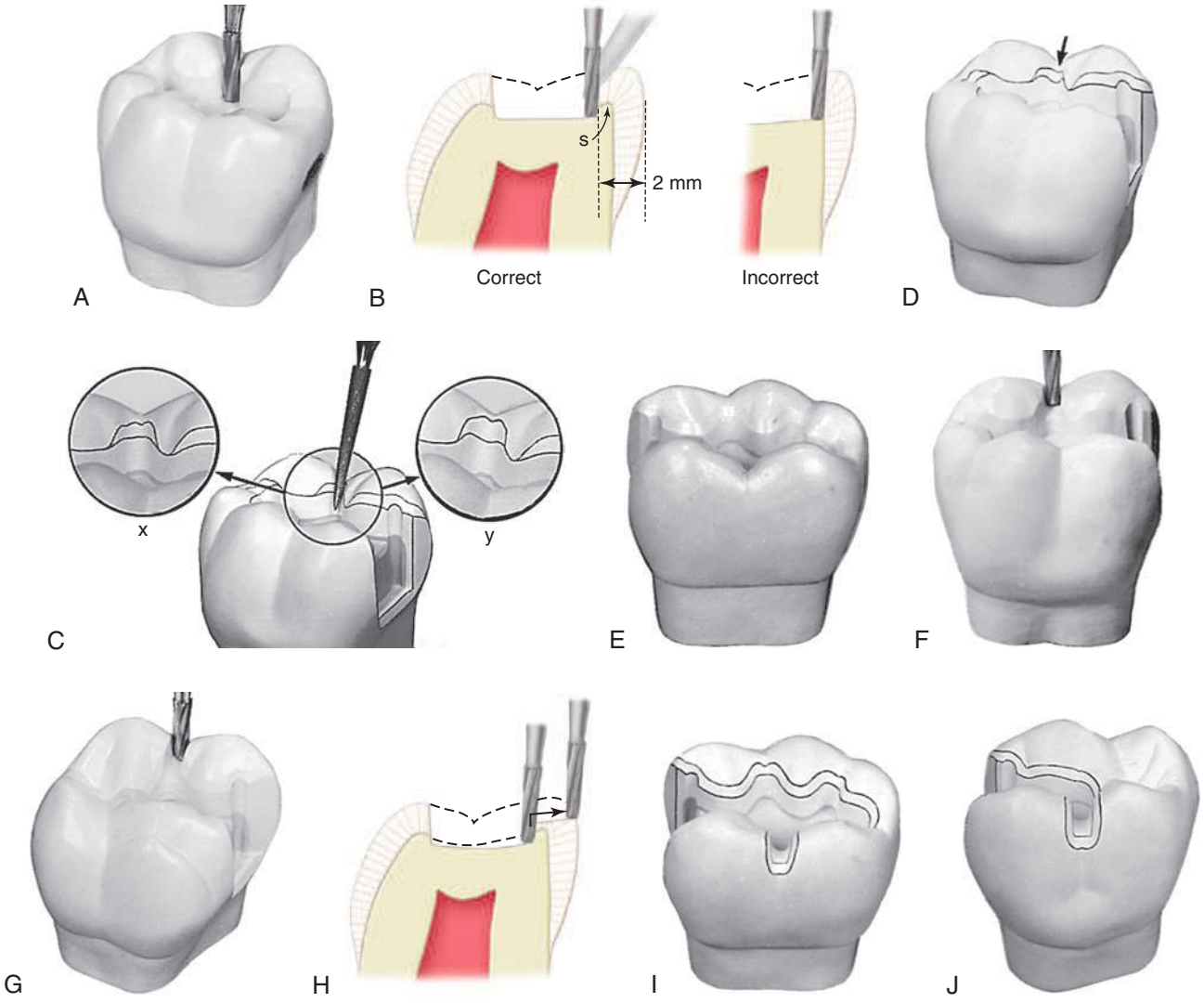
Fissures in the Facial and Lingual Cusp Ridges or Marginal Ridges

In the preparation of Class II preparations for inlays, facial and lingual occlusal fissures may extend nearly to, or through, the respective facial and lingual cusp ridges but not onto the facial or lingual surface. The proper outline form dictates that the preparation margin should not cross such fissures but should be extended to include them. For the occlusal step portion of the preparation, the dentist initially extends along the lingual fissure with the No. 271 carbide bur until only 2 mm of tooth structure remains between



• **Fig. 18.19** A and B, Mesioocclusal preparation on the maxillary molar having an unaffected oblique ridge. C, Preparing the lingual groove extension of the distoocclusolingual preparation. D and E, Cutting retention grooves in the lingual surface extension (D) and the distal box (E). F and G, Completed distoocclusolingual preparation on the maxillary molar having an unaffected oblique ridge. H and I, Preparations for treating both proximal surfaces of the maxillary molar having a strong, unaffected oblique ridge.

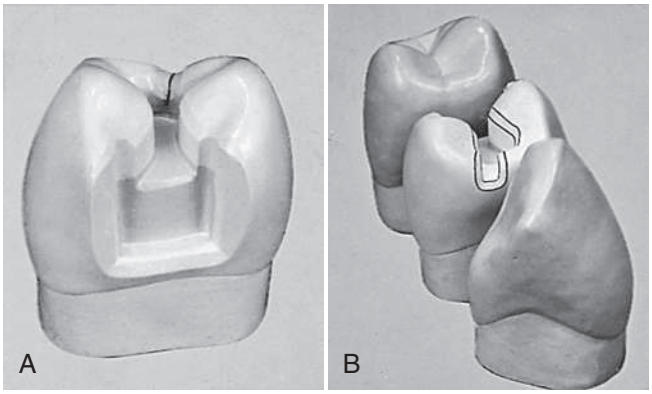
the bur and the lingual surface of the tooth. Additional lingual extension at this time is incorrect because it may remove the supporting dentin unnecessarily (Fig. 18.20A and B). If this extension almost includes the length of the fissure, additional extension is achieved later by using the occlusal bevel; this bevel may be wider than conventional if the remaining fissure can be eliminated by such a wider bevel (see Fig. 18.20C). Enameloplasty sometimes may eliminate the end portion of the fissure and provide a smooth enamel surface where previously a fault was present, thus reducing the extent of the required extension (see Fig. 18.20D).



• **Fig. 18.20** A, Extending to include the lingual (occlusal) fissure. B, Section of *A*. The dentinal support *(s)* of the lingual cusp ridge should not be removed. A bevel can provide additional extension to include the fissure that does not extend to the crest of the ridge. C, Completed preparations with standard width bevel *(x)* and with wider bevel to include a groove defect that nearly extends to the ridge height *(y)*. D, Completed preparation illustrating enameloplasty for the elimination of a shallow fissure extending to or through the lingual ridge height. (Compare the smooth, saucer-shaped lingual ridge contour with *C*, in which no enameloplasty has been performed.) E, Fissure remaining through the lingual ridge after unsuccessful enameloplasty. This indicates procedures subsequently illustrated. F and G, Extending the preparation if enameloplasty has not eliminated the fissure in the lingual ridge (F) or the facial ridge (G). H, Section of F. I and J, Completed preparations after beveling the margins of the extensions through the lingual ridge (I) and the facial ridge (J).

If possible, the fissure should be included in the preparation outline without extending the margin to the height of the ridge. If the occlusal bevel places the margin on the height of the ridge, however, the marginal enamel likely is weak because of its sharpness and because of the inclination of the enamel rods in this region. The preparation outline should be extended just onto the facial or lingual surface (see Fig. 18.20I and J). Such extension onto the facial or lingual surface also would be indicated if the fissure still remains through the ridge after enameloplasty (see Fig. 18.20E).

When necessary, extension through a cusp ridge is accomplished by cutting through the ridge at a depth of 1 mm with the No. 271 carbide bur (see Fig. 18.20F and G). The dentist bevels the margins of the extension with the flame-shaped, fine-grit diamond instrument to provide for the desired 40-degree marginal metal on the occlusal, mesial, and distal margins and for 30-degree marginal metal on the gingival margin (see Fig. 18.20C, D, I, and J). In the same manner, the operator should manage the fissures that may extend into or through a proximal marginal ridge, assuming that the proximal surface otherwise was not to be included in the outline form and that such fissure management does not extend the preparation outline near the adjacent tooth contact. This treatment particularly applies to a mesial fissure of the maxillary



• **Fig. 18.21** The fissure that remains on the mesial marginal ridge after unsuccessful enameloplasty (A) is treated (B) in the same manner as lingual or facial ridge fissures (see Fig. 18.20I and J).

first premolar (Fig. 18.21). If this procedure extends the margin near or into the contact, the outline form on the affected proximal surface must be extended to include the contact, as for a conventional proximal surface preparation.

Cusp-Capping Partial Onlay

The term *partial onlay* is used when a cast-metal restoration covers and restores at least one but not all of the cusp tips of a posterior tooth (Fig. 18.22). The facial and lingual margins on the occlusal surface frequently must be extended toward the cusp tips to the extent of the existing restorative materials and to uncover a caries lesion (see Fig. 18.22B and C). Undermined occlusal enamel should be removed because it is weak; removing such enamel provides access for the proper excavation of the lesion. When the occlusal outline is extended up, the cusp slopes more than half the distance from any primary occlusal groove (central, facial, or lingual) to the cusp tip, covering (capping) the cusp should be considered. If the preparation outline is extended two thirds of this distance or more, capping the cusp is usually necessary to (1) protect the weak, underlying cuspal structure from fracture caused by masticatory force and (2) remove the occlusal margin from a region subjected to heavy stress and wear (see Fig. 18.22A and B). At this point in the preparation of the pulpal floor, depth can be increased from 1.5 mm to 2 mm. This additional pulpal depth ensures sufficient reduction in an area that is often underreduced and results in imparting greater strength and rigidity to the wax pattern and cast restoration.

Reduce the cusps for capping as soon as the indication for such capping is determined because this improves access and visibility for the subsequent steps in the preparation. If a cusp is in infraocclusion of the desired occlusal plane before reduction, the amount of cusp reduction is less and needs to be only that which provides the required clearance with the desired occlusal plane. Before reducing the surface, the operator prepares depth gauge grooves (depth cuts) with the side of the No. 271 carbide bur (see Fig. 18.22D). Such depth cuts should help to prevent thin spots in the restoration. With the depth cuts serving as guides, the operator completes the cusp reduction with the side of the carbide bur (see Fig. 18.22E). The reduction should provide for a uniform 1.5 mm of metal thickness over the reduced cusp. On maxillary premolars and first molars, the reduction should be minimal (i.e., 0.75–1 mm) on the facial cusp ridge to decrease the display of metal. This reduction should increase progressively to 1.5 mm toward the center of the tooth to help impart rigidity to the capping metal (Fig. 18.23A and C).

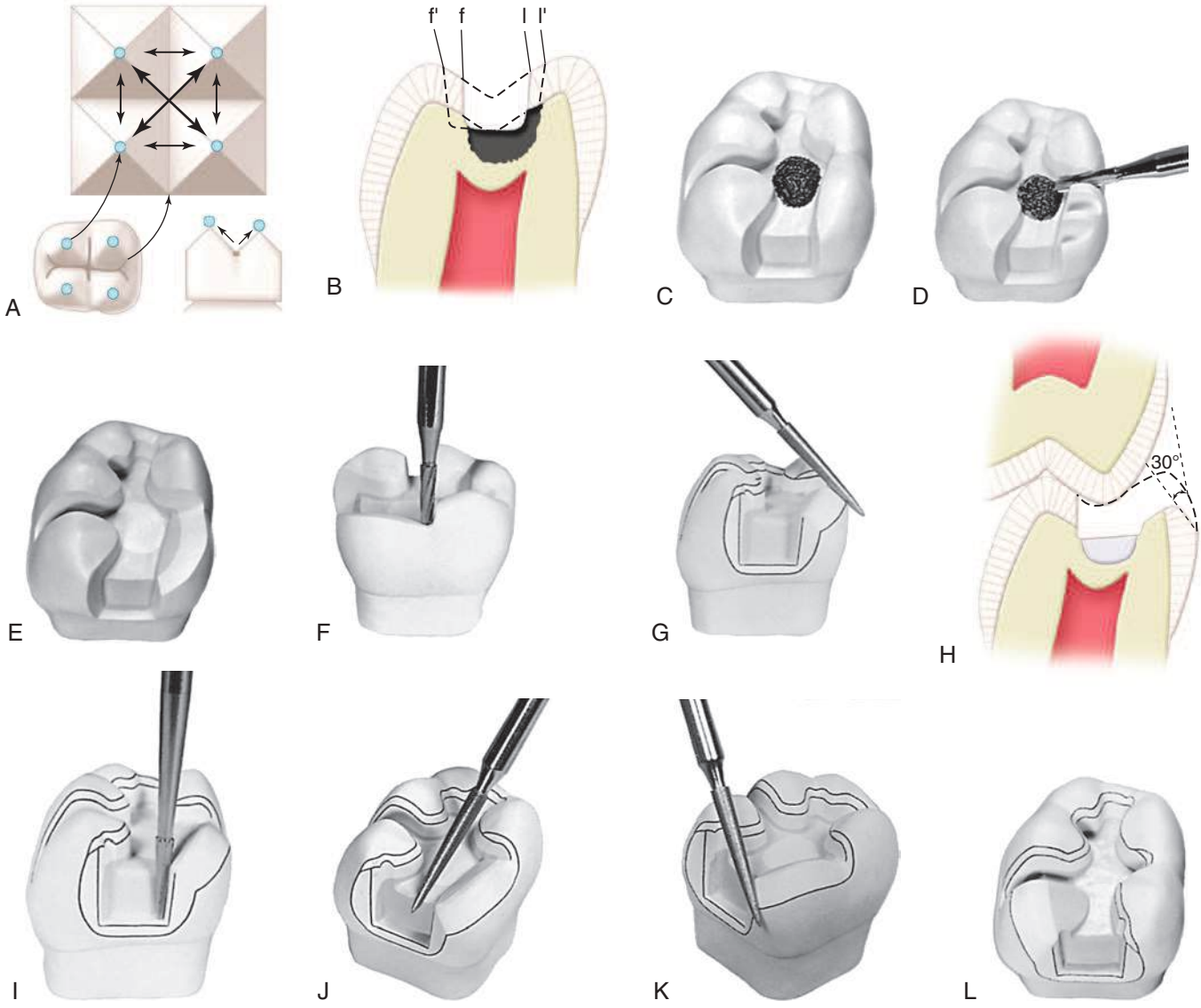
If only one of the two lingual cusps of a molar is reduced for capping, the reduction must extend to include just the lingual groove between the reduced and unreduced cusps. This reduction should terminate with a distinct vertical wall that has a height that is the same as the prescribed cusp reduction. Applying the bur vertically (see Fig. 18.22F) should help establish a vertical wall of proper depth and direction. Similar principles apply when only one of the facial cusps is to be reduced (see Figs. 18.22L and 18.23B).

A bevel of generous width is prepared on the facial (lingual) margin of a reduced cusp with the flame-shaped, fine-grit diamond instrument (with the exception of esthetically prominent areas). This bevel is referred to as *reverse bevel* or *counterbevel*. The width varies because it usually should extend beyond any occlusal contact with opposing teeth, either in maximum intercuspation or during mandibular movements (see Fig. 18.24C later in the chapter). It should be at an angle that results in 30-degree marginal metal (see Fig. 18.22G and H). The exception is the facial margin on maxillary premolars and the first molar, where esthetic requirements dictate only a blunting and smoothing of the enamel margin (a stub margin) by the light application of a fine-grit sandpaper disk or the fine-grit diamond instrument (flame-shaped) held at a right angle to the facial surface (see Fig. 18.23C). Any sharp external corners should be rounded slightly to strengthen them and reduce the problems they may generate in future steps (see Fig. 18.22J and K).

Cusp reduction appreciably decreases the retention form because it decreases the height of the vertical walls. Therefore proximal retention grooves usually are recommended (see Fig. 18.22I). It may be necessary to increase the retention form by extending facial and lingual groove regions of the respective surfaces or by collar and skirt features (see later). These additional retention features also provide the desired resistance form against forces tending to split the tooth (see Fig. 18.22K; see also Fig. 18.28 later in the chapter).

The principles stated in the preceding paragraphs may be applied in the treatment of the distal cusp of the mandibular first molar when preparing a mesioocclusodistal preparation (see Fig. 18.23D). Proper extension of the distofacial margin usually places the occlusal margin in a region subjected to heavy masticatory forces and wear. Satisfactory treatment usually dictates either extending the distofacial margin (and wall) slightly mesial of the distofacial groove (see Fig. 18.23E) or capping the remaining portion of the distal cusp (see Fig. 18.23F).

After cusp reduction, the dentist visually verifies that the occlusal clearances are sufficient. A wax interocclusal record is helpful when checking the occlusal clearances, especially in areas that are difficult to visualize, for example, in the central groove and lingual cusp regions. To make a wax "bite," the dentist first dries the preparation free of any visible moisture; however, dentin should not be desiccated (Fig. 18.24A). Next the dentist lightly presses a portion of softened, low-fusing inlay wax over the prepared tooth; the dentist immediately requests the patient to close into the soft wax and slide the teeth in all directions (see Fig. 18.24B–F). During the mandibular movements, the dentist observes to verify that (1) the patient performs right lateral, left lateral, and protrusive movements; (2) the adjacent unprepared teeth are in contact with the opposing teeth; (3) the wax in the preparation is stable (not loose and rocking); and (4) the wax is not in infraocclusion. The dentist cools the wax and carefully removes it, holds it up to a light, and notes the degree of light transmitted through it. With experience, this is a good indicator of the thickness of the wax. An alternative method is to

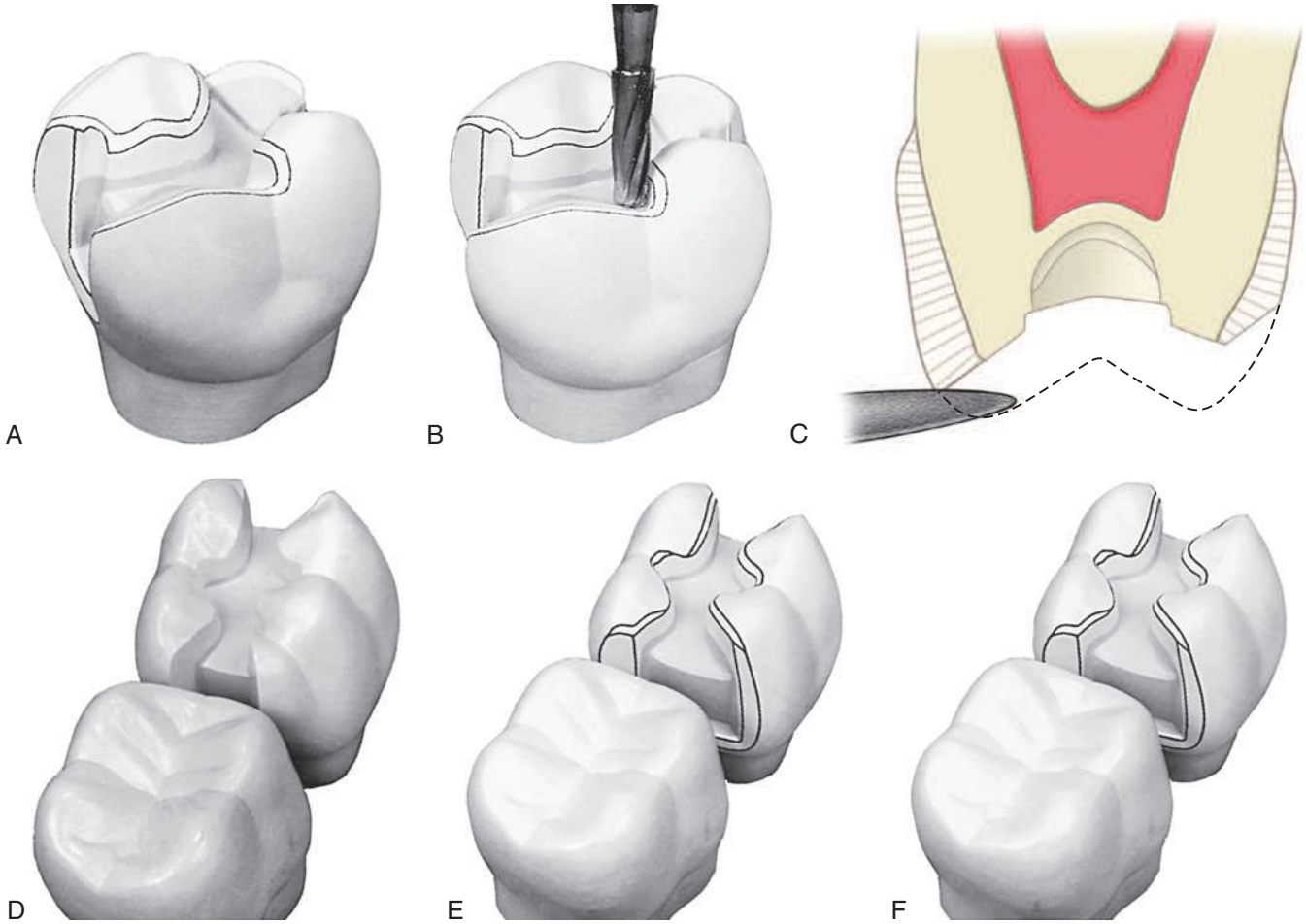


• **Fig. 18.22** A, When the extension of the occlusal margin is one half the distance from any point on the primary grooves *(cross)* toward the cusp tip *(dot),* capping of the cusp should be considered; when this distance is two thirds or more, capping of the cusp is usually indicated. B, *l* is midway between the central groove and the lingual cusp tip; *f* is midway between the central groove and the facial cusp tip. When enamel at *l* and *f* is undermined by a caries lesion, the respective walls must be extended to the dotted lines *l* and *f* to uncover the lesion. Cusps should be reduced for capping. C, Extension to uncover the caries lesion indicates that the mesiolingual cusp should be reduced for capping. D, Depth cuts. E, Reduced mesiolingual cusp. The caries lesion has been removed, and the base has been placed. F, Applying the bur vertically helps establish the vertical wall that barely includes the lingual groove. G, Counterbeveling reduced cusp. H, Section of the counterbevel. I, Improving the retention form by cutting the proximal retention grooves. J and K, The preparation is complete except for the rounding of the axiopulpal line angle (J) and the rounding of the junction of the counterbevel and the secondary flare (K). Facial surface groove extension improves the retention and resistance forms. L, Preparation when reducing one of two facial cusps on the mandibular molar.

use wax calipers or to section the wax to verify its thickness. Insufficient thickness calls for more reduction in the indicated area before proceeding. As an alternative to wax, an interocclusal record can be made in maximum intercuspation with a quick-setting PVS impression material. Once set, this interocclusal record can be measured with wax calipers to evaluate the reduction. If wax calipers are not available, the interocclusal record can be sectioned with a knife to see the thickness in cross section. However, a PVS interocclusal record will not offer as much information as would the softened inlay wax technique, since the lateral and protrusive paths are not registered in the former.

Including Portions of the Facial and Lingual Smooth Surfaces Affected by Caries or Other Injury

When portions of a facial (lingual) smooth surface and a proximal surface are affected by caries or some other factor (e.g., fracture)

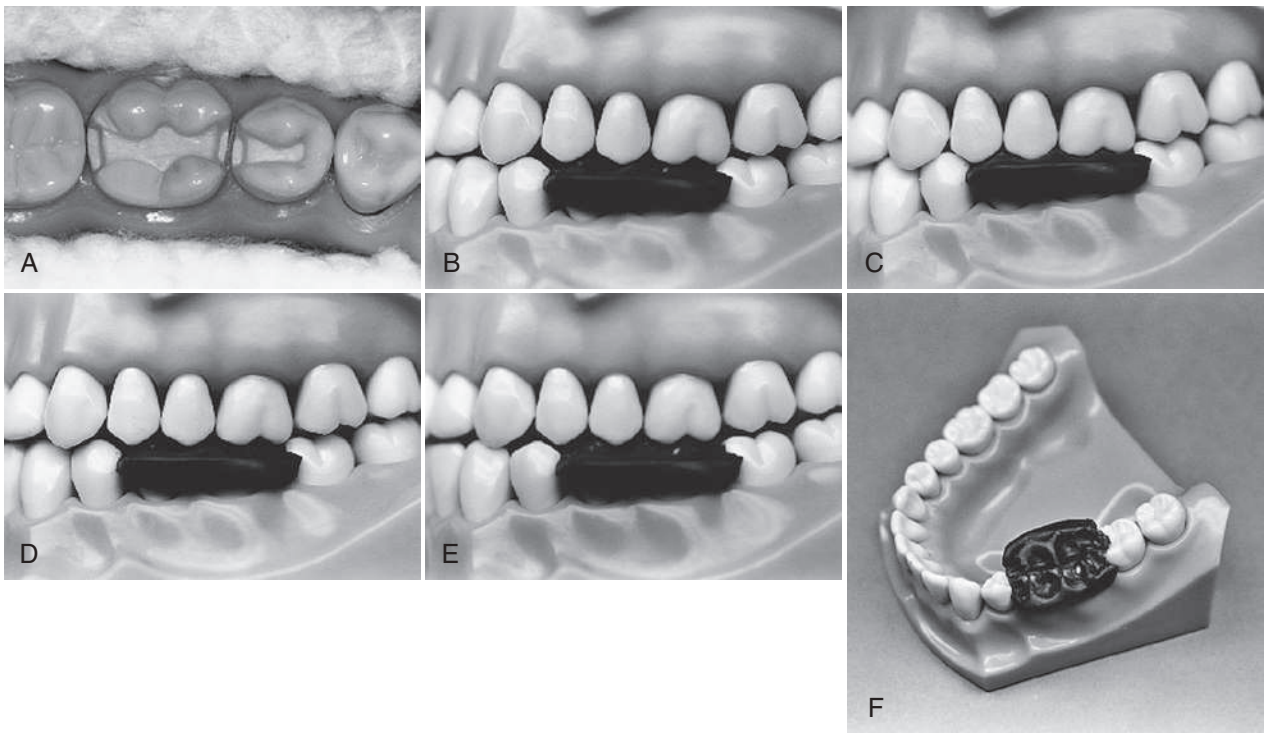


• **Fig. 18.23** A and B, Capping one of two facial cusps on the maxillary molar. C, Blunting the margin of the reduced cusp when esthetics is a major consideration. D–F, The margin shown crossing the distal cusp in D indicates treatment illustrated in E or F.

(Fig. 18.25A and I), the treatment may be a large inlay, an onlay, a three-quarter crown, a full crown, or multiple amalgam or composite restorations. Generally, if carious portions are extensive, the choice between the previously listed cast-metal restorations is determined by the degree of tooth circumference involved. A full crown is indicated if the lingual and the facial smooth surfaces are defective, especially if the tooth is a second or third molar. When only a portion of the facial smooth surface is carious, and the lingual surfaces of the teeth are conspicuously free of caries, a mesioocclusal, distofacial, and distolingual inlay or onlay with a lingual groove extension is chosen over the crown because the former is more favorable to the health of the gingival tissues and more conservative in the removal of tooth structure. Often this is the treatment choice for the maxillary second molar, which may exhibit a caries lesion on the distofacial surface as a result of poor oral hygiene (owing to poor access) in this region.

In the preparation of the maxillary molar referred to in the preceding paragraph, the mesiofacial and distolingual cusps and the distofacial cusp are usually reduced for capping. If the distofacial cusp defect is primarily shallow decalcification, the flame-shaped diamond instrument is used to reduce the involved facial surface and distofacial corner approximately to the depth of enamel and to establish the gingival margin of this reduction apical to the affected area (see Fig. 18.25D). This instrument also is used to terminate the facial surface reduction in a definite facial margin running gingivoocclusally and in a manner to provide for 40-degree metal at this margin (see Fig. 18.25E).

If the distofacial defect is more extensive and deeper into the tooth (see Fig. 18.25I), eliminating the opportunity for an effective distal box or groove (no facial wall possible), the No. 271 carbide bur should be used to cut a gingival shoulder extending from the distal gingival floor around to include the affected facial surface. This shoulder partially provides the desired resistance form. (A gingival floor, perpendicular to occlusal force, has been provided in lieu of the missing pulpal wall in the distofacial cusp region.) The No. 271 bur is used to create a nearly vertical wall in the remaining facial enamel (see Fig. 18.25J). The width of the shoulder should be the diameter of the end of the cutting instrument. The vertical walls should have the appropriate degree of draft to contribute to retention form. Then the faciogingival and facial margins are beveled with the flame-shaped, fine-grit diamond instrument to provide 30-degree metal at the gingival margin (see Fig. 18.25K) and 40-degree metal along the facial margin (see Fig. 18.25L). These two bevels should blend together (see *x* in Fig. 18.25M), and the faciogingival bevel should be continuous with the gingival bevel on the distal surface. Additional retention and



• **Fig. 18.24** Verifying sufficient cusp reduction by forming a wax interocclusal record. A, The walls of the preparations (distoocclusal for the second premolar, and mesioocclusodistal for the first molar) are air-dried of visible moisture. The low-fusing inlay wax that is the same length as the mesiodistal length of the inlay preparations is softened and pressed over the prepared teeth. B–E, The patient moves the mandible into all occlusal positions, left lateral (B), through maximum intercuspation (C), to right lateral (D), and protrusive (E). F, Completed interocclusal record.

resistance forms are indicated for this preparation and can be developed by an arbitrary lingual groove extension (see Fig. 18.25N) or a distolingual skirt extension (see Fig. 18.25O and P). These preparation features resist forces normally opposed by the missing distofacial wall and help protect the restored tooth from fracture injury.

Tooth Preparation for Full Cast-Metal Onlays

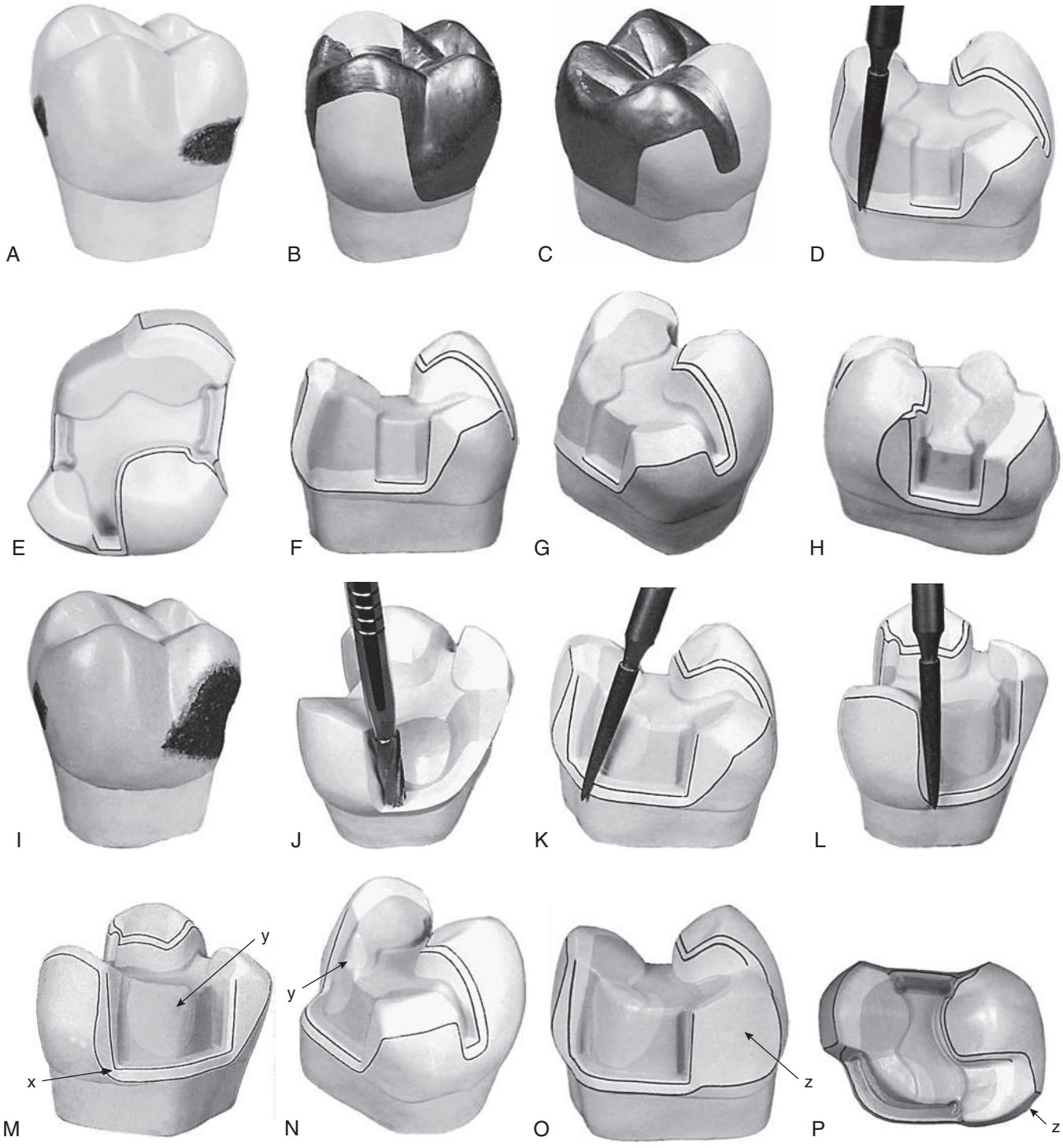
The preceding sections have presented basic tooth preparation principles and techniques for small, simple cast-metal inlays and for partial onlays that cap less than all the cusps. This section presents the tooth preparation principles and techniques for *full onlay* restorations that cover the entire occlusal surface. Onlay restorations have many clinical applications and may be desired by many patients. These restorations have a well-deserved reputation for providing excellent service.

The cast-metal onlay restoration spans the gap between the inlay, which is primarily an intracoronal restoration, and the full crown, which is a totally extracoronal restoration. The full onlay by definition caps all of the cusps of a posterior tooth and can be designed to help strengthen a tooth that has been weakened by caries or previous restorative experiences. It can be designed to distribute occlusal loads over the tooth in a manner that greatly decreases the chance of future fracture.[4,6] It is more conservative of the tooth structure than the full crown preparation, and its supragingival margins, when possible, are less irritating to the gingiva. Usually an onlay diagnosis is made preoperatively because of the tooth's status. Sometimes the diagnosis is deferred until the extension of the occlusal step of an inlay preparation facially and lingually to the limits of the caries lesion shows that cusp reduction is mandatory. The mandibular first molar is used to illustrate one mesioocclusodistal preparation for a full cast-metal onlay.

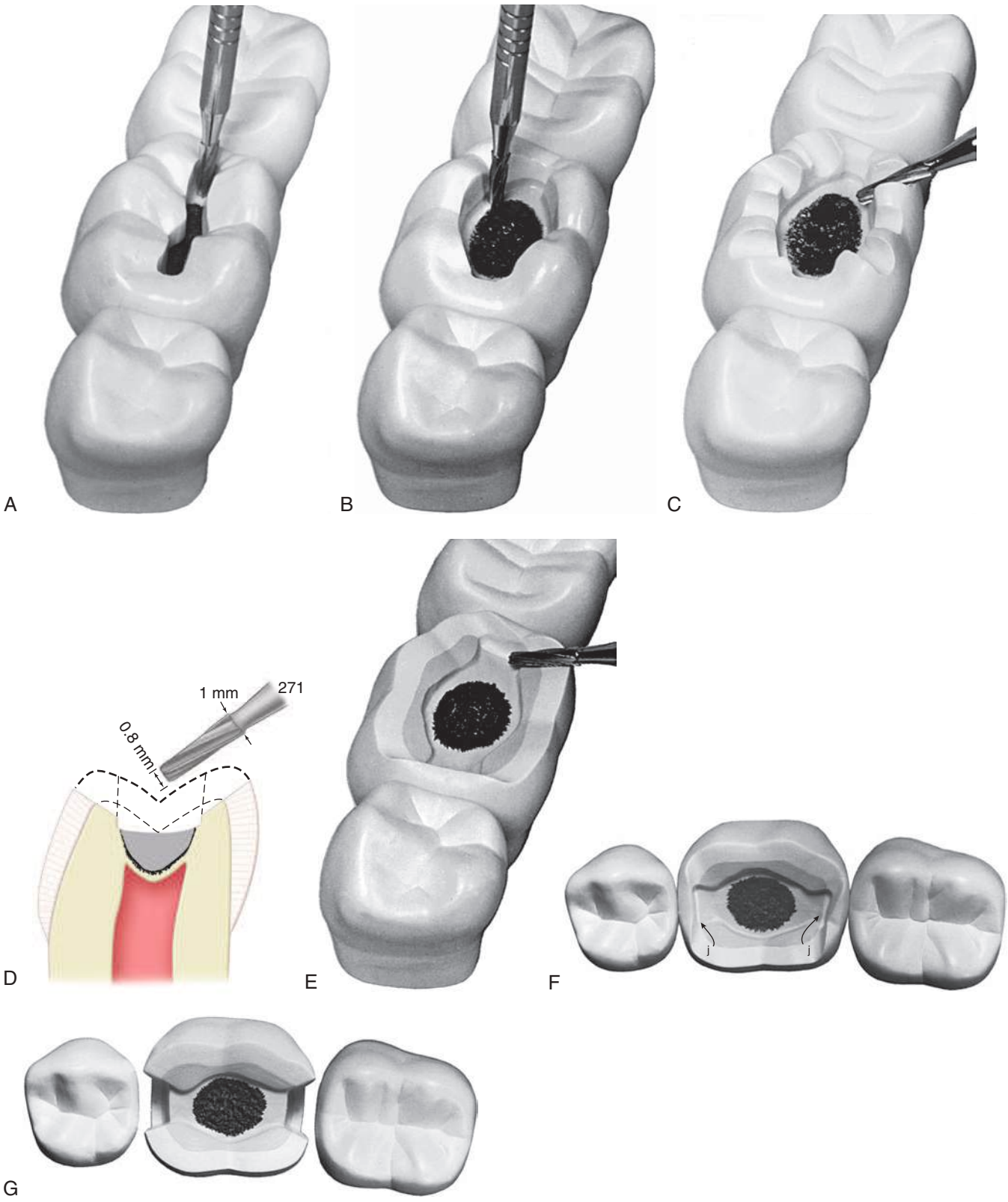
Initial Preparation

Occlusal Reduction

As soon as the decision is made to restore the tooth with a full cast-metal onlay, the cusps should be reduced because this improves the access and the visibility for subsequent steps in tooth preparation. With the cusps reduced, the efficiency of the cutting instrument and the air-water cooling spray is improved. Also when the cusps are reduced, it is easier to assess the height of the remaining clinical crown of the tooth, which determines the degree of occlusal divergence necessary for adequate retention form. Using the No. 271 carbide bur held parallel to the long axis of the tooth crown, a 2-mm deep pulpal floor is prepared along the central groove (Fig. 18.26A). To verify the preoperative diagnosis for cusp reduction, this occlusal preparation is extended facially and lingually just beyond the caries to sound tooth structure (see Fig. 18.26B). The groove should not be extended farther, however, than two thirds the distance from the central groove to the cusp tips because the need for cusp reduction is verified at this point. With the side of the No. 271 carbide bur, uniform 1.5-mm deep depth cuts are prepared on the remaining occlusal surface (see Fig. 18.26C and D). Depth cuts usually are placed on the crest of the triangular ridges and in the facial and lingual groove regions. These depth cuts help prevent thin spots in



• **Fig. 18.25** A, Maxillary molar with caries lesions on the distofacial corner and the mesial surface. B and C, Completed mesiooccclusal, distofacial, and distolingual inlay for treating lesions shown in A, facio-occlusal view (B) and distolinguoocclusal view (C). D–H, Preparation for treating caries lesions illustrated in A, distoocclusal view with diamond instrument being applied (D), occlusal view (E), distal view (F), distolinguoocclusal view (G), and mesioocclusal view (H). I, Maxillary molar with a deeper caries lesion on the distofacial corner and with a mesial caries lesion. J, Preparation (minus bevels and flares) for mesiooc-clusal, distofacial, and distolingual inlay to restore the carious molar shown in I. A No. 271 carbide bur is used to prepare the gingival shoulder and the vertical wall. K and L, Beveling margins. M and N, Completed preparation for treating the caries lesion shown in I. Gingival and facial bevels blend at *x*, and *y* is the cement base. O and P, When the lingual surface groove has not been prepared and when the facial wall of the proximal box is mostly or totally missing, forces directed to displace the inlay facially can be opposed by lingual skirt extension *(z).*



• **Fig. 18.26** A, Cutting a 2-mm deep central groove. B, Extending the central groove cut facially and lingually to verify any need for cusp capping. C, Depth cuts. D, Section of *C*. E, Completion of cusp reduction. Small portions of the mesial and distal marginal ridges are left unreduced to avoid scarring the adjacent teeth. F, The occlusal step is extended facially and lingually past any carious areas and is extended to expose the proximal dentinoenamel junction (DEJ) *(j)* in anticipation of proximal boxing. G, Preparation with proximal boxes prepared. Note the clearances with the adjacent teeth.

the final restoration. If a cusp is in infraocclusion of the desired occlusal plane before reduction, the amount of cusp reduction is less and needs only that which provides the required clearance with the desired occlusal plane. Carious dentin and old restorative material that is deeper in the tooth than the desired clearance are not removed at this step in preparation.

With the depth cuts serving as guides for the amount of reduction, the cusp reduction is completed with the side of the No. 271 bur. When completed, this reduction should reflect the general topography of the original occlusal surface (see Fig. 18.26E). The operator should not attempt to reduce the mesial and distal marginal ridges completely at this time to avoid hitting an adjacent tooth. The remainder of the ridges are reduced in a later step when the proximal boxes are prepared.

Throughout the next steps in the initial preparation, the cutting instruments used to develop the vertical walls are oriented continually to a single draw path, usually the long axis of the tooth crown, so that the completed preparation has draft (i.e., no undercuts). For mandibular molars and second premolars whose crowns tilt slightly lingually, the bur should be tilted slightly (5–10 degrees) lingually to help preserve the strength of the lingual cusps (see Fig. 18.4D). The gingival-to-occlusal divergence of these preparation walls may range from 2 to 5 degrees from the line of draw, depending on their heights. If the vertical walls are unusually short, a minimum of 2 degrees occlusal divergence is desirable for retentive purposes. Cusp reduction appreciably decreases the retention form because it decreases the height of the vertical walls, so this minimal amount of divergence is often indicated in the preparation of a tooth for a cast-metal onlay. As the gingivooccusal height of the vertical walls increases, the occlusal divergence should increase, allowing 5 degrees in the preparation of the greatest gingivooccusal length. The latter preparations present difficulties during pattern withdrawal, trial seating and withdrawal of the casting, and cementing, unless this maximal divergence is provided.

Occlusal Step

After cusp reduction, a 0.5-mm deep occlusal step should be present in the central groove region between the reduced cuspal inclines and the pulpal floor. Maintaining the pulpal depth (0.5 mm) of the step, it is extended facially and lingually just beyond any carious areas, to sound tooth structure (or to sound base or restorative material if certain conditions, discussed subsequently, have been met). Next the operator extends the step mesially and distally far enough to expose the proximal DEJ (see Fig. 18.26F). The step is extended along any remaining facial (and lingual) occlusal fissures as far as they are faulty (fissured). The facial and lingual walls of the occlusal step should go around the cusps in graceful curves, and the isthmus should be only as wide as necessary to be in sound tooth structure or sound base or restorative material. Old restorative material or caries that is deeper pulpally than this 0.5-mm step is not removed at this stage of tooth preparation.

As the occlusal step approaches the mesial and distal surfaces, it should widen faciolingually in anticipation of the proximal box extensions (see Fig. 18.26F). This 0.5-mm occlusal step contributes to the retention of the restoration and provides the wax pattern and cast-metal onlay with additional bulk for rigidity.[7]

Proximal Box

Continuing with the No. 271 carbide bur held parallel to the long axis of the tooth crown, the proximal boxes are prepared as described in the inlay section. Fig. 18.26G illustrates the preparation after the proximal boxes are prepared.

Final Preparation

Removal of Infected Carious Dentin and Defective Restorative Materials and Pulp Protection

If the occlusal step and the proximal boxes have been extended properly, any carious dentin or previous restorative materials remaining on the pulpal and axial walls should be visible. They should be removed as described previously.

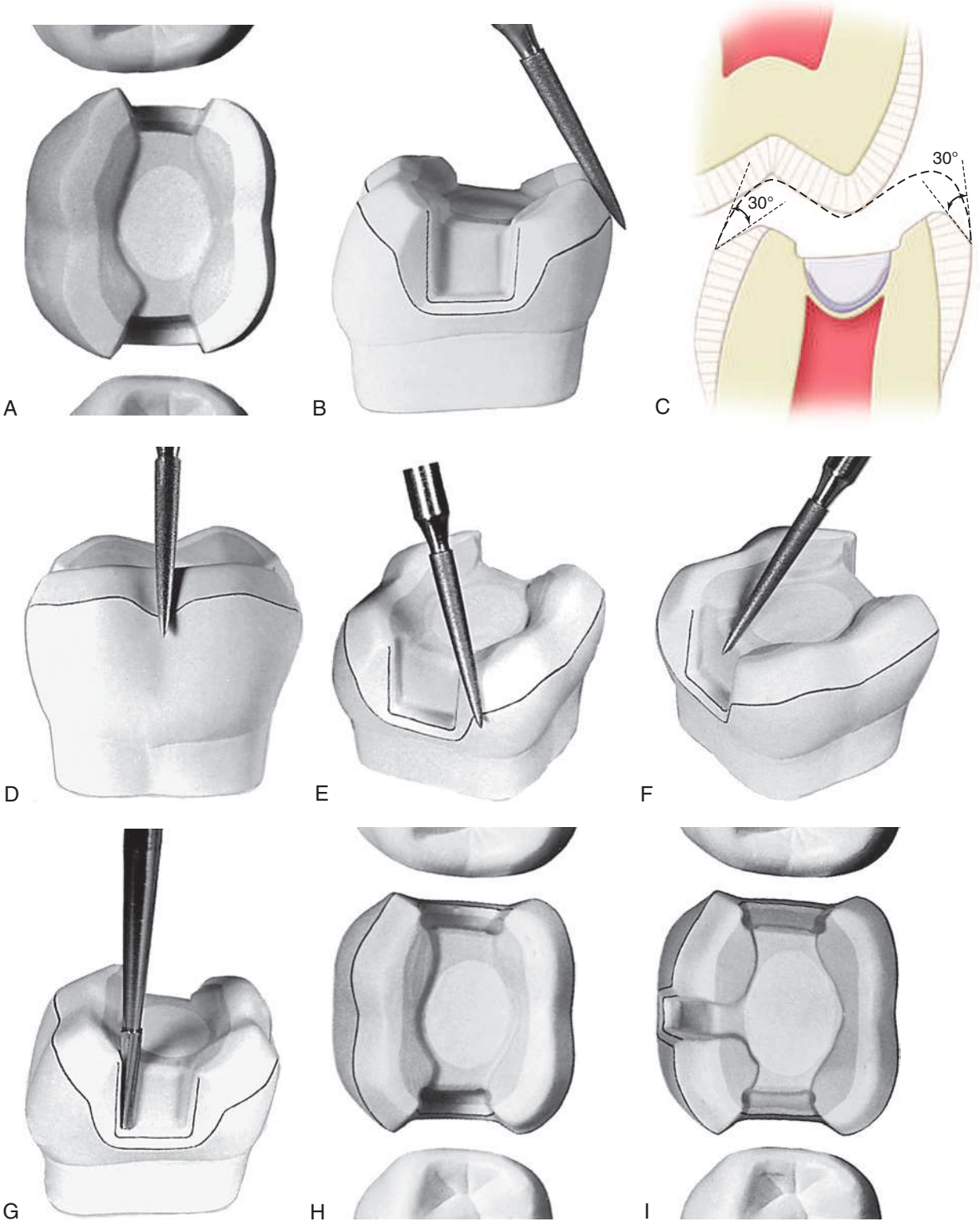
Preparation of Bevels and Flares

After the cement base (when indicated) is completed (Fig. 18.27A), the slender, flame-shaped, fine-grit diamond instrument is used to place counterbevels on the reduced cusps, to apply the gingival bevels, and to create secondary flares on the facial and lingual walls of the proximal boxes. First a gingival retraction cord is inserted, as described in the previous inlay section. During the few minutes required for the cord's effect on the gingival tissues, the diamond instrument is used to prepare the counterbevels on the facial and lingual margins of the reduced cusps. The bevel should be of generous width and should result in 30-degree marginal metal. The best way to judge this is to always maintain a 30-degree angle between the side of the instrument and the external enamel surface beyond the counterbevel (see Fig. 18.27B and C). The counterbevel usually should be wide enough so that the cavosurface margin is beyond (gingival to) any contact with the opposing dentition. If a facial (lingual) surface fissure extends slightly beyond the normal position of the counterbevel, it may be included (removed) by deepening the counterbevel in the region of the fissure (see Fig. 18.27D). If the fissure extends gingivally more than 0.5 mm, however, the fissure is managed as described later.

A counterbevel is not placed on the facial cusps of maxillary premolars and first molars where esthetic considerations may dictate using a stubbed margin by blunting and smoothing the enamel margin by the light application of a fine-grit sandpaper disk or the fine-grit diamond instrument (flame-shaped) held at a right angle to the facial surface (see Fig. 18.23C). The surface created by this blunting should be approximately 0.5 mm in width. For beveling the gingival margins and flaring (secondary) the proximal enamel walls, refer to the inlay section.

After beveling and flaring, any sharp junctions between the counterbevels and the secondary flares are rounded slightly (see Fig. 18.27E). The fine-grit diamond instrument also is used to bevel the axiopulpal line angles lightly (see Fig. 18.27F). Such a bevel produces a stronger wax pattern at this critical region by increasing its thickness. Any sharp projecting corners in the preparation are rounded slightly because these projections are difficult to reproduce without voids when developing the working cast and often cause difficulties when seating the casting. The desirable metal angle at the margins of onlays is 40 degrees except at the gingivally directed margins, where the metal angle should be 30 degrees.

When deemed necessary, shallow (0.3-mm deep) retention grooves may be cut in the facioaxial and the linguoaxial line angles with the No. 169L carbide bur (see Fig. 18.27G). These grooves are especially important for retention when the prepared tooth is short, which is often the case after reducing all the cusps. When properly positioned, the grooves are entirely in dentin near the DEJ and do not undermine enamel. The direction of cutting (translation of the bur) is parallel to the DEJ. The long axis of the No. 169L bur must be held parallel to the line of draw, and the tip of the bur must be positioned in the gingival box internal point angles. If the axial walls are deeper than ideal, however, the correct



• **Fig. 18.27** A, The caries lesion has been removed, and the base has been inserted. B, Counterbeveling facial and lingual margins of reduced cusps. C, Section of B. D, The fissure that extends slightly gingival to the normal position of the counterbevel may be included by slightly deepening the counterbevel in the fissured area. E, The junctions between the counterbevels and the secondary flares are slightly rounded. F, The axiopulpal line angle is lightly beveled. G, Improving the retention form by cutting proximal grooves. H, Completed mesioocclusodistal onlay preparation. I, Completed mesioocclusodistofacial onlay preparation showing the extension to include the facial surface groove or fissure.

reference for placing retention grooves is just inside the DEJ to minimize pulpal impacts but avoids undermining enamel. The model showing the completed preparation is illustrated in Fig. 18.27H.

Modifications in Full Onlay Tooth Preparations

Facial or Lingual Surface Groove Extension

A facial surface fissure (mandibular molar) or a lingual surface fissure (maxillary molar) is included in the outline in the same manner as described in the section on inlays. This extension sometimes is indicated to provide additional retention form, even though the groove is not faulty. A completed mesiooccludodistofacial onlay preparation on a mandibular first molar is illustrated in Fig. 18.27I.

Inclusion of Portions of the Facial and Lingual Smooth Surfaces Affected by Caries, Fractured Cusps, or Other Injury

For inclusion of shallow to moderate lesions on the facial and lingual smooth surfaces, refer to the section on inlays. A mandibular molar with a fractured mesiolingual cusp is used to illustrate the treatment of a fractured cusp of a molar (Fig. 18.28). The dentist uses a No. 271 carbide bur to cut a shoulder perpendicular to occlusal force by extending the proximal gingival floor (adjacent to the fracture) to include the affected surface. This shoulder partially provides the desired resistance form by being perpendicular to gingivally directed occlusal force. This instrument also is used to create a vertical wall in the remaining lingual enamel (see Fig. 18.28B). The width of the gingival floor should be the diameter of the end of the cutting instrument. The vertical walls should have the degree of draft necessary for the retention form. If the clinical crown of the tooth is short, it is advisable to cut proximal grooves for additional retention with the No. 169L bur. The linguogingival and lingual margins are beveled with the flame-shaped, fine-grit diamond instrument to provide 30-degree metal at the gingival margin (see Fig. 18.28C) and 40-degree metal along the lingual margin (see Fig. 18.28D).

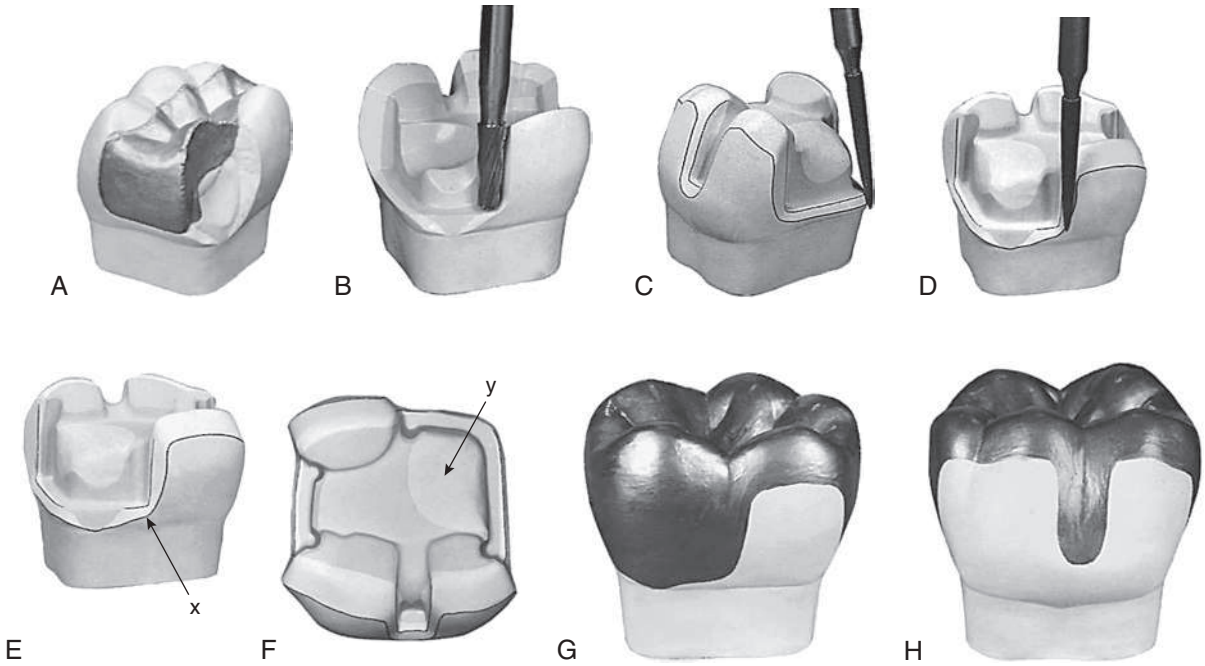
These two bevels should blend together (see *x* in Fig. 18.28E), and the linguogingival bevel is continuous with the gingival bevel on the mesial surface. Additional features to improve the retention and resistance forms are indicated and can be developed by a mesiofacial skirt extension or by a facial groove extension. These preparation features (discussed in the following section) improve the retention form, resist forces normally opposed by the missing mesiolingual wall, and help protect the restored tooth from further fracture injury.

Enhancement of Resistance and Retention Forms

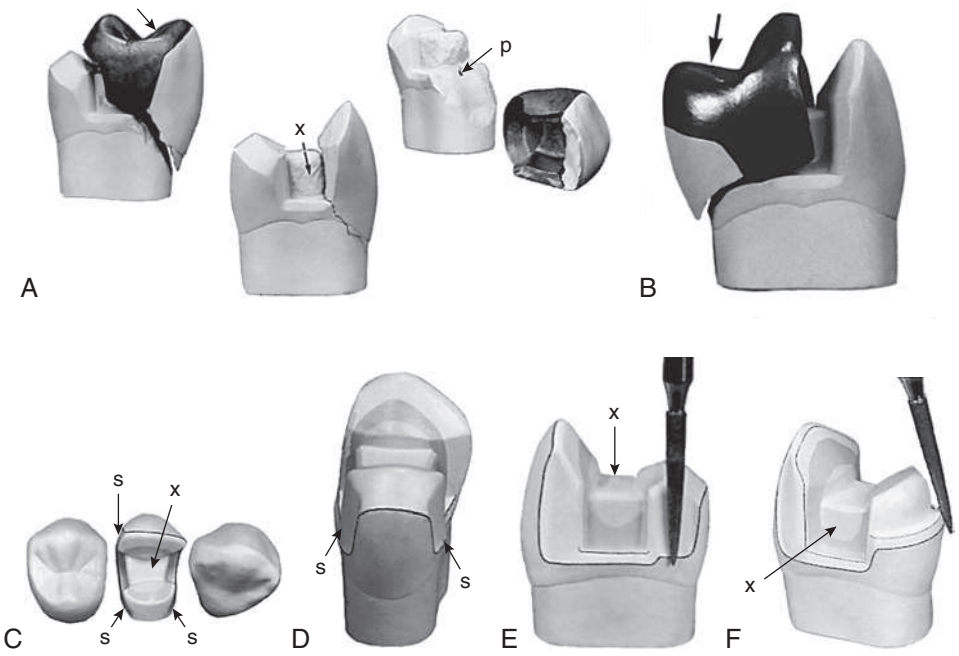
When the tooth crown is short (which is often the case when all cusps are reduced), the operator must strive to maximize the retention form in the preparation. Retention features that already have been presented are as follows:

1. Minimal amount of taper (2 degrees per wall) on the vertical walls of the preparation
2. Addition of proximal retention grooves
3. Preparation of facial (or lingual) surface groove extensions

In the preparation of a tooth that has been grossly weakened by caries or previous filling material and is judged to be prone to fracture under occlusal loads, the resistance form that cusp capping provides should be augmented by the use of skirts, collars, or facial (lingual) surface groove extensions. When properly placed, these features result in onlays that distribute the occlusal forces over most or all of the tooth and not just a portion of it, reducing the



• **Fig. 18.28** A, Mandibular first molar with large mesioocclusodistal amalgam and fractured mesiolingual cusp. B, Preparation (minus bevels and flares) for mesioocclusal, distofacial, and distolingual onlay to restore the fractured molar shown in A. A No. 271 carbide bur is used to prepare the gingival shoulder and the vertical lingual wall. Reducing cusps for capping and extending out the facial groove improve the retention and resistance forms. C and D, Beveling of margins. E and F, Completed preparation. The gingival and lingual bevels blend at *x*, and *y* is the base. G and H, Completed onlay.



• **Fig. 18.29** The large cement base *x* indicates severely weakened tooth crown. Occlusal force *(thick arrow)* may fracture the facial cusp (A) or the lingual cusp (B), which may expose the pulp *(p)*. C and D, Skirt extensions *(s)* on the mesiolingual, distolingual, and distofacial transitional line angles prevent the fractures shown in A and B. Esthetic consideration contraindicates skirting the mesiofacial line angle. E, Distal view of the preparation shown in *D*. Skirt extensions are prepared with a fine-grit diamond instrument. F, A collar preparation around the lingual cusp prevents the fracture shown in *A*.

likelihood of fractures of teeth (Fig. 18.29A and B). The lingual "skirt" extension (see Fig. 18.29C–E), the lingual "collar" preparation (see Fig. 18.29F), or the lingual surface groove extension on a maxillary molar protects the facial cusps from fracture. The facial skirt extension, the facial collar preparation, or the facial surface groove extension on a mandibular molar protects the lingual cusp from fracture.

Skirt Preparation

Skirts are thin extensions of the facial or lingual proximal margins of the cast-metal onlay that extend from the primary flare to a termination just past the transitional line angle of the tooth. A skirt extension is a conservative method of improving the retention and resistance forms of the preparation. It is relatively atraumatic to the tooth because it involves removing very little (if any) dentin. Usually the skirt extensions are prepared entirely in enamel.

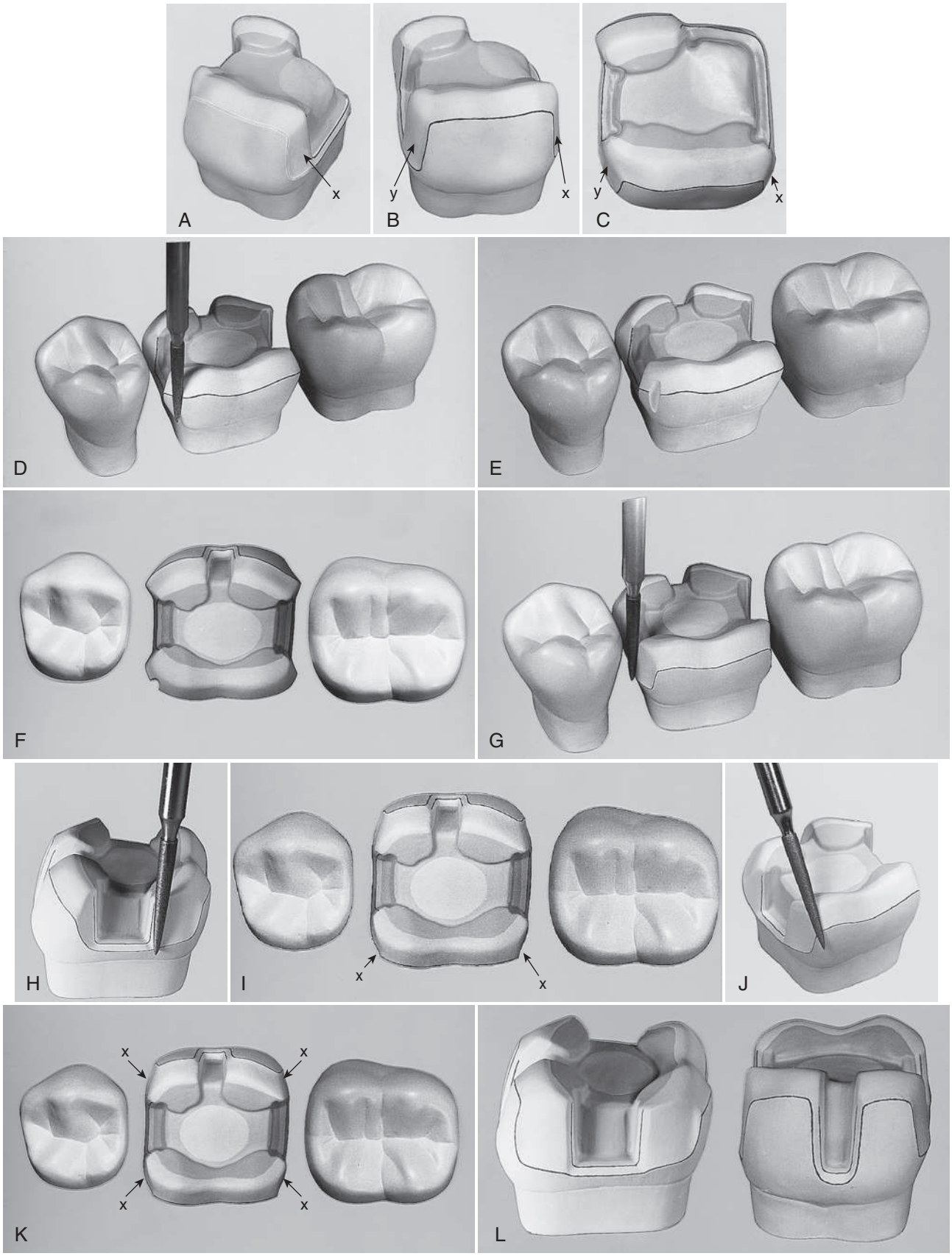
When the proximal portion of a Class II preparation for an onlay is being prepared and the lingual wall is partially or totally missing, the retention form normally provided by this wall can be developed with a skirt extension of the facial margin (Fig. 18.30A–C). Similarly, if the facial wall is not retentive, a skirt extension of the lingual margin supplies the desired retention form (see Fig. 18.25O and P). When the lingual and facial walls of a proximal box are inadequate, skirt extensions on the respective lingual and facial margins can satisfy the retention and resistance form requirements. The addition of properly prepared skirts to three of four line angles of the tooth virtually eliminates the chance of postrestorative fracture of the tooth because the skirting onlay is primarily an extracoronal restoration that encompasses and braces the tooth against forces that might otherwise split the tooth. The skirting onlay is often used successfully for many teeth that exhibit split-tooth syndrome.

The addition of skirt extensions also is recommended when the proximal surface contour and contact are to be extended more than the normal dimension to develop a proximal contact. Extending these proximal margins onto the respective facial and lingual surfaces aids in recontouring the proximal surface to this increased dimension. Also, when improving the occlusal plane of a mesially tilted molar by a cusp-capping onlay, reshaping the mesial surface to a satisfactory contour and contact is facilitated when the mesiofacial and mesiolingual margins are extended generously.

Skirting also is recommended when splinting posterior teeth together with onlays. The added retention and resistance forms are desirable because of the increased stress on each unit. Because the facial and lingual proximal margins are extended generously, the ease of soldering the connector and finishing of the proximal margins is increased.

A disadvantage of skirting is that it increases the display of metal on the facial and lingual surfaces of the tooth. For this reason, skirts are not placed on the mesiofacial margin of maxillary premolars and first molars. Skirting the remaining three line angles of the tooth provides ample retention and resistance forms.

The preparation of a skirt is done entirely with the slender, flame-shaped, fine-grit diamond instrument. Skirt preparations follow the completion of the proximal gingival bevel and primary flares. Experienced operators often prepare the skirt extensions at the same time that the gingival bevel is placed, however, working from the lingual toward the facial or vice versa. Maintaining the long axis of the instrument parallel to the line of draw, the operator translates the rotating instrument into the tooth to create a definite vertical margin, just beyond the line angle of the tooth, providing at the same time a 140-degree cavosurface enamel angle (40-degree metal angle) (see Fig. 18.30D–F). The occlusogingival length of this entrance cut varies, depending on the length of the clinical



• **Fig. 18.30** A, When the lingual wall of the proximal box is inadequate or missing, the retention form can be improved by facial skirt extension *(x).* B, Facioocclusal view of *A*. Maximal resistance form is developed by skirting the distofacial *(y)* and mesiofacial *(x)* transitional line angles. C, Occlusal view of *B*. D–F, The initial cut for the skirt is placed just past the transitional line angle of the tooth. G and H, Blending the skirt into the primary flare. I, Occlusal view showing the mesiolingual and distolingual skirts. Caution is exercised to prevent the overreduction of transitional line angles *(x).* Facial surface groove extension also improves the retention and resistance forms. J, The junction of the skirt and the counterbevel is slightly rounded. K, Skirting all four transitional line angles of the tooth further enhances the retention and resistance forms. Caution is exercised to prevent the overreduction of transitional line angles *(x).* L, Mesial and facial views of the preparation shown in *K*.

crown and the amount of extracoronal retention and resistance forms desired. Extending into the gingival third of the anatomic crown is usually necessary for an effective resistance form. In most instances the gingival margin of the skirt extension is occlusal to the position of the gingival bevel of the proximal box (see Fig. 18.30H and L).

The operator should use less than half the tip diameter of the flame-shaped diamond instrument to avoid creating a ledge at the gingival margin of the skirt extension. Using high speed and maintaining the long axis of the diamond instrument parallel with the line of draw, the operator translates the instrument from the entrance cut toward the proximal box to blend the skirt into the primary flare and the proximal gingival margin (see Fig. 18.30G and H). The operator must ensure that the line angle of the tooth is not overreduced when preparing skirt extensions (see *x* in Fig. 18.30I and K). If the line angle of the tooth is overreduced, the bracing effect of the skirt is diminished. Holding the diamond instrument at the same angle that was used for preparing the counterbevel, the operator rounds the junction between the skirt and the counterbevel (see Fig. 18.30J). Any sharp angles that remain after preparation of the skirt need to be rounded slightly because these angles often lead to difficulties in the subsequent steps of the restoration.

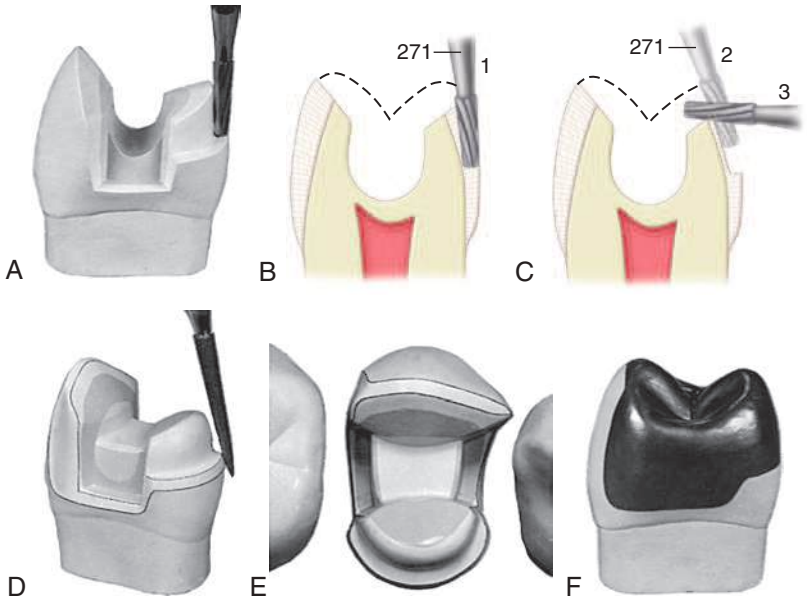
Collar Preparation

To increase the retention and resistance forms when preparing a weakened tooth for a mesioocclusodistal onlay to cap all cusps, a facial or lingual "collar" or both may be provided (Fig. 18.31). To reduce the display of metal, however, the facial surfaces of maxillary premolars and first molars usually are not prepared for a collar. The operator uses a No. 271 carbide bur at high speed parallel to the line of draw to prepare a 0.8-mm–deep shoulder (equivalent to the diameter of the tip end of the bur) around the lingual (or facial) surface to provide for a collar about 2 to 3 mm high occlusogingivally (see Fig. 18.31A and B). To provide for a uniform thickness of metal, the occlusal 1 mm of this reduction should be prepared to follow the original contour of the tooth (see Fig. 18.31C), and any undesirable sharp line angle formed by the union of the prepared lingual and occlusal surfaces should be rounded. This aspect of the preparation is completed by lightly beveling the gingival margin of the shoulder with the flame-shaped, fine-grit diamond instrument to achieve a 30-degree metal angle at the margin (see Fig. 18.31D).

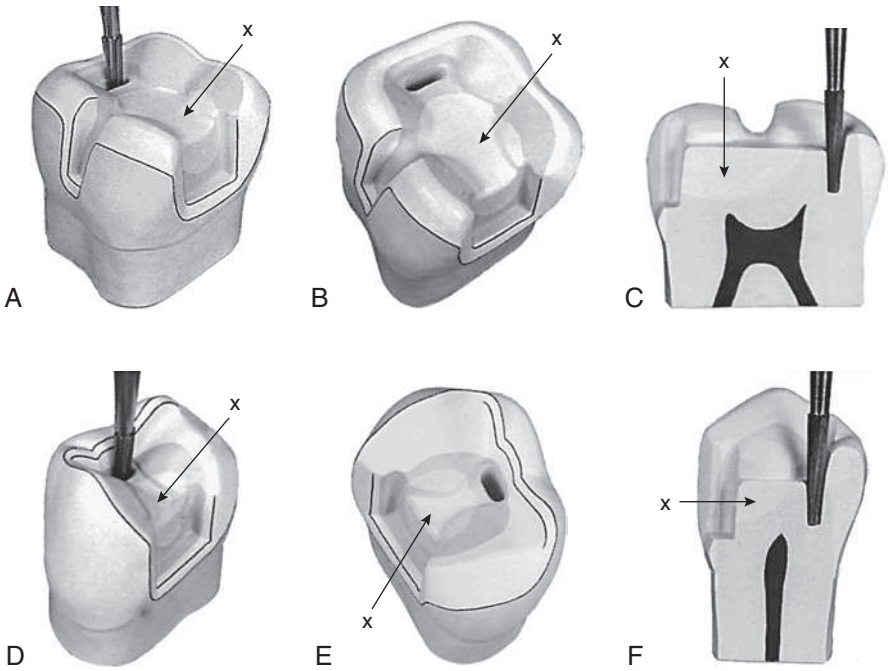
Slot Preparation

Occasionally the use of a slot in dentin is helpful in creating the necessary retention form. An example is the mandibular second molar that has no molar posterior to it and requires a mesioocclusal onlay restoration that caps all of the cusps (Fig. 18.32A–C). The distal, facial, and lingual surfaces are free of caries or other injury, and these surfaces also are judged not to be prone to caries. After cusp reduction, the vertical walls of the occlusal step portion of the preparation have been reduced so as to provide very little retention form. The necessary retention can be achieved by cutting a distal slot. Such a slot is preferred over preparing a box in the distal surface because (1) the former is more conserving of the tooth structure and of the strength of the tooth crown, and (2) the linear extent of marginal outline is less.

To form this slot, the dentist uses a No. 169L carbide bur with its long axis parallel to the line of draw (this must be reasonably close to a line parallel with the long axis of the tooth) (see Fig. 18.32A). The slot is cut in dentin so that it would pass midway between the pulp and the DEJ if it were to be extended gingivally (see Fig. 18.32C). The position and direction of the slot thus avert (1) the exposure of the pulp, (2) the removal of the dentin supporting the distal enamel, and (3) the perforation of the distal surface of the tooth at the gingival termination of the slot. The slot should have the following approximate dimensions: (1) the width (diameter) of the bur mesiodistally, (2) 2 mm faciolingually, and (3) a depth of 2 mm gingival of the normally positioned



• **Fig. 18.31** A, First position of the bur in preparing for the lingual collar on a weakened maxillary premolar. B and C, Section drawings of the first position of the bur (B) and the second and third positions (C). D, Beveling the lingual margin. Note the distofacial skirt extension. E, Completed preparation. F, Completed onlay.



• **Fig. 18.32** A and B, Cutting a distal slot for the retention for the mesioocclusal onlay to treat the terminal molar having a large base *(x)* resulting from extensive occlusal and mesial caries. C, Section of A. D and E, Preparing a mesial slot for the retention for the distoocclusal onlay to treat the maxillary first premolar that has a large base *(x)*. F, Section of *D*.

pulpal wall. To be effective, the mesial wall of the slot must be in sound dentin; otherwise, the retention form obtained is insufficient.

A comparable situation occurs occasionally: The maxillary first premolar requires a distoocclusal onlay restoration to cap the cusps, and the mesial surface is noncarious and deemed not prone to caries (see Fig. 18.32D–F). To reduce the display of metal and to conserve the tooth structure, a slot similar to that described in the preceding paragraph (except that it is mesially positioned and 1.5 mm wide faciolingually) may be used for the production of adequate retention. The mesioocclusal marginal outline in this preparation should be distal of the height of the mesial marginal ridge.

Modifications for Esthetics on Maxillary Premolars and First Molars

To minimize the display of metal on maxillary premolars and first molars, several modifications for esthetics are made to the basic onlay preparation. On the facial cusps of maxillary premolars and on the mesiofacial cusp of the maxillary first molar, the occlusal reduction should be only 0.75 to 1 mm on the facial cusp ridge to minimize the display of metal. This thickness should increase progressively to 1.5 mm toward the center of the tooth to provide rigidity to the capping metal. These cusps do not receive a counterbevel but are "stubbed" or blunted by the application of a sandpaper disk or the fine-grit diamond instrument held at a right angle to the facial surface (see Fig. 18.23C). The surface created by this blunting should be approximately 0.5 mm in width.

To further decrease the display of metal on maxillary premolars and first molars, the mesiofacial margin is minimally extended facially of the contact to such a position that the margin is barely visible from a facial viewing position. To accomplish this, the secondary flare is omitted, and the wall and margin are developed with a chisel or enamel hatchet. Final smoothing with the fine-grit paper disk is recommended when access permits. The cavosurface margin should result in a gold angle of 40 to 50 degrees, if possible.

When more than ideal extension of the mesiofacial margin is necessary because of a caries lesion or previous restoration, and as dictated by the esthetic desires of the patient, the operator may choose to place a composite insert at this margin. This is a more conservative option than preparing the tooth to receive a porcelain-veneered metal crown. When preparing the mesiofacial margin, no attempt is made to develop a straight mesiofacial wall past the point of ideal extension. After caries lesion excavation, a resin-modified glass ionomer base is inserted to temporarily form the missing portion of the wall. The base is contoured to the ideal form, and the preparation can continue, terminating the mesiofacial onlay margin in the ideal position in the base. After placement, the operator removes (with small round burs) the glass ionomer material to a depth of 1 mm for a composite insert. Small undercuts should be prepared in the wall formed by the cast-metal onlay (see Fig. 18.57A later in the chapter). (It is best to carve the undercut in the wall formed by the onlay during the wax pattern stage.) After beveling the enamel cavosurface margin and preparing a gingival retention groove where, and if, enamel is thin or missing, the composite veneer is inserted (see Fig. 18.62A later in the chapter).

Endodontically Treated Teeth

Routinely, teeth that have had endodontic treatment are weak and subject to fracture from occlusal forces. These teeth require restorations designed to provide protection from this injury (see Fig. 18.30K and L). This particularly applies to posterior teeth, which are subjected to greater stress. The need for such protection is accentuated when much of the strength of the tooth has been lost

because of extensive caries or previous restorations. When the facial and lingual surfaces of an endodontically treated tooth are sound, it is more conservative, for the health of the facial and lingual gingival tissue, to prepare the tooth not for a full crown, but for a full occlusal coverage onlay that has been designed with adequate resistance form to prevent future tooth fracture. Such features include skirt extensions and collar preparations. These features make the onlay more of an extracoronal restoration that encompasses the tooth such that the tooth is better able to resist lateral forces that otherwise might fracture the tooth.

Before starting the preparation of an endodontically treated posterior tooth, the pulp chamber should be excavated to the chamber floor and usually into the canals (1–2 mm), and an amalgam or composite foundation should be placed; this gives the onlay a firm base on which to rest. In the preparation of an endodontically treated premolar that has had extensive damage, the root canal may be prepared for a cast-metal or fiber-reinforced composite post, which is cemented before the onlay preparation is completed. This post helps the tooth resist forces that otherwise might cause a horizontal fracture of the entire tooth crown from the root. The post should extend roughly two thirds the length of the root and should terminate, leaving at least 5 mm of the root canal filling material at the apical portion of the root.

Restoring the Occlusal Plane of a Tilted Molar

An onlay is excellent for restoring the occlusal plane of a mesially tilted molar (Fig. 18.33). When the unprepared occlusal surface (mesial portion) is less than the desired occlusal plane, a corresponding decrease in occlusal surface reduction is indicated. To facilitate increasing the height of the tooth, while maintaining the desirable faciolingual dimension of the restored occlusal surface and good contour of the facial and lingual surfaces, the counterbevels on the latter surfaces often should be extended gingivally more than usual (see Fig. 18.33B).

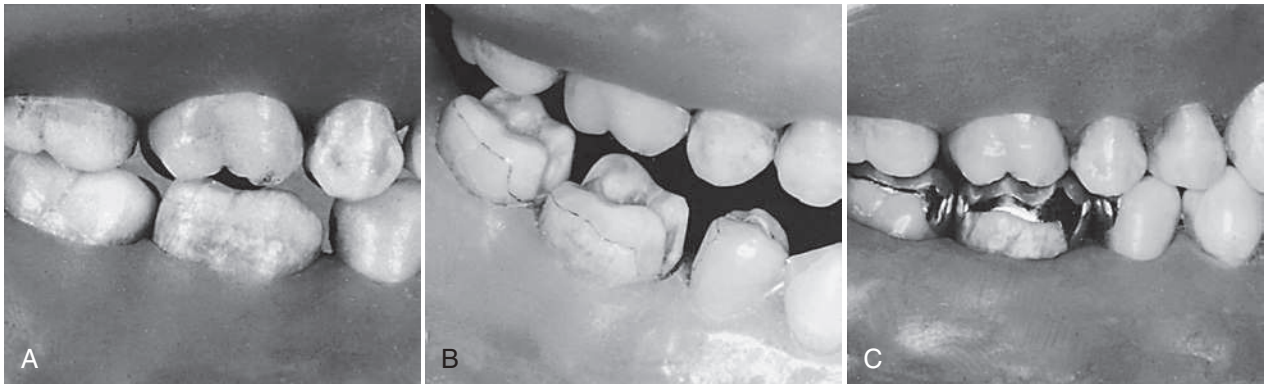
Often the mesiofacial and mesiolingual margins (on the "submerged" proximal surface) should be well extended onto the respective facial and lingual surfaces to help in recontouring the mesial surface to desirable proximal surface contour and contact. This extension can be accomplished with a minimal loss of the tooth structure by preparing facial and lingual skirt extensions on the respective proximal margins, which improves retention and resistance forms. In contrast, achieving extension by preparing the mesiofacial and mesiolingual walls facially and lingually does not improve retention or resistance forms and is less conservative of the tooth structure. Verification of appropriate cusp reduction is the same as presented for the inlay tooth preparation, as illustrated in Fig. 18.24.

Restorative Techniques for Cast-Metal Restorations

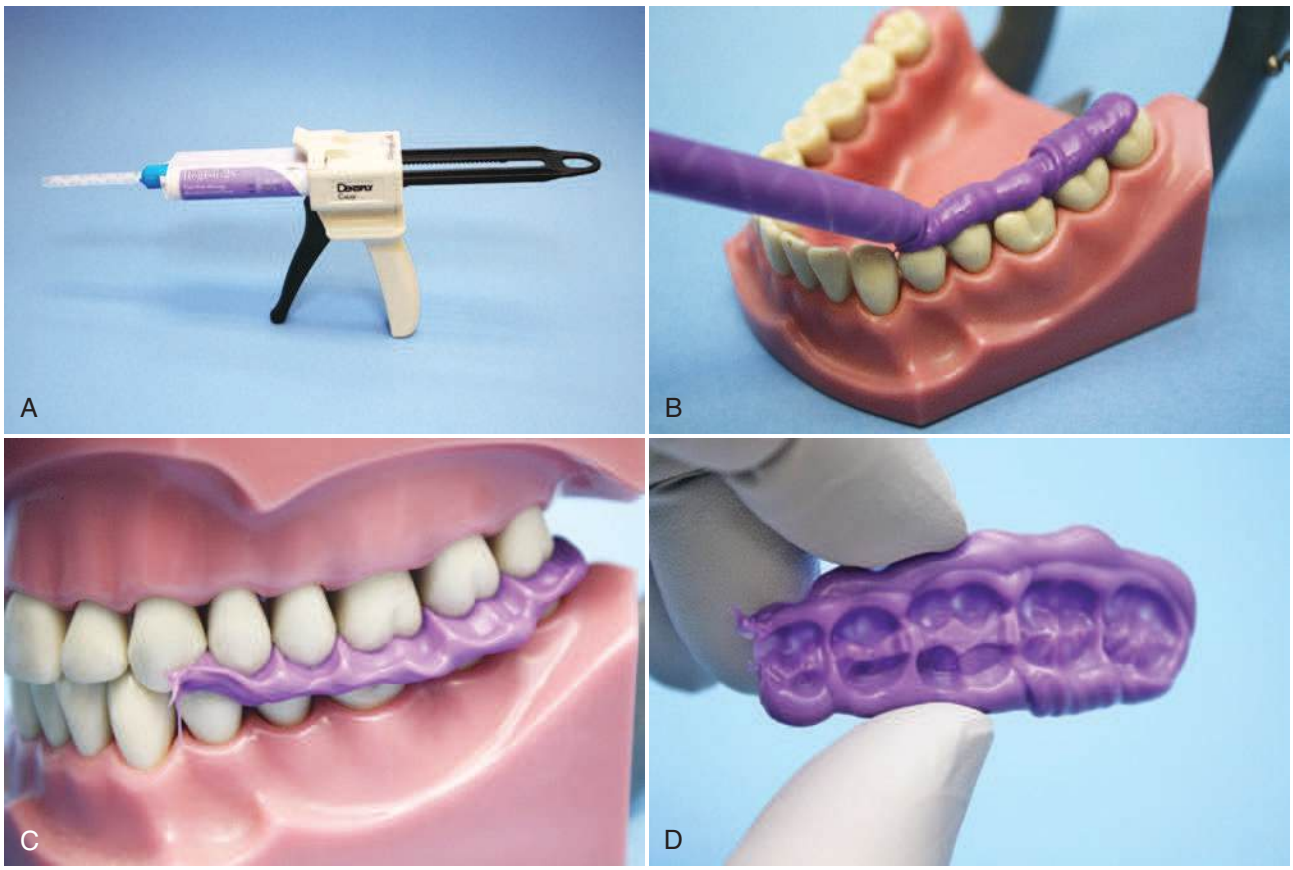
Interocclusal Record

Before preparation of the tooth, the occlusal contacts in maximum intercuspation and in all lateral and protrusive movements should have been carefully evaluated. If the patient has sufficient canine guidance to provide disocclusion of posterior teeth, the necessary registration of the opposing teeth can be obtained by (1) making a maximum intercuspation interocclusal record with commercially available bite registration pastes or (2) making full-arch impressions and mounting the casts made from these impressions on a simple hinge articulator. The interocclusal record works well when preparing one tooth; the full-arch casts are preferred when two or more prepared teeth are involved.

The maximum intercuspation interocclusal record can be made from one of several commercially available bite registration pastes. The most commonly used bite registration pastes are composed of heavily filled PVS impression materials. Several materials are available in cartridge systems that automatically mix the base and accelerator pastes together as they are expressed through a special disposable mixing tip (Fig. 18.34A). The mixed impression material is dispensed directly onto the prepared teeth and their opponents, then the patient closes completely (see Fig. 18.34B and C). The dentist observes teeth not covered by the bite registration paste to verify that teeth are in maximum intercuspation. When the material has set, the dentist removes the interocclusal record and inspects it for completeness. When held up to a light, areas where the



• **Fig. 18.33** A, The mandibular second and third molars being tilted mesially often is the result of failure to replace a lost first molar by bridgework. Note the poor contact relationship between the molars and between the molar and the second premolar. B, The second premolar is prepared for an inlay, and the molars are prepared for onlays. The margins of the preparations are well extended on the facial and lingual surfaces to aid in recontouring teeth to improve the occlusal relationship and to improve the proximal contours and contacts. C, Completed restorations. Note the improvement in the occlusal plane and in the proximal contacts.



• **Fig. 18.34** Maximum intercuspation interocclusal record made with PVS bite registration paste. A, One of many commercially available bite registration materials used in this technique. B, Using a cartridge dispenser and a disposable automixing tip, the base and accelerator pastes are automatically mixed and applied to the prepared teeth, their neighbors, and the opposing teeth. C, Have the patient close into maximum intercuspation position. Be sure that the adjacent unprepared teeth are touching in their normal relationships. D, Remove the maximum intercuspation interocclusal record carefully after it has set, and inspect it for completeness. Areas where the adjacent, unprepared teeth have penetrated through paste should be seen.

adjacent unprepared teeth have penetrated through the material should be seen (see Fig. 18.34D). The interocclusal record is set aside for later use in the laboratory.

The maximum intercuspation interocclusal records described in the previous paragraph provide information on the shape and position of the opposing teeth in maximum intercuspation. Such records give the laboratory technician some information about how to form the occlusal surface and position the occlusal contacts on the restoration, but they supply no data on how these structures and contacts might function during mandibular movements. This is also true when full-arch casts are mounted on a simple hinge articulator. Cast-metal restorations made with these simple bite registration techniques often require adjustments in the mouth to alleviate interferences during mandibular movements.

If information is desired in the laboratory about the pathways of cusps during mandibular movements (such as when the tooth is to be restored in group function), an excellent technique involves making full-arch impressions and mounting casts from these impressions on a properly adjusted semiadjustable articulator (Fig. 18.35). The use of full-arch casts mounted on a semiadjustable articulator is recommended when restoring a large portion of the patient's posterior occlusion with cast-metal restorations. It involves only a little extra chair time and gives the laboratory technician much more information about the general occlusal scheme, pathways of cusps, opposing cusp steepness and groove direction, and the anatomy of other teeth in the mouth. The technique uses a full-arch tray when making the final impression, which requires mixing more material, especially when using stock trays. The opposing arch is impressed with impression material, and the appropriate mandibular movement and face-bow transfer records are made. The reader is referred to Chapter 1 for principles regarding the use of the semiadjustable articulator in developing proper occlusal relationships for cast-metal restorations.

Temporary Restoration

Between the time the tooth is prepared and the cast-metal restoration is delivered, it is important that the patient be comfortable and the tooth be protected and stabilized with an adequate temporary restoration. The temporary restoration should satisfy the following requirements:

1. It should be nonirritating and protect the prepared tooth from injury.
2. It should protect and maintain the health of the periodontium.
3. It should maintain the position of the prepared, adjacent, and opposing teeth.

4. It should provide for esthetic, phonetic, and masticatory function, as indicated.
5. It should have adequate strength and retention to withstand the forces to which it will be subjected.

When properly made, the custom temporary restoration can satisfy these requirements and is the preferred temporary restoration. Temporaries can be fabricated intraorally directly on the prepared teeth (direct technique) or outside of the mouth using a postoperative cast of the prepared teeth (indirect technique). The indirect technique is not as popular as the direct technique because of the increased number of steps and complexity in the former; however, it is useful when making temporaries that might become "locked on" (e.g., intracoronal inlays) when using the direct technique.



• **Fig. 18.35** Full-arch casts mounted via a facebow transfer on a semi-adjustable articulator provide maximal information in the laboratory on how to position cusps to prevent undesirable contacts.

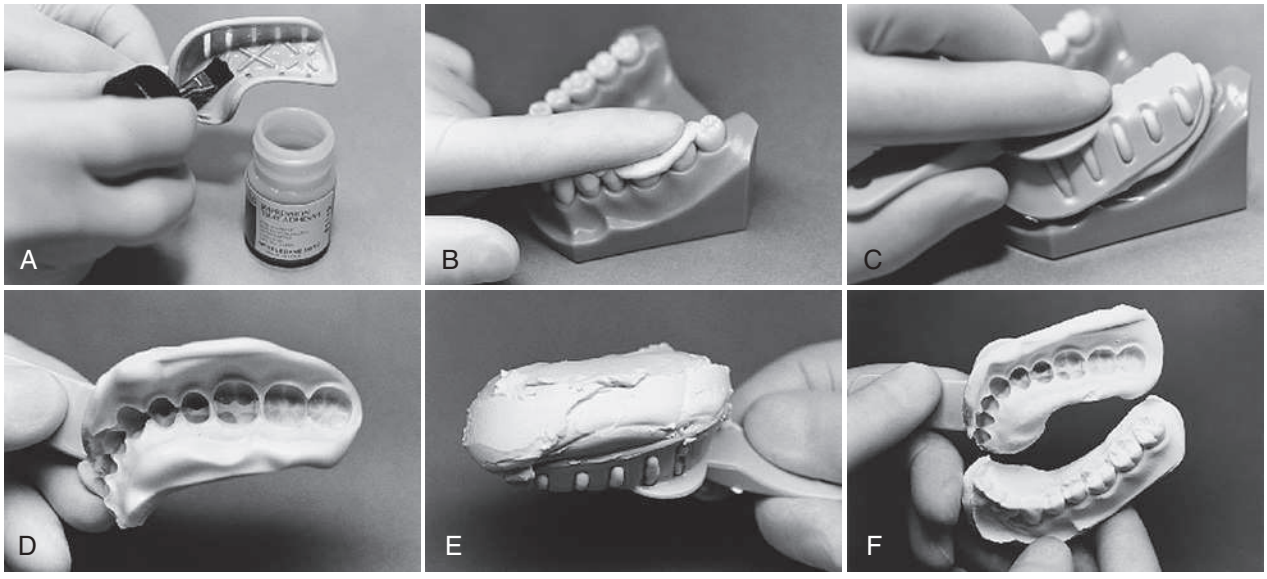
Technique for Indirect Temporary Restoration

The indirect temporary technique has the following advantages:

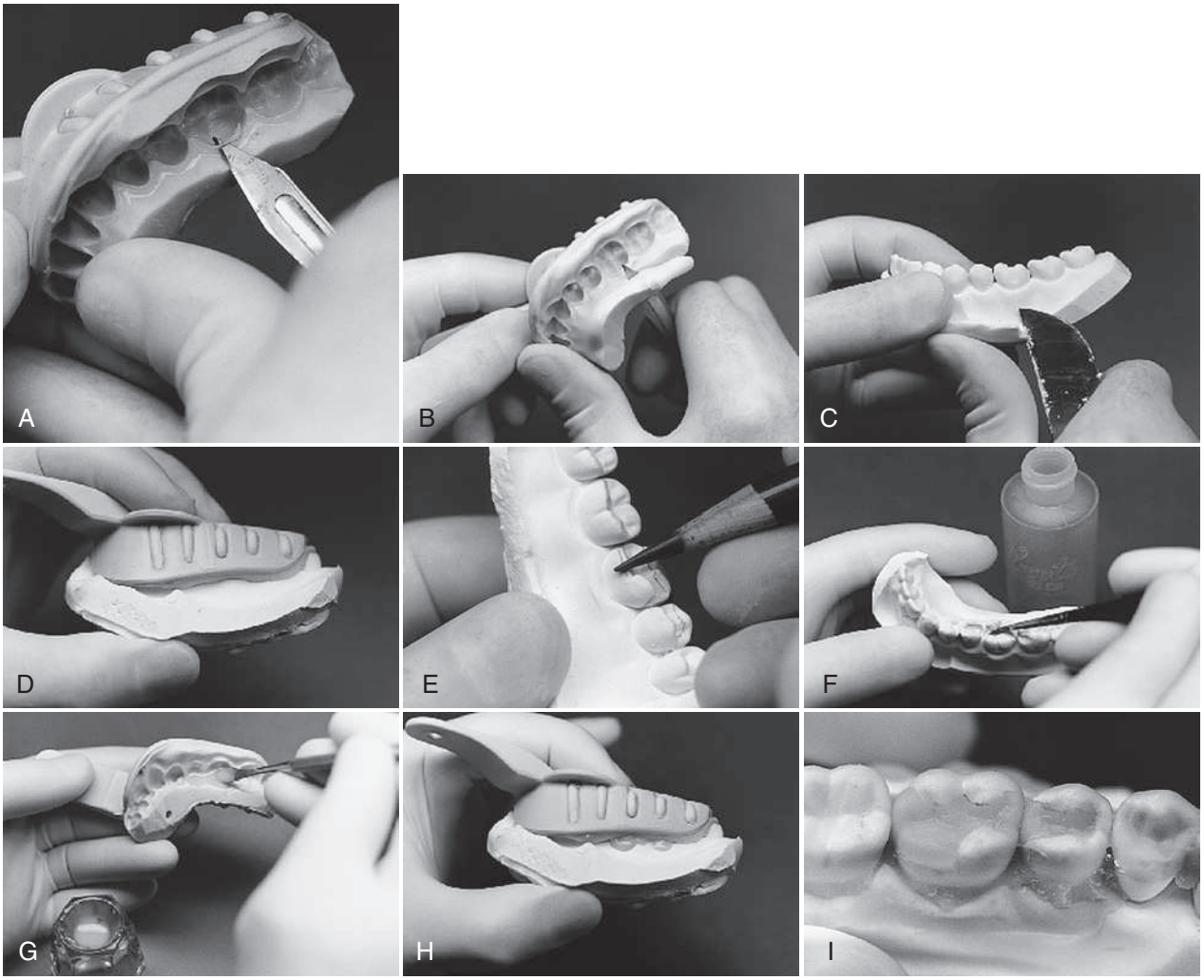
1. The indirect technique avoids the possibility of "locking on" the set temporary material into undercuts on the prepared tooth or the adjacent teeth.
2. The indirect technique avoids placing polymerizing temporary material directly on freshly prepared dentin and investing soft tissue, reducing potential irritation to these tissues.[8-10]
3. The postoperative cast made in the indirect technique affords an opportunity to evaluate the preparation (before the final impression) and serves as an excellent guide when trimming and contouring the temporary restoration.
4. Fabrication of the temporary restoration can be delegated to a well-trained dental auxiliary.

To form the indirect temporary, first an impression of the prepared tooth is made with fast-setting impression material. A stock, plastic impression tray that has been painted with tray adhesive is used (Fig. 18.36A). If using alginate, it is ensured that teeth are slightly moistened by saliva, then some alginate is applied over and into the preparation with a fingertip to avoid or to minimize trapping air (see Fig. 18.36B); then the tray is seated over the region (see Fig. 18.36C). After the material has become elastic, the impression is removed with a quick pull in the direction of the draw of the preparation and is inspected for completeness (see Fig. 18.36D). The impression is poured with fast-setting plaster or stone (see Fig. 18.36E).

As soon as the postoperative cast has been recovered from the impression, the dentist inspects the cast for any negative or positive defects (see Fig. 18.36F). Small voids on the cast may be filled in



• **Fig. 18.36** Making a postoperative plaster cast for indirectly forming a temporary restoration. A, The interior of the tray is coated with alginate tray adhesive. B, Some alginate is applied over and into the preparations with the fingertip to avoid trapping air. C, Alginate-filled tray in place. D, Alginate impression. E, The alginate impression is poured with fast-setting plaster. F, Plaster cast of the preparations shown in Fig. 18.24A.



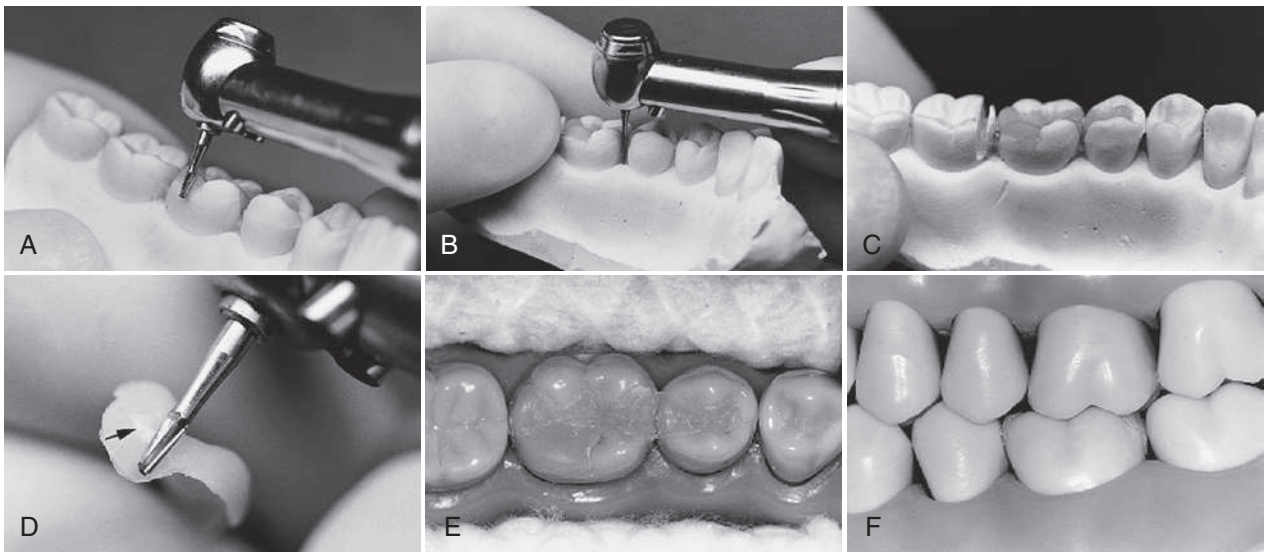
• **Fig. 18.37** Forming indirect temporary restorations for the preparations initially shown in Fig. 18.24A. A, Thin edges of the preoperative impression material that record the gingival sulcus should be cut away because these are apt to tear when seating the postoperative cast in the impression. B and C, Trimming away much of the soft tissue areas recorded by the impression and the cast also facilitates seating. D, Trial seating the postoperative cast into the preoperative impression. E, Marking the margins with red pencil. F, Applying the release agent to the cast. G, Filling the preoperative impression with temporary material in the area of the tooth preparation. H, Seating the cast into impression, taking care not to overseat or tilt the cast. I, Formed temporary restoration.

with utility wax. Large voids indicate repouring the impression. Positives (blebs) on the cast should be removed carefully with a suitable instrument.

The postoperative cast is seated into the preoperative impression (Fig. 18.37A–D). The thin edges of the postoperative impression material that record the gingival sulcus can be cut away for more thickness of the temporary in this area (see Fig. 18.37A). The postoperative cast is trial-seated into the preoperative impression to verify that it seats completely. Soft tissue around the perimeter of the impression or the cast, or in both areas, may have to be relieved to allow full seating (see Fig. 18.37B and C).

When satisfied that the gypsum cast seats completely in the preoperative impression (see Fig. 18.37D), the dentist removes the cast and marks the margins of the preparations on the cast with a red pencil to facilitate trimming (see Fig. 18.37E). A release agent is brushed on the preparations and adjacent teeth (see Fig. 18.37F). Tooth-colored temporary resin is mixed and inserted into the preoperative impression in the area of the prepared teeth (see Fig. 18.37G). When adjacent teeth are prepared, the temporary material may be continuous from one tooth to the next. The cast is seated into the preoperative impression (see Fig. 18.37H). The formed temporary restoration is shown in Fig. 18.37I.

With suitable burs the excess temporary material along the facial and lingual margins is trimmed away. The red line previously placed helps in the trimming, especially if it is performed by an auxiliary (Fig. 18.38A). On multiple-unit temporary restorations, a thin diamond instrument or the slender No. 169L bur or diamond can be used to refine the interproximal embrasures (see Fig. 18.38B). After the excess temporary material has been removed from the facial and lingual embrasures, a cut is made through the adjacent unprepared tooth 1 mm away from the proximal contact (see Fig. 18.38C). A knife is inserted into the cut and the temporary restoration is pried off from the cast. Access to improve the contour of the proximal surface that will contact the adjacent unprepared



• **Fig. 18.38** Trimming and adjusting the indirect temporary restorations. A, Trimming the excess material back to the accessible facial and lingual margins (marked by red line on the plaster cast). B, On multiple-unit temporaries, a slender bur or diamond instrument can be used to refine the interproximal embrasure form. C, On the cast, any tooth adjacent to the temporary restoration is cut away. D, Trimming the proximal surface of the temporary restoration to the proper contour. Care should be taken to avoid removing the proximal contact *(arrow)*. E and F, After the final impression is made, the temporary restoration is cemented with temporary cement. Note the anatomic contour and fit (E) and the functional occlusion of the temporary restoration (F).

tooth is now available (see Fig. 18.38D). The contact area on the temporary restoration that was accurately formed should not be disturbed.

Trial-fit the temporary restoration on the patient's teeth (see Fig. 18.38E). It should fit well, make desirable contact with the adjacent teeth, and meet occlusal requirements with minimal adjustments (see Fig. 18.38F). If occlusal adjustments are indicated, the prematurities are marked with articulating paper and reduced with an appropriate rotary instrument. After correcting the occlusion, any roughness or undesirable sharp edges are smoothed with a rubber point or wheel. The temporary restoration is removed from the mouth and set aside for cementation with temporary cement after the final impression has been made.

Technique for Direct Temporary Restoration

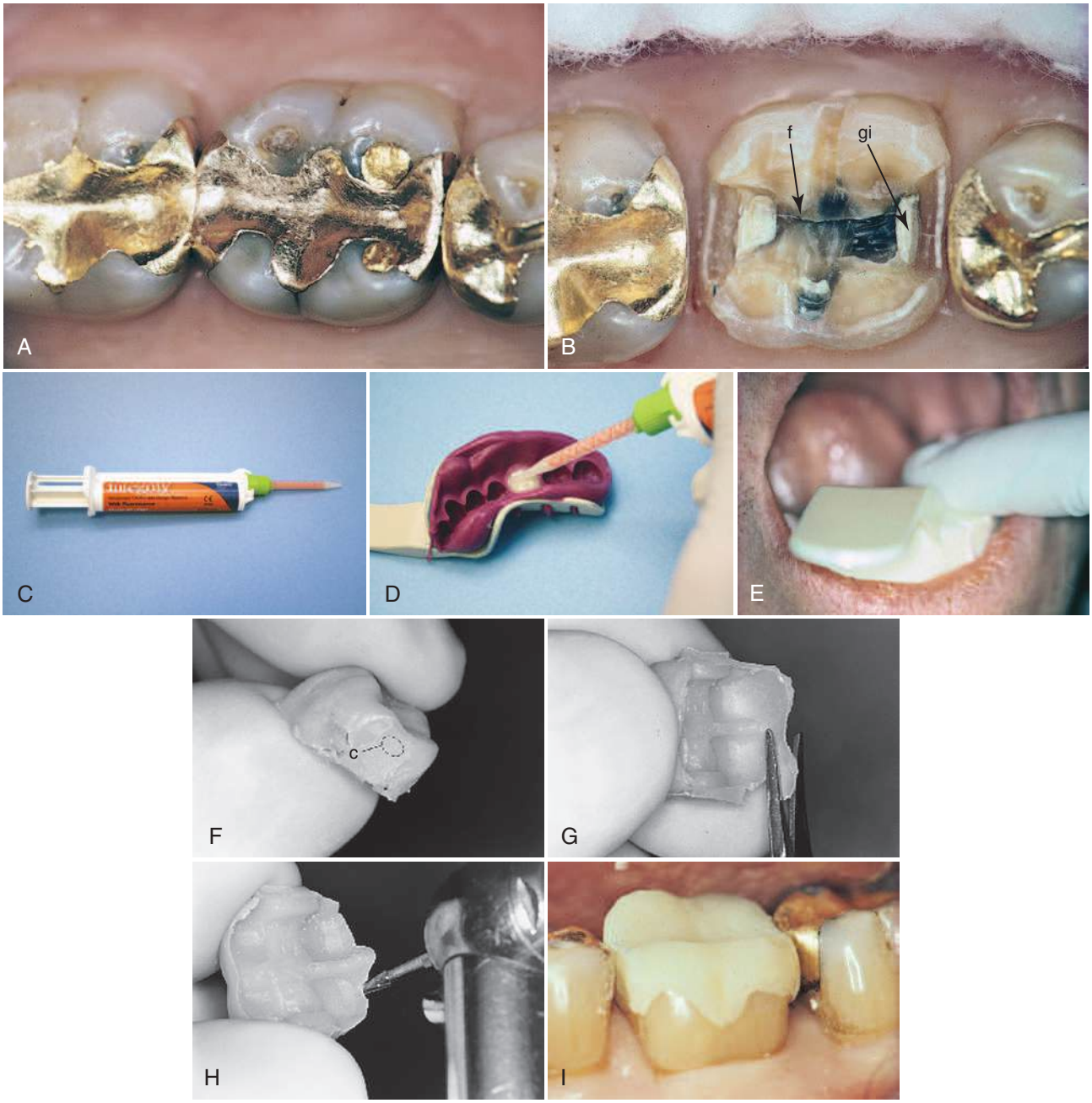
The direct temporary technique involves forming the temporary restoration directly on the prepared tooth and has the following advantages (Fig. 18.39): (1) The direct technique involves fewer steps and materials because no postoperative impression and gypsum cast are required, and (2) it is much faster than the indirect technique. The main disadvantages of the direct temporary technique include the following: (1) There is a chance of locking hardened temporary materials into small undercuts on the prepared tooth and the adjacent teeth, (2) the marginal fit may be slightly inferior to the indirect technique, and (3) it is more difficult to contour the temporary restoration without the guidelines offered by the postoperative cast.[11]

Forming the temporary restoration directly on the prepared tooth requires the preoperative impression (see Fig. 18.2C). Trial-fitting seats the preoperative impression onto teeth to verify that it seats completely. Because a potential for locking the temporary restoration on the tooth exists, it is necessary to eliminate undercuts in the preparation and occasionally in the proximal areas. Undercuts in the preparation should be "blocked out" using a resin-modified glass ionomer base (see Fig. 18.39B). A light film of a water-based lubricant over any exposed base prevents adherence and facilitates removal.

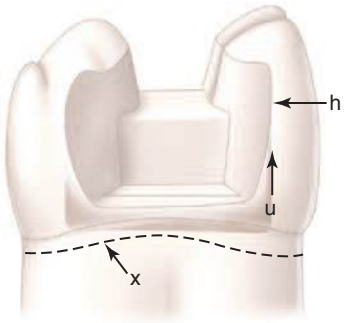
When using the direct technique with inlay and onlay preparations (preparations that gain their retention primarily through internal retention features), it is helpful to select temporary material systems that become elastic before the final set, allowing removal from undercuts without permanent distortion. The temporary material is mixed, following the manufacturer's instructions. Temporary materials that use automixing tips are especially convenient (see Fig. 18.39C). The dentist places the material into the preoperative impression in the area of the prepared tooth, taking care not to entrap any air (see Fig. 18.39D). The impression is placed on teeth, and the dentist ensures that it seats completely (see Fig. 18.39E). The manufacturer's instructions for gauging the setting time should be followed. Most temporary systems recommend monitoring the setting by rolling some excess material into a small ball and holding it between two fingers. When the temporary material has set to a firm stage, the impression is removed. When the material is sufficiently strong, the operator removes it from the tooth (see Fig. 18.39F). Excess material is trimmed away (see Fig. 18.39G). The cavosurface margins of the preparation can be seen inside the temporary restoration and are used as a guide for trimming the critical external areas near the margins (see Fig. 18.39H). The techniques for try-in, adjustment, and finishing the direct temporary restoration are identical to those described in the previous section (see Fig. 18.39I).

Final Impression

Fabrication of cast-metal restorations occurs in dental laboratories using replicas of the prepared and adjacent unprepared teeth. Classic



• **Fig. 18.39** Forming the direct temporary restoration with preoperative impression. Mesioocclusodistofacial onlay preparation for the mandibular first molar is used for illustration. A, Preoperatively, this patient had symptoms indicating an incomplete fracture of a vital tooth. B, After preparation for an onlay, an incomplete fracture *(f)* of dentin is seen extending mesiodistally along the pulpal floor. To maximize the retention and resistance forms, all the cusps are reduced for capping, a facial surface groove extension is prepared, and all four transitional line angles have skirt extensions. Resin-modified glass ionomer bases are inserted into excavations on axial walls *(gi).* Bases should have a light coat of water-soluble lubricant to prevent adhesion. C, Automixing the temporary resin material. D, The mixed temporary material is poured into the preoperative impression of the prepared tooth. E, The preoperative impression is seated with the temporary material onto the prepared tooth. F, The formed temporary restoration is removed from the preparation (note the contact area *c*, which must not be removed during trimming). G, Thin excess can be removed by using scissors. H, The internal surface of the temporary restoration has record of the cavosurface margin that is used as guide for final trimming. I, After the final impression is made, the temporary restoration is cemented with temporary cement. The temporary material over the skirt extensions is left slightly overcontoured for additional strength.



• **Fig. 18.40** The shaded area on the prepared tooth is undercut in relation to the line of withdrawal of the impression. The impression material that is in the position of greatest undercut *(u)* must be withdrawn in the direction of the vertical arrow and flexed over the greatest heights of contour *(h)*. The position of the gingival attachment is indicated by *x*.

techniques involving impressions and gypsum casts are described in this chapter. The reader is referred to Chapter 12 for alternative methods in digital dentistry.

The material used for the final impression should have the following qualities:

1. It must become elastic after placement in the mouth because it must be withdrawn from undercut regions that usually exist on the prepared and adjacent teeth. Note the shaded portions in Fig. 18.40, which are undercut areas with regard to the line of draw of the preparation. A satisfactory impression must register some of this undercut surface to delineate the margin sharply and to signify the desirable contour of the restoration in regions near the margin.
2. It must have adequate strength to resist breaking or tearing on removal from the mouth.
3. It must have adequate dimensional accuracy, stability, and reproduction of detail so that it is an exact negative imprint of the prepared and adjacent unprepared teeth.
4. It must have handling and setting characteristics that meet clinical requirements.
5. It must be free of toxic or irritating components.
6. It must be possible to disinfect it without distorting it.

In addition to the absolute requirements listed, the choice of impression material is usually made by comparisons of cost; ease of use; working time; shelf life; and pleasantness of odor, taste, and color. The most common impression material used for the indirect casting technique is PVS. The technique for the use of this material is discussed in detail in the following sections.

Tissue Retraction

Final impression materials make accurate impressions only of tooth surfaces that are visible, clean, and dry. When margins are subgingival, retraction cords can be used to displace the free gingiva temporarily away from the tooth and to control the flow of any gingival hemorrhage and sulcular fluids. The objective of gingival retraction is to widen the gingival sulcus to provide access for the impression material to reach the subgingival margins in adequate bulk to resist tearing during impression withdrawal (see Fig. 18.40). The objective of control of hemorrhage and moisture is met by the use of retraction cord impregnated with appropriate styptics (e.g., aluminum chloride), vasoconstrictors (e.g., epinephrine), or both. The use of vasoconstrictors in retraction cord is contraindicated in some patients, especially those who have cardiac arrhythmias, severe cardiovascular disease, uncontrolled hyperthyroidism, or diabetes and patients taking drugs such as β-blockers, monoamine oxidase inhibitors, or tricyclic antidepressants.[12]

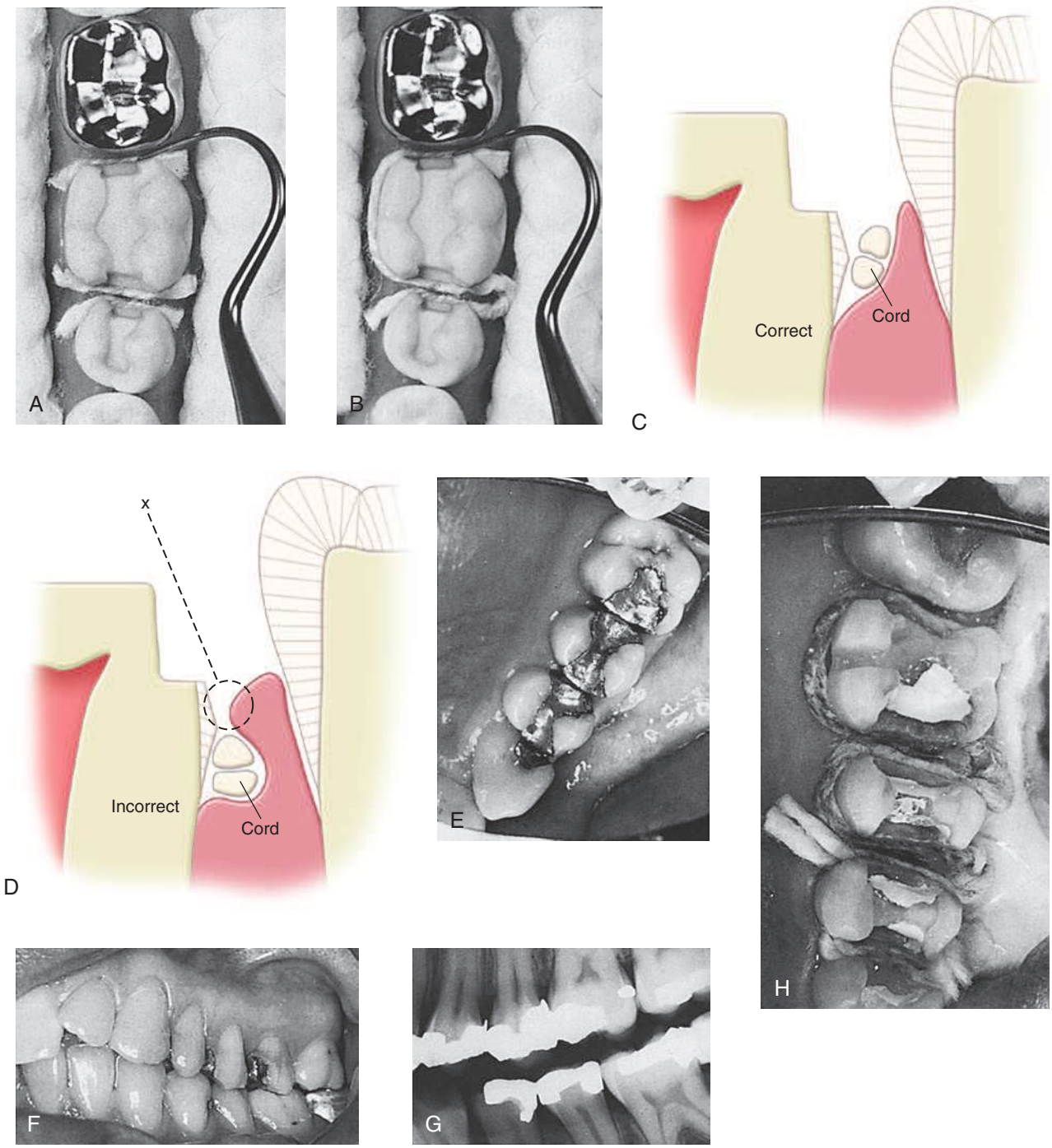
All sensory nerves to the region should be anesthetized, cotton rolls applied, and the saliva ejector inserted. Profound local anesthesia substantially reduces salivation to facilitate a dry field and allows tissue retraction without causing discomfort to the patient. The dentist selects and cuts a retraction cord of suitable diameter that is slightly longer than the length of the gingival margin. The cord may be cut long enough to extend from one gingival margin to another if they are on the same tooth or on adjacent teeth. In Fig. 18.41A and B, the cord is inserted into the gingival sulcus only in areas where the cavosurface margin is prepared subgingivally. Using the edge of a paddle-tipped instrument or the side of an explorer, one end of the cord is gently placed into the sulcus, about 2 mm facial to the point where the facial margin passes under the free gingiva. Then the cord is inserted progressively into the remainder of the sulcus, with the end of the cord left exposed, to be grasped with tweezers later in the technique (see Fig. 18.41A–C, H). The cord is placed to widen the sulcus and not to depress soft tissue gingivally (although some temporary retraction does occur apically).

Occasionally when the gingival margin is deep, it is helpful to insert a second cord of the same or larger diameter over the first. When the free gingiva is thin and the sulcus is narrow (e.g., facial surface of the maxillary or mandibular canine), a cord of very small diameter must be selected to prevent undue trauma to the tissue. In instances when a small diameter cord is used, layering a second cord on top of the first may be necessary to keep the sulcus from narrowing at the gingival crest.

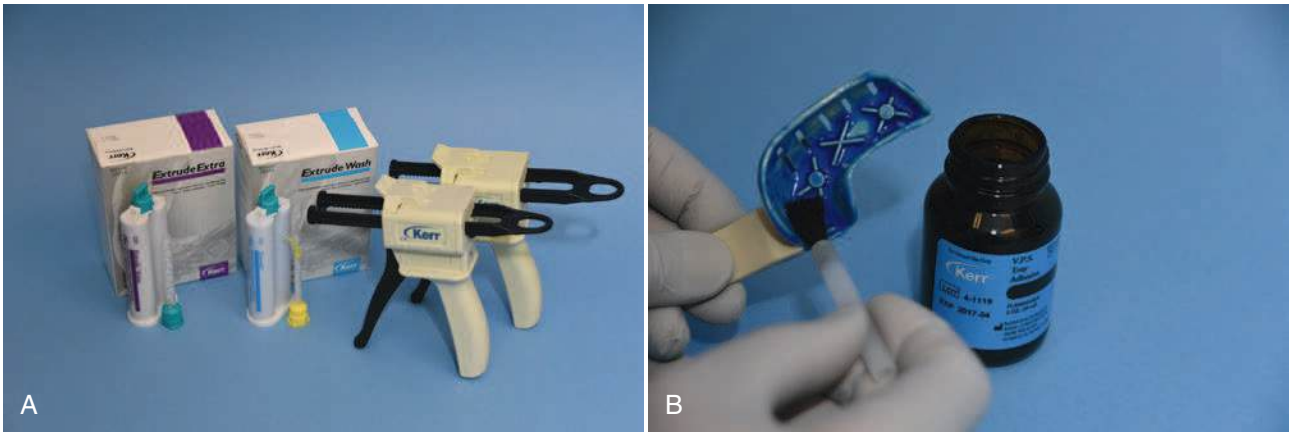
In Fig. 18.41D, the cord is incorrectly placed because it is tucked too deeply into the sulcus, as its depth permitted such positioning. When the cord is withdrawn before the injection of the impression material, the sulcus is wide at the bottom but narrow at the top. If the impression material is injected successfully into such a sulcus, the material is likely to tear in the region of *x* during the removal of the impression from the mouth. Correct application of the retraction cord is shown in Fig. 18.41C.

Occasionally the retraction cord becomes displaced from the sulcus during its insertion in the presence of slight hemorrhage or seepage, but this can be controlled if an assistant repeatedly touches the cord with dry cotton pellets or dries the area with a gentle stream of air. When excessive hemorrhage from the interproximal tissue occurs, first a cotton pellet is moistened with aqueous aluminum chloride solution, and the pellet is wedged between teeth so that it presses on the bleeding tissue. This pellet is left in for several minutes before it is removed and the cord is inserted. The widening or opening of the gingival sulcus by the earlier insertion of the retraction cord before the beveling of the gingival margin also should minimize or eliminate hemorrhage of the gingiva. For retracting a large mass of tissue, first a suitably shaped, large diameter cotton pack is made by rolling cotton fibers between fingertips, and the pack is then moistened with a drop or two of aqueous aluminum chloride and inserted into the sulcus.

The cords remain in place for several minutes. When hemorrhage or excessive tissue is present, more time is recommended. The region must remain free of saliva during this interval, and the patient should be cautioned not to close or allow the tongue to wet the teeth. Placing cotton rolls over teeth and having the patient close lightly to relax while the teeth remain isolated is sometimes helpful.



• **Fig. 18.41** A and B, Inserting the retraction cord to widen the gingival sulcus to expose the gingival margin. Separate lengths of cord can be inserted (one for each gingival margin) (A), or a cord long enough to run from one gingival margin to another can be inserted (B). Where the margin is not subgingival, as on the lingual surface of the molar, the cord should not be in the sulcus. C, Correct application of the retraction cord. D, Incorrect application of the retraction cord causing the impression material to tear at *x*. E, Maxillary quadrant before preparing teeth for onlays. Note the fracture of the mesiofacial cusp of the molar. F, Facial view of *E*. G, Bitewing radiograph of *E*. H, Teeth prepared for onlays and ready for making the final impression. The lingual and distofacial transitional line angles of premolars are prepared for skirting.



• **Fig. 18.42** A, Light-bodied (low viscosity), heavy-bodied (high-viscosity) polyvinyl siloxane impression materials; dispensers, automixing tips. B, Painting adhesive on stock tray.

Fig. 18.41 E-H illustrates correct use of gingival retraction on the maxillary quadrant.

Polyvinyl Siloxane Impression

The PVS impression is discussed in detail here because it is widely used, and the technique for its use can be readily applied to most other impression materials. PVS impression materials have many advantages over other impression materials used for final impressions. They have excellent reproduction of detail and dimensional stability over time. They are user friendly because they are easy to mix and have no unpleasant odor or taste. PVS impressions can withstand disinfection routines without significant distortion. These impression materials come in the form of two pastes (base and catalyst) that are mixed in disposable, automix, cartridge-dispensing systems. These automix systems provide excellent mixing of the accelerator and base pastes (Fig. 18.42A).

Tray Selection and Preparation

The impression tray must be sufficiently rigid to avoid deformation during the impression technique. If the tray bends or flexes at any time, the accuracy of the impression may be affected. Two types of trays, commercial stock and custom made, are suitable. Use of stock plastic trays is convenient and saves time. The custom resin tray made over a 2- to 3-mm wax spacer on the study cast is an excellent tray. A thickness of impression material greater than 3 mm increases shrinkage and the chance of voids; a thickness less than 2 mm may lead to distortion or tear of the impression material or to breakage of narrow or isolated teeth on the cast during withdrawal from the impression. Adequate bonding of impression material to the tray is accomplished with the application of a special adhesive to the tray (see Fig. 18.42B).

Impression Technique

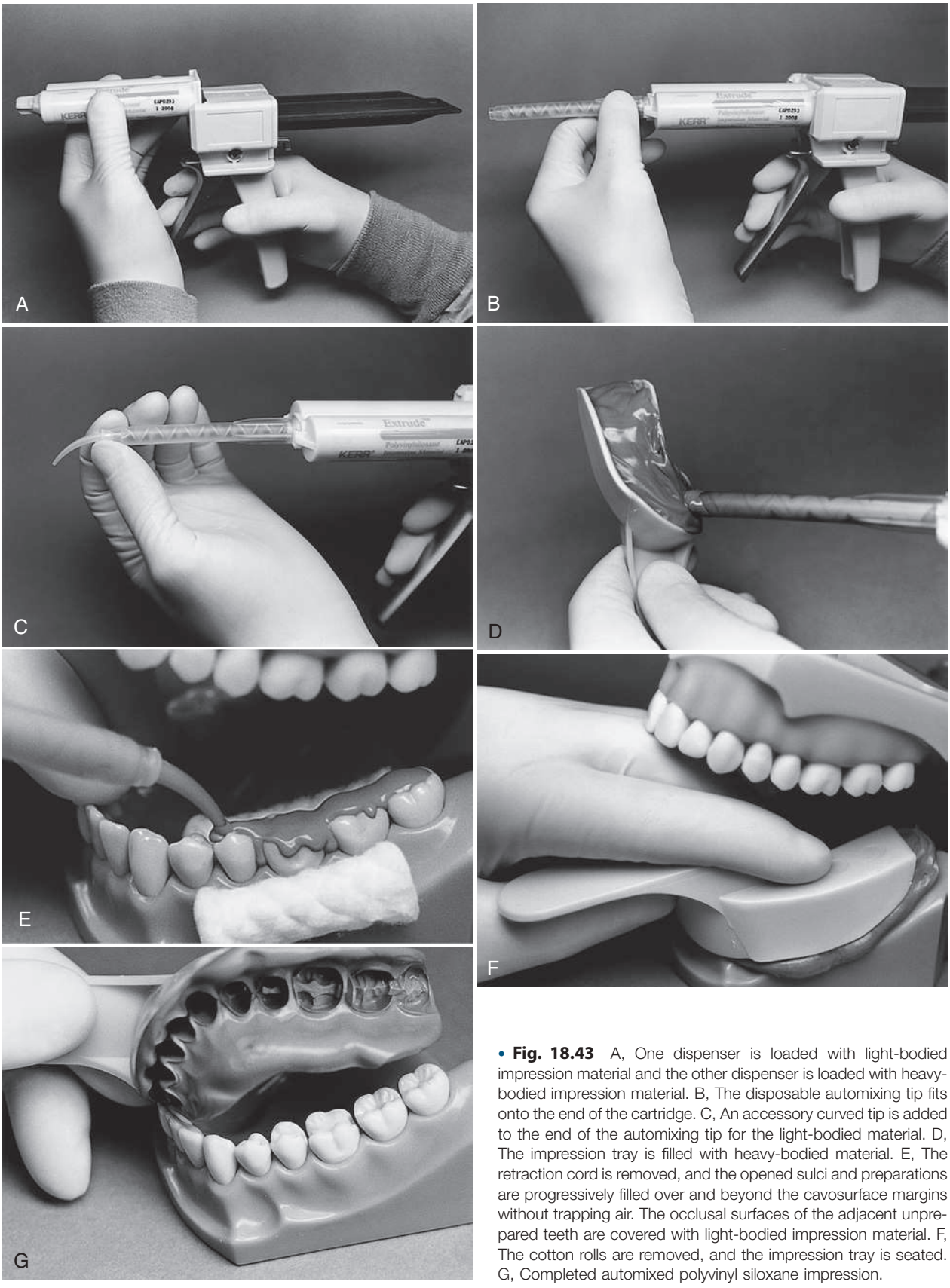
Most dental manufacturers offer their PVS impression materials in automix dispensing systems. The automixing systems have many advantages, including (1) speed, (2) consistent and complete mixing of accelerator and base pastes, and (3) incorporation of very few air voids during mixing and delivery to teeth. The technique demonstrated illustrates the use of two viscosities of impression material, a light-bodied material to inject around the preparation and a heavy-bodied material to fill the tray. Two dispensing guns are needed (Fig. 18.43A). The dispensers are loaded with cartridges that contain the accelerator and base pastes (see Fig. 18.43B). A disposable automixing tip fits onto the end of each cartridge (see Fig. 18.43C). The light-bodied mixing tip has an accessory curved tip that is small enough to gain access to the smallest, most remote areas of the preparation (see Fig. 18.43D).

The first dispenser is used to mix and fill the impression tray with the heavy-bodied impression material (see Fig. 18.43E). The dispensing tip should be kept embedded in the impression material as it is expressed into the tray so that the chance of trapping air is decreased. The second dispenser is then used to mix and inject the light-bodied impression material on the prepared teeth (see Fig. 18.43F). Teeth should be examined to ensure that the field is still clean and dry. Any visible moisture on teeth is removed with compressed air. The retraction cord is gently removed with operative pliers. All preparation surfaces should be clean, dry, and exposed to view. Next the opened gingival sulci and preparations are deliberately and progressively (moving from distal to mesial) filled over and beyond the margins with material from the syringe. To avoid trapping air, the tip is kept directly on the gingival and pulpal walls, filling the preparations from the gingival to the occlusal aspect, and the flow is regulated so that the material is not extruded too fast ahead of the tip. Light-bodied material also is injected on the occlusal surfaces of the unprepared adjacent teeth to eliminate the trapping of air on the occlusal grooves.

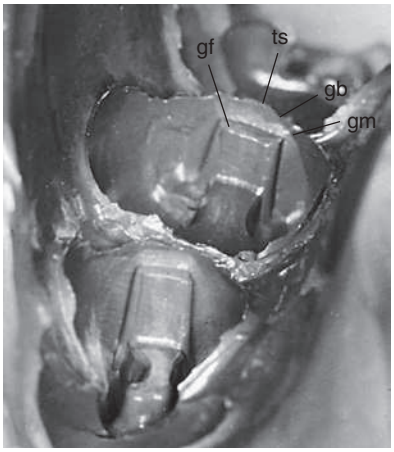
After filling and covering teeth with material from the syringe, the cotton rolls are immediately removed and the loaded tray seated over the region. The manufacturer's product instructions should be followed with regard to how long the material should be allowed to set before removal. As an additional safeguard, the operator should test the set of the impression material wherever it is accessible at the periphery of the tray. When it recovers elastically from an indentation made by the tips of the operative pliers, it is ready for removal (see Fig. 18.43G).

Removing and Inspecting the Impression

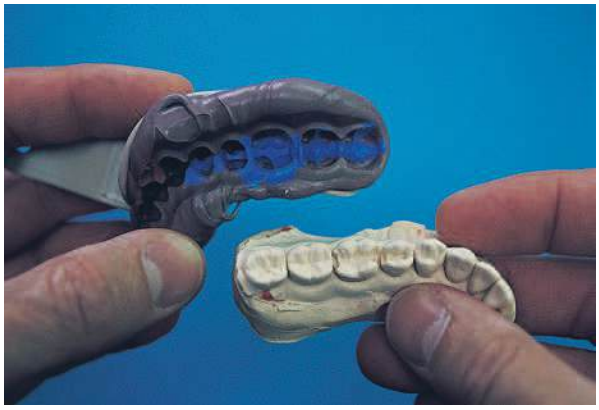
After the PVS impression has properly polymerized, it is removed from the mouth by a quick, firm pull that is directed as much as possible in line with the draw of the preparation. Removal is aided by inserting a fingertip at the junction of the facial border of the impression and the vestibule fornix, disrupting the vacuum that occasionally occurs during withdrawal, especially with full-arch impressions. The impression should be inspected carefully with good lighting and magnification. It should register every detail of the teeth and the preparation (Fig. 18.44).



• **Fig. 18.43** A, One dispenser is loaded with light-bodied impression material and the other dispenser is loaded with heavy-bodied impression material. B, The disposable automixing tip fits onto the end of the cartridge. C, An accessory curved tip is added to the end of the automixing tip for the light-bodied material. D, The impression tray is filled with heavy-bodied material. E, The retraction cord is removed, and the opened sulci and preparations are progressively filled over and beyond the cavosurface margins without trapping air. The occlusal surfaces of the adjacent unprepared teeth are covered with light-bodied impression material. F, The cotton rolls are removed, and the impression tray is seated. G, Completed automixed polyvinyl siloxane impression.



• **Fig. 18.44** Close-up view of the impression shows sharp details of record of the gingival floor *(gf),* gingival bevel *(gb),* and margin *(gm)* and a small amount of unprepared tooth surface *(ts)* beyond the margin.



• **Fig. 18.45** Cast poured from the die stone is inspected for completeness.

Working Casts and Dies

The working cast is an accurate replica of the prepared and adjacent unprepared teeth that allows the cast-metal restoration to be fabricated in the laboratory. During this fabrication procedure, it is most helpful if the replicas of the prepared teeth and of the adjacent unprepared teeth, called *dies*, are individually removable. The most used methods for creating a working cast with removable dies from an elastic impression require two pours. The first pour is made to produce the removable dies and the second pour is made to establish intraarch relationships. Working casts made in this manner are called *split casts*. Several satisfactory methods are available for making a split cast with removable dies. The Pindex system (Coltene/Whaledent Inc., Cuyahoga Falls, OH) is illustrated because it offers many advantages, as follows:

1. The first pour becomes the die segment and can be made quickly and easily.
2. Dowel pins can be positioned precisely, where needed.
3. Dowel pins are automatically positioned parallel, which facilitates die removal.

Pouring the Final Impression

A mix of high-strength die stone is made using a vacuum mechanical mixer, and the dies are poured with the aid of a vibrator and a No. 7 spatula. The first increments are applied in small amounts, allowing the material to flow into the remote corners and angles of the preparation without trapping air. Surface tension–reducing agents that allow the stone to flow more readily into the deep, internal corners of the impression are available. The impression should be sufficiently filled so that the dies are approximately 15 to 20 mm tall occlusogingivally after trimming. This may require surrounding the impression with boxing wax before pouring. After the die stone has set, the cast is removed from the impression and inspected for completeness (Fig. 18.45). This first pour (die segment) becomes the removable dies.

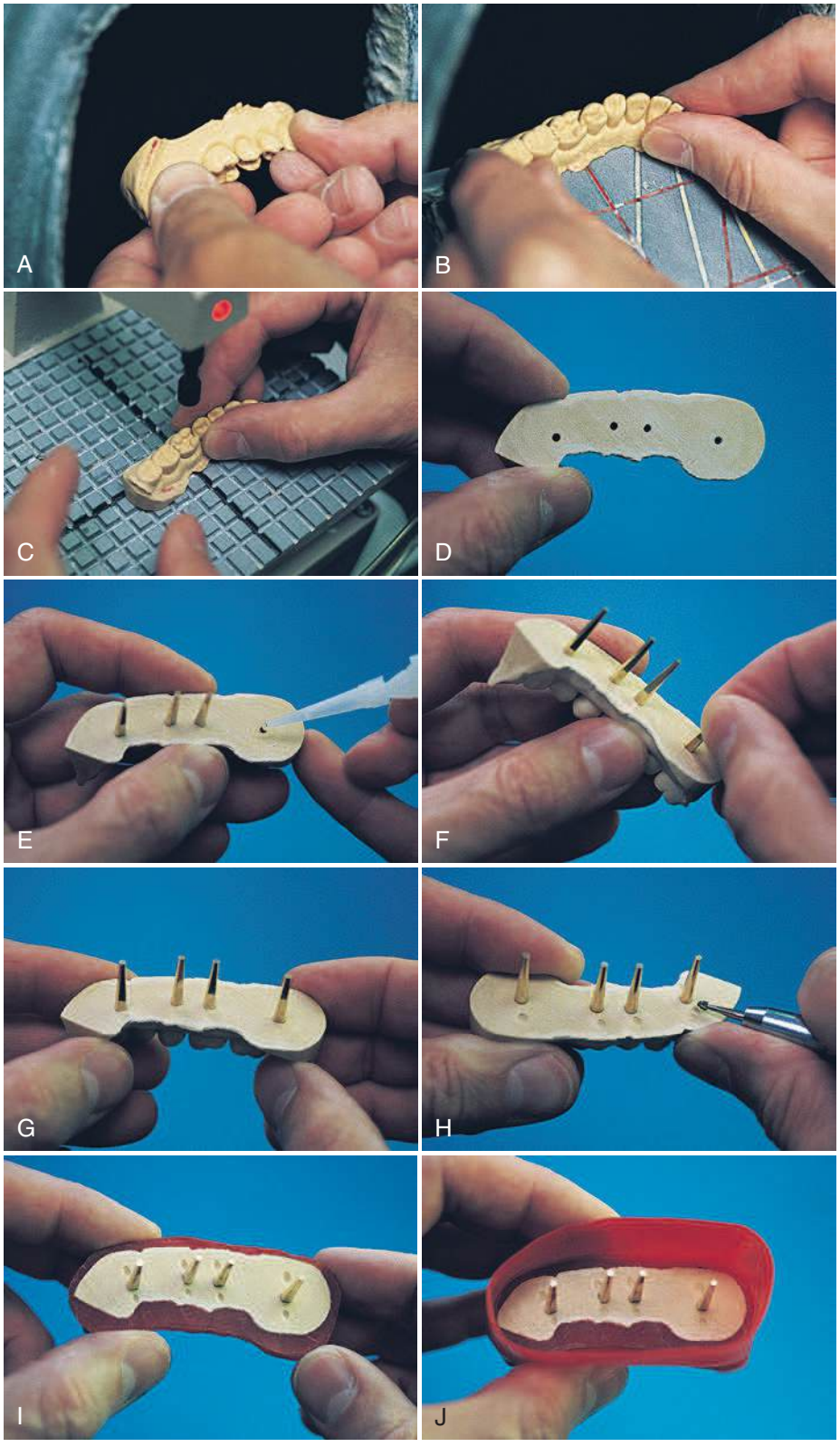
Completing the Working Cast

The base of the die segment is trimmed flat on a model trimmer (Fig. 18.46A). This trimming is approximately parallel to the occlusal surfaces of teeth. The operator must take care while doing this so that no grinding slurry is allowed to splash onto the dies. The dies should be approximately 15 mm occlusogingivally. When the base of the die segment is flat, the sides closer to the facial and lingual aspects of teeth should be trimmed (see Fig. 18.46B). Deep scratches left by the model trimmer are removed by wet sanding the base of the die segment with 220-grit wet or dry sandpaper.

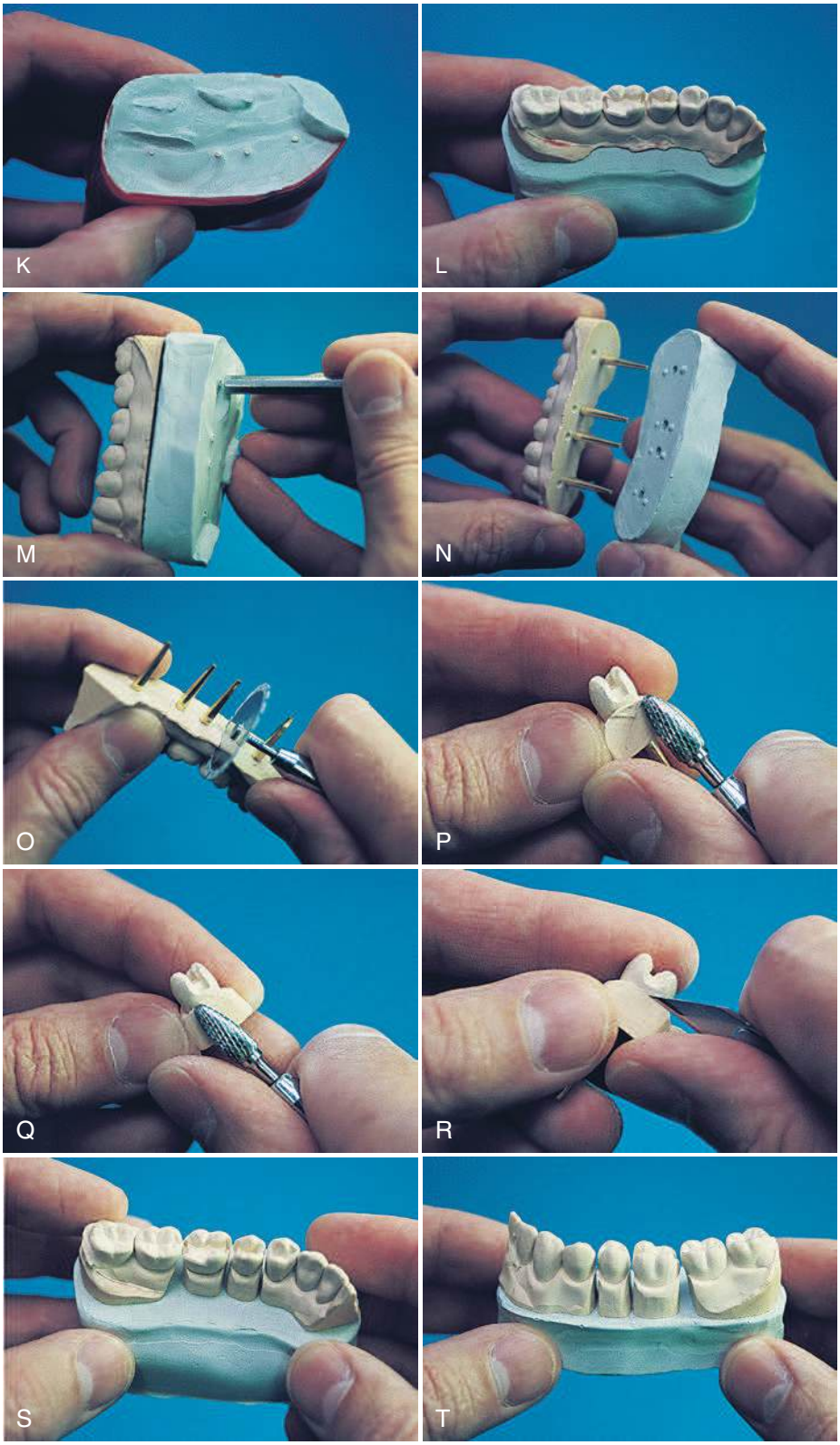
General rule: Teeth that will be removable are the prepared teeth with proximal gingival margins and any unprepared teeth adjacent to the prepared proximal surfaces. The two main advantages to making removable dies of unprepared teeth adjacent to prepared proximal surfaces are as follows: (1) The adjacent tooth will not interfere with removing the die that has the preparation, as occasionally may happen otherwise; and (2) adjusting the contacts is easier and more accurate when waxing and finishing the castings.

One dowel pin usually is placed in each prepared tooth and each adjacent tooth. When long sections of teeth are to be removable, the operator may wish to place more than one pin to increase stability and prevent rotation of the die. The cast is placed on the Pindex drilling machine, and one hole is drilled into the die base precisely in the middle of each tooth that is to be removable (see Fig. 18.46C and D). A small light beam helps position the cast correctly. When all the holes are drilled, a small drop of cyanoacrylate glue is placed in each hole, and a dowel pin is inserted (see Fig. 18.46E and F). The cast must be dry before cementing the pins, or the cement may not adhere. Any excess glue should be removed, and the operator should ensure that the dowel pins are parallel to one another (see Fig. 18.46G). To prevent rotation of the dies on the model base, small dimples may be placed just facial and lingual to each dowel pin with one third the diameter of a No. 6 round bur (see Fig. 18.46H). A bead of rope wax is placed around the die segment level with the base of the dies (see Fig. 18.46I). Then boxing wax is added around this to form a container for the base pour (see Fig. 18.46J). A separating medium is applied on the die segment, and a mix of dental stone is vibrated into the boxing wax container (see Fig. 18.46K). At least 1 mm of the ends of the dowel pins should be allowed to protrude. To provide adequate strength, the base of the cast should not be less than 10 mm thick.

After the stone has hardened, the boxing and rope wax are removed. Then the cast is removed from the impression (see Fig. 18.46L). The operator taps the end of each dowel pin lightly with the end of an instrument handle until a different sound is heard; this indicates that the die segment has moved slightly from its seating (see Fig. 18.46M). Next the ends of the pins are carefully pushed conjointly, causing the die segment to move equally away from its seating (see Fig. 18.46N). After the die segment is removed in this manner, the teeth that are to be individually removable



• **Fig. 18.46** A, The base of the die segment is trimmed flat and approximately parallel to the occlusal surfaces with a model trimmer. Dies should be approximately 15 mm high occlusogingivally. B, The die segment is trimmed on the facial and lingual surfaces to reduce the need for trimming in later steps. C, The die segment on the Pindex machine, ready to drill hole for first molar die. A small red dot of light helps position the cast. D, Holes drilled for removable dies. E, A drop of cyanoacrylate glue is poured into each hole. The cast must be dry for the glue to adhere. F, Immediately insert a dowel pin into the hole, being sure it is fully seated. G, The dowel pins must be parallel to one another and fully seated, and no excess glue must be present. H, To aid in indexing, small dimples are cut in the base of the dies, using one third the diameter of a large (No. 6) round bur. Typically these are positioned facial and lingual to the dowel pin. I, Rope wax is placed around the cast, flush with the die bases. J, Boxing wax is placed around the rope wax to create a container for the base pour. A separating agent must be painted on the die bases to prevent adherence to the base pour.



• **Fig. 18.46, cont'd** K, Base pour is completed. At least 1 mm of the dowel pin should be left protruding. L, Cast after removing boxing and rope wax. M, Tapping on the end of each dowel pin until the die segment moves. N, Removing the die segment from base. O, The dies are carefully cut apart by using a saw, bur, or thin diamond abrasive disk. Eye protection and dust collection are essential. P, Excess die stone around the gingival margins usually prevents good access for later steps in fabrication. Q, Removing the excess die stone with a large crosscut carbide bur in a slow-speed handpiece. Trimming across slightly gingival of the recorded gingival contour of the tooth weakens the excess, causing it to fall away. R, Final trimming is completed with a sharp scalpel. S, Cast completed, lingual view. Note how each prepared tooth and the adjacent unprepared teeth are removable now. T, Cast completed, facial view. Note the full seating of the dies.

must be cut apart from one another (see Fig. 18.46O). This requires the use of a saw, bur, or disk. To aid in carving the wax pattern and polishing the casting, the gingival aspect of the dies is carefully trimmed to expose the gingival margins properly (see Fig. 18.46P–R). The trimmed dies should have a positive and complete seating in the base portion of the cast (see Fig. 18.46S and T).

Caution: Do not allow any debris between the die portion and the base, or the accuracy will be compromised. This is especially true for the walls of the dowel pin holes. A small bit of wax or gypsum can be carelessly pressed onto the wall and prevent complete seating of the pin. Such debris is difficult to detect and remove to regain accuracy.

Use of Interocclusal Records

A maximum intercuspation interocclusal record is made before making the final impression. From this interocclusal record, a gypsum cast of the opposing teeth is made; this cast can be related accurately to the working cast, when forming the occlusal surface of the wax pattern. This step can be omitted if full-arch casts are to be used in waxing. See Chapter 1 for the principles of developing occlusion when using full-arch casts.

When using this type of interocclusal record, the working cast is mounted on a simple hinge articulator. The working cast is attached to one member of the articulator with fast-setting plaster. The interocclusal record is carefully fitted on the dies of the working cast (Fig. 18.47A and B). The interocclusal record should seat completely without rocking. Interocclusal bite records must never touch the registrations of soft tissue areas on the cast because these contacts usually interfere with complete seating. Such areas of contact on the interocclusal record can be trimmed away easily with a sharp knife. After ensuring that the interocclusal record is completely seated, lute the record adjacent to the unprepared teeth with sticky wax to prevent dislodgment when dental stone is poured into the record. Dental stone is then poured into the record (see Fig. 18.47C). This gypsum is attached to the opposite arm of the hinge articulator (see Fig. 18.47D), it is allowed to set, and then the interocclusal record is removed (see Fig. 18.47E–G).

Wax Patterns

Forming the Pattern Base

The operator lubricates the die and incrementally adds liquid wax from a No. 7 wax spatula by the "flow and press" method to form the proximal, facial, and lingual surface aspects of the pattern. A thin layer of wax should be added on the occlusal surface (Fig. 18.48A). Wax shrinks as it cools and hardens and tends to pull away from the die. This effect can be minimized and pattern adaptation improved by applying finger pressure for at least several seconds on each increment of wax soon after surface solidification and before any subsequent wax additions (see Fig. 18.48B). In this incremental technique, the wax that is flowed on the previously applied wax must be hot enough, as otherwise voids are formed.

Forming the Proximal Contour and Contact

The proximal contour and contact of the pattern are now formed on the pattern base (Figs. 18.49 and 18.50). The normal proximal contact relationship between teeth is that of two curved surfaces touching one another. The contact on each curved proximal surface is a point inside a small area of near-approach. Soon after eruption and the establishment of proximal contact, wear of the contact point from physiologic movements of teeth creates a contact surface. Lack of a proximal contact is usually undesirable because it creates the risk of proximal drifting of teeth, shifting occlusion, food impaction, and damage to the supporting tissues. Total lack of a proximal contact is often referred to as an *open contact* and is to be avoided.

Drawings of two maxillary premolars (see Fig. 18.50) are used to illustrate forms of contact and mesiodistal widths of interproximal spaces. Fig. 18.50A–C shows normal conditions. In Fig. 18.50A, the position of the contact is marked with an *x*, the area of near-approach of the two surfaces is indicated with a broken line, and the position of the crest of the gingiva is indicated with a continuous line. Fig. 18.50B is a mesiodistal section through teeth at the point of contact, and Fig. 18.50C is an occlusal view.

A broad contact faciolingually is illustrated in Fig. 18.50D–F. In the proximal view (see Fig. 18.50D), the position of a normal contact is marked *x*, the contact of this tooth is the outlined oblong area, and the area of near approach is the broken line. The crest of the gingiva is less arched, being almost horizontal along the area of near-approach. Viewed from the facial in the mesiodistal section (see Fig. 18.50E), the contact appears to be the same as in Fig. 18.50B, but a comparison of the occlusal views (see Fig. 18.50F and C) shows the extra breadth of this contact at the expense of the lingual embrasure. Fig. 18.50H shows a contact that is too far to the gingival. Its position in comparison with normal is shown by the relation of the circle to the *x* in Fig. 18.50G. The problem with such a contact is in the inclinations of the proximal surfaces from the occlusal marginal ridges to the contact. Stringy food is likely to become packed into this space, and the contact may impinge on the interproximal tissue.

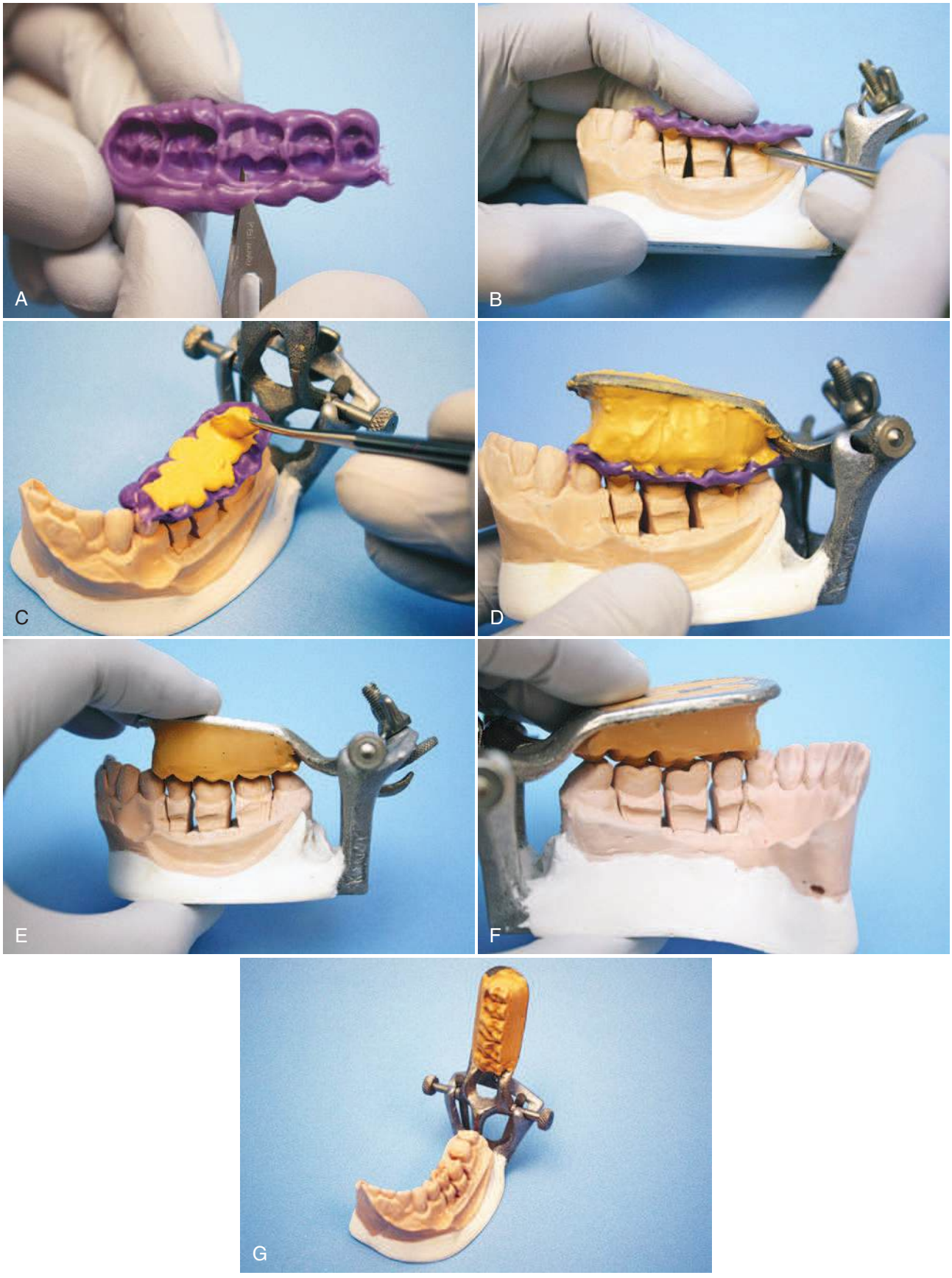
See Fig. 18.50I and J for an illustration of a contact too close to the occlusal. This form is frequently observed in restorations (especially amalgams), seldom in virgin teeth. Such a contact allows food to fill the gingival embrasure and invites proximal caries.

A contact that is too broad in the occlusogingival direction, but narrow faciolingually, is illustrated in Fig. 18.50K and L. The principal objections to this form of contact are that stringy foods are likely to be caught and held; also, if proximal recurrent caries occurs, it is farther to the gingival, requiring a tooth preparation close to the cementoenamel junction (CEJ). In cases of excessive proximal wear of teeth, the condition of the contact areas is similar to the combination of the areas illustrated in Fig. 18.50D and K, resulting in a facet of considerable dimensions.

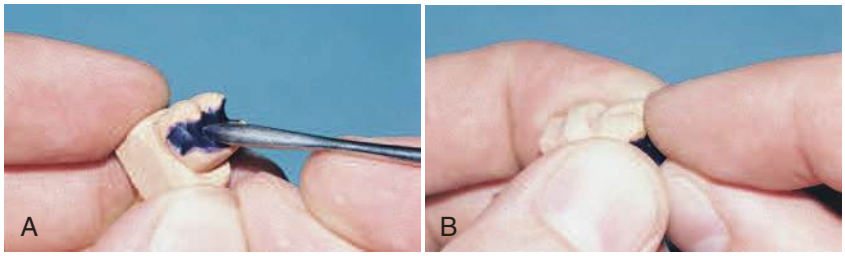
Forming the Occlusal Surface

Payne developed the fundamental principles in the following method of waxing.[7] The technique is particularly applicable when capping cusps. With practice, it has proven to be faster than the old method of building up wax, cutting away, building up again, and so on. The amount of wax desired is added in steps until the occlusal surface of the pattern is completed (Fig. 18.51).

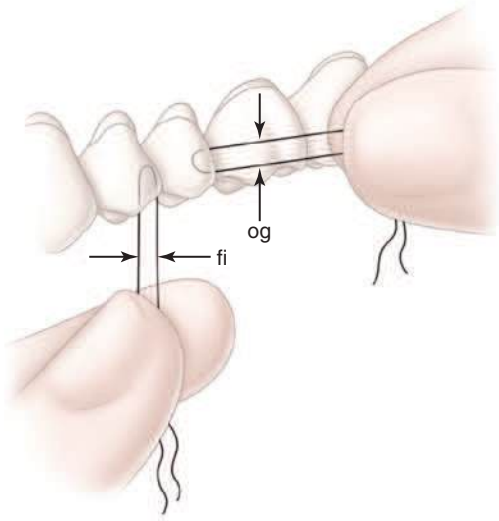
To obtain the faciolingual position of the cusp tips, the faciolingual width of the tooth is divided in quarters. Facial cusps are located on the first facial quarter line. Lingual cusps fall on the first lingual quarter line (see Fig. 18.51B). To obtain the mesiodistal position of the cusp tips, one notes the regions in the opposing tooth that should receive the cusp tips. The operator waxes small cones of inlay wax to the pattern to establish the cusp tips one at a time (see Fig. 18.51C and D). Next the operator waxes the inner and outer aspects of each cusp, being careful not to generate premature occlusal contacts (see Fig. 18.51D–F). It is suggested that only one aspect of each cusp be waxed into occlusion at a time. On the maxillary molar (illustrated in Fig. 18.51D), where all of the cusps are being restored, each of the nine aspects present



• **Fig. 18.47** Pouring the interocclusal record made with bite registration paste. A, Trimming away some of the interocclusal record (on the preparation side) with a sharp knife is often necessary to allow complete seating on the working cast. B, Fastening the seated interocclusal record to the working cast of preparations first shown in Fig. 18.33A with small amounts of sticky wax. C, Pouring stone into the interocclusal record. D, Attaching gypsum to the upper member of the hinge articulator. E–G, Three views of the completed mounting.



• **Fig. 18.48** To ensure optimal wax adaptation to the preparation walls, a thin layer of wax is first poured (A), and then finger pressure is applied for several seconds while the wax cools (B).



• **Fig. 18.49** Measuring the diameters of the proximal contact faciolingually *(fl)* and occlusogingivally *(og)* with dental floss. Two parallel strands should not be more than 1 to 2 mm apart. (Modified from Black GV: *Operative dentistry*, vol 2, ed 8, Woodstock, IL, 1947, Medico-Dental.)

are waxed one at a time. The operator should follow the proper angle on the inner and outer aspects (see Fig. 18.51E).

Next the distal slopes of the cusps are waxed (one at a time) into occlusal relation with the opposing teeth. The mesial slopes of the cusps are waxed next (one at a time) (see Fig. 18.51G). After the cusps are formed, the operator waxes in the proximal marginal ridge areas (see Fig. 18.51H). The same level to adjacent proximal marginal ridges should be developed, even though occasionally this may sacrifice a contact on one of the two ridges. Restoring marginal ridges to the same level avoids a "food trap" that otherwise would be created. The mesial and distal pit regions also should be carved enough to have them deeper than the respective marginal ridges. This provides appropriate spillways for the removal of food from the occlusal table and helps prevent food impaction in the occlusal embrasure area of the proximal surface.

To complete the occlusal wax-up, wax is added (where appropriate) to the fossae until they contact the opposing centric-holding cusps (see Fig. 18.51I). Spillways for the movement of food are established by carving appropriately placed grooves. Flat-plane occlusal relationships are not desired.

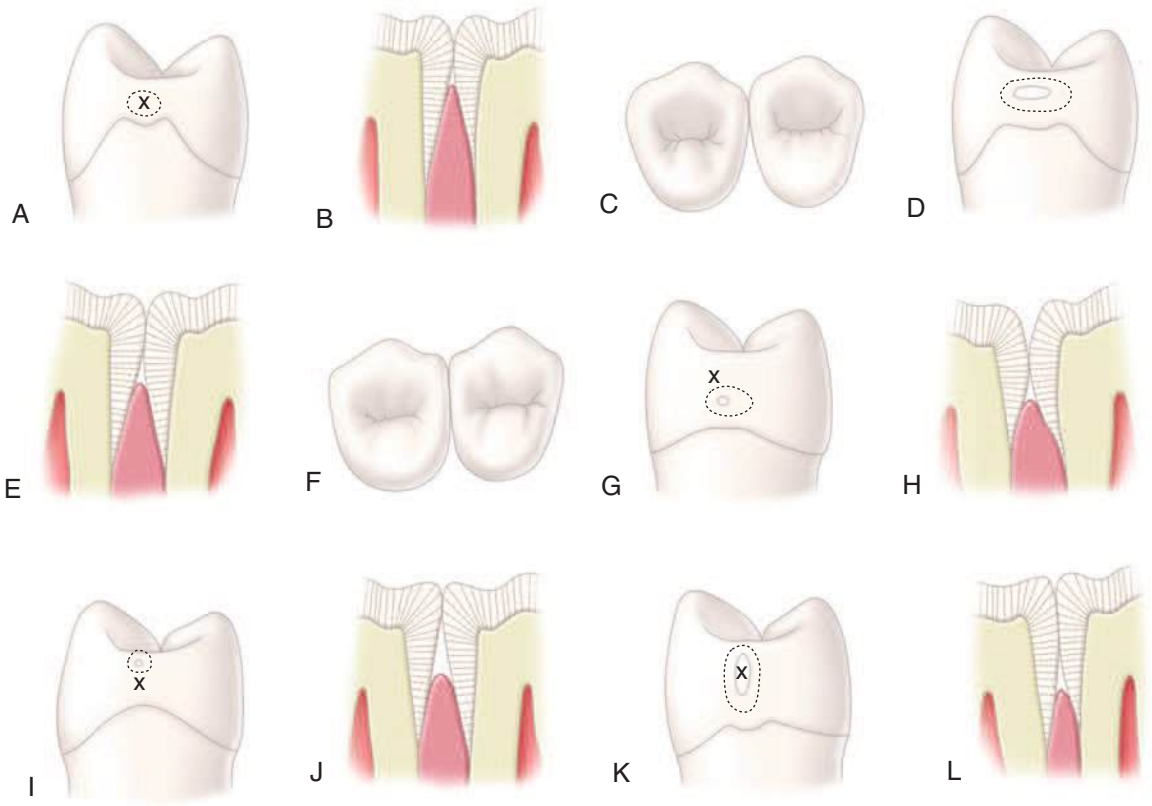
This technique is a systematic and practical method of waxing the occlusal aspect of the pattern into proper occlusion. Forming one small portion at a time results in waxing each portion into proper occlusion before adding another, which simplifies the procedure. Building the occlusal aspect by such small increments should help develop a pattern with minimal stress and distortion. Whenever a large portion of wax is added, it creates a potential for pattern distortion caused by the large shrinkage of such an addition.

For establishing stable occlusal relationships, the operator should take care to place the cusp tips against flat plateaus or into fossae on the stone cast of the opposing teeth. In other areas, the wax is shaped to simulate normal tooth contours, using adjacent teeth as references. Some relief between the opposing cusp inclines should be provided because these incline contacts often interfere during mandibular movements. The maximum intercuspation record provides only information regarding the position of the opposing teeth in maximum intercuspation. Some adjustment to the casting may be necessary in the mouth to eliminate interferences during mandibular movements. See Chapter 1 for the principles of cusp and fossa placement when using full-arch casts mounted on a semiadjustable articulator.

Finishing the Wax Pattern

Careful attention to good technique is required for waxing the margins of the wax pattern. There must be a continuous adaptation of wax to the margins, with no voids, folds, or faults. If adaptation is questionable, the marginal wax should be remelted to a distance into the pattern of approximately 2 mm. Finger pressure is applied immediately after surface solidification and before subsequent cooling of the wax, with pressure maintained for at least 4 seconds. This finger pressure helps develop close adaptation to the die by offsetting the cooling shrinkage of the wax. Additional wax should be added during the remelting procedure to ensure a slight excess of contour and extension beyond the margin.

Wax that is along the margins is now carved back to the cavosurface outline with a warmed No. 7 wax spatula (Fig. 18.52A–E). This warming of the spatula permits carving of the marginal wax with light pressure so that the stone margins are not damaged. A little practice helps the operator determine how much to heat the instrument for easy and effective carving. The No. 7 spatula should not have sharp edges; when it touches the die lightly, it should not abrade or injure the die surface. The operator uses the die surface just beyond the cavosurface margin to guide the position and direction of the carving instrument. The direction of the instrument movement is not dictated by the margin but by the contour of the unprepared tooth (die) surface just beyond the margin. The instrument blade is held parallel to this surface and used as a guide for the contour of the pattern near the margin; this should result in a continuity of contour across the margin. This principle of carving is too often neglected, resulting in the contour errors (see *x* in Fig. 18.52B–D); correct application of



• **Fig. 18.50** A–C, Correct contact. Note the position and form of the contact and the form of the embrasures around the contact. The mesial and distal pits are below (gingival of) the proximal marginal ridges. D–F, Contact too broad faciolingually. G and H, Contact positioned too far gingivally. I and J, Contact too close to the occlusal surface. K and L, Contact too broad occlusogingivally. (Modified from Black GV: *Operative dentistry*, vol 2, ed 8, Woodstock, IL, 1947, Medico-Dental.)

the carving instrument results in correct contours (exemplified by *y*). The completed patterns are shown in Fig. 18.52F–I.

On accessible surfaces of the carved pattern, satisfactory smoothness can be imparted by a few strokes with the end of a finger if surfaces have been carefully carved with the No. 7 spatula. Rubbing with cotton that has been twisted onto a round toothpick may smooth less accessible surfaces such as grooves.

Initial Withdrawal and Reseating of the Wax Pattern

Care must be exercised when initially withdrawing the wax pattern from the die. The wax can be dislodged by holding the die and pattern as shown in Fig. 18.53. When the pattern has been dislodged, it should be removed gently from the preparation. The operator should inspect the preparation side of the pattern to see if any wrinkles or holes are present. Such voids indicate poor wax adaptation and should be corrected if they are (1) in critical regions of the preparation designed to provide the retention form, (2) numerous, or (3) closer than 1 mm to the margin. To eliminate these voids, the operator first relubricates the die and reseats the pattern on the die. Then a hot instrument is passed through the wax to the unadapted area. This usually results in the air (void) rising through the liquid wax to the pattern's surface as the wax takes the place of the air. A consequence of this correcting procedure on the occlusal surface is the obliteration of the occlusal carving in the affected region, requiring the addition of wax, recarving, and rechecking the occlusion.

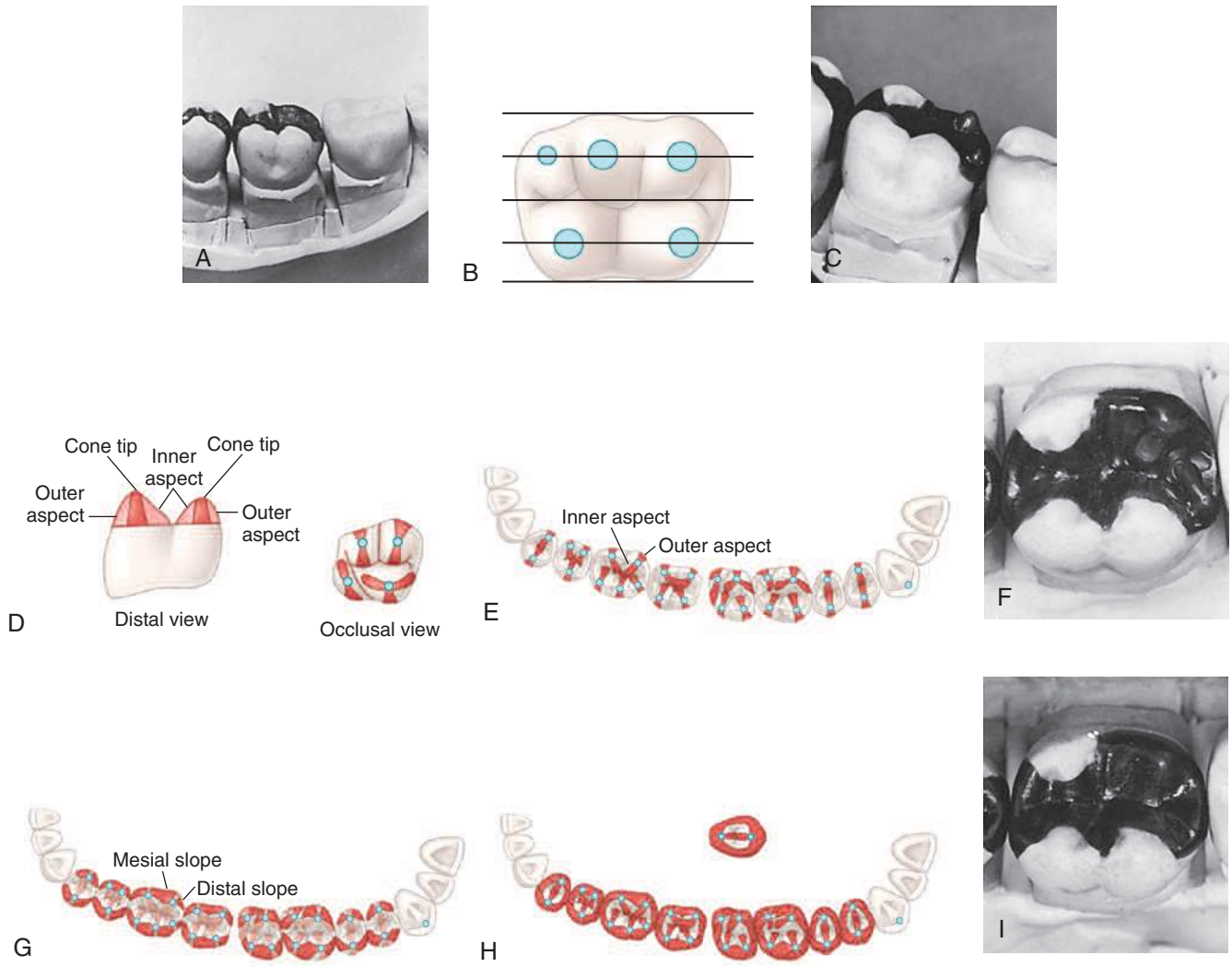
Spruing, Investing, and Casting

If a delay of several hours or more occurs between the forming of the wax pattern and the investing procedure, the pattern should remain on the die and the margins should be inspected carefully again before spruing and investing. When such a delay is contemplated, it is suggested that the sprue be added to the pattern before the delay period. If the addition of the sprue caused the induction of enough stress to produce pattern distortion, such a condition is more evident after the rest period, and corrective waxing can be instituted before investing. The reader is referred to textbooks on dental materials for the principles and techniques of spruing, investing, casting, and cleaning the casting.

Seating, Adjusting, and Polishing the Casting

It is crucial to examine the casting closely, preferably under magnification, before testing the fit on the die. The internal and external surfaces should be examined with good lighting to identify any traces of investment, positive defects (blebs), or negative defects (voids). Voids in critical areas indicate rejection of the casting, unless they can be corrected by soldering. Any small positive defects on the internal surface should be carefully removed with an appropriately sized round bur in the high-speed handpiece.

The casting is then trial-fitted on the die before removing the sprue and sprue button, which serve as a handle to remove the

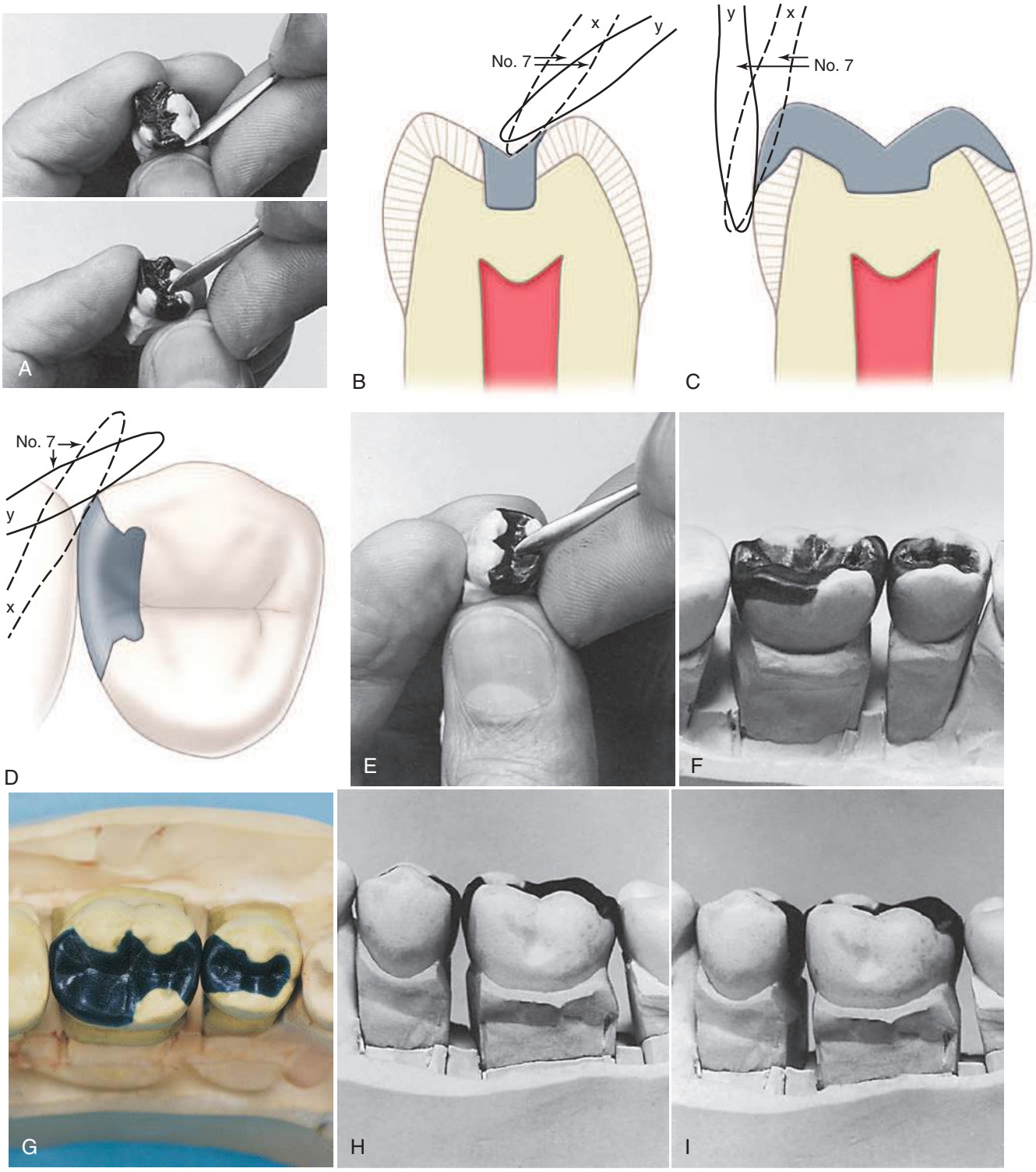


• **Fig. 18.51** A, The pattern base is completed and ready for waxing two reduced cusps (distolingual and distal) into occlusion by using Payne's waxing technique. B, The facial cusps are located on the first facial quarter line, and the lingual cusps fall on the first lingual quarter line. C, The distolingual and distal cusp tips are waxed into occlusion in the form of small cones. D, Cone tips and inner and outer aspects. E, Cone tips and inner and outer aspects of the cusps of teeth. F, The inner and outer aspects of the distolingual and distal cusps have been added to the pattern base. G, The mesial and distal slopes of the cusps of teeth. H, Marginal ridges of teeth. I, After the marginal ridge is added to the pattern base, fossae are waxed in and grooves are carved to complete the wax pattern. (Modified from Payne E: Reproduction of tooth form, *Ney Tech Bull* 1, 1961.)

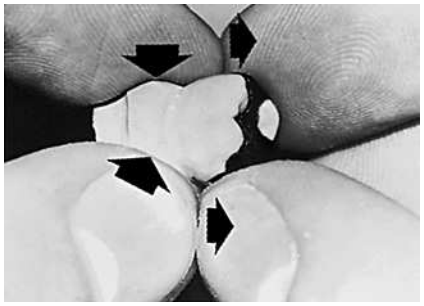
casting, if removal is necessary. The casting should seat with little or no pressure (Fig. 18.54A). Ideally when being placed on the die, it should have the same feel as that of the wax pattern when it was seated on the die. If the casting fails to seat completely, it should be removed and the die surface should be inspected for small scratches to see where it is binding. Usually failure to seat is caused by small positive defects not seen on the first inspection. Attempts at forcing the casting into place cause irreparable damage to the die and difficulties when trial-seating the casting in the mouth.

After the accuracy of the casting is found to be satisfactory, the casting is separated from the sprue as close to the inlay as possible using a carborundum separating disk. The cut should be made twice as wide as the thickness of the disk to prevent binding and should not cut completely through the sprue (a small uncut portion should be left) (see Fig. 18.54B). If the cut is made completely through, control of the disk is sometimes lost, often resulting in damage to the casting or to the operator's fingers. The uncut portion should be so small that bending with the fingers breaks it with very little effort (see Fig. 18.54C).

Having seated the casting on the die, the technician hand burnishes the marginal metal using a ball or beaver-tail burnisher (see Fig. 18.54D). An area approximately 1 mm in width is burnished, using strokes that increasingly approach the marginal metal and are directed parallel to the margin. Burnishing improves marginal adaptation and begins the smoothing process, almost imparting a polish to this rubbed surface. While burnishing, the adaptation of the casting along the margin is continually assessed by using magnification, as needed, to see any marginal opening 0.05 mm in size. Moderate pressure during burnishing is indicated



• **Fig. 18.52** A, Wax is carved to margins with a warm No. 7 spatula. B–D, Incorrect application of No. 7 spatula to carve the contour of the marginal wax is shown by *x*; the correct manner is labeled *y*. E, Carving the occlusal groove and pit anatomy. F, The adjacent marginal ridges should be on the same level as much as possible. G, Occlusal view of completed patterns. Note the shape of the facial and lingual embrasures and the position of the contact. H and I, Facial view of the completed patterns. Note the gingival and occlusal embrasures and the position of the contact.



• **Fig. 18.53** Removing the wax pattern by using indirect finger pressure. Arrows indicate the direction of the pressure. Care must be exercised not to squeeze and distort the wax pattern as it is initially withdrawn.

during closure of small marginal gaps. When the casting is well adapted, pressure is reduced to a gentle rubbing for continued smoothing of the metal surface. At this stage, marginal openings and irregularities should not be detectable even under (1.5× or 2×) magnification (see Fig. 18.54E and F). Care must be taken not to overburnish the metal because this can crush and destroy the underlying die surface. Overburnished metal prevents complete seating of the casting on the prepared tooth. Proper burnishing usually improves the retention of the casting on the die so that the casting does not come loose during subsequent polishing steps. A casting must not be loose on the die if the inlay is to be polished properly.

The remaining sprue metal is carefully removed with a heatless stone or a carborundum disk (see Fig. 18.54G and H). The grooves are accentuated by lightly applying a dull No. 1 round bur (see Fig. 18.54I) or other appropriate rotary instrument. Next a knife-edge rubber polishing wheel is used on accessible surfaces (see Fig. 18.54J) (Flexie rubber disk, Dedico International Inc., Long Eddy, NY). The operator should guard against the polishing wheel touching the margins and the die because both can be unknowingly and quickly polished away, resulting in "short" margins on the tooth. Also at this time, the proximal contacts are adjusted one at a time. If the distal surface of a mesiooccludodistal casting on the first molar is being adjusted, only the first and second molar dies are on the cast. Proximal contacts are deemed correct when they are the correct size, correctly positioned, and passive. If a temporary restoration was made properly, these contact relationships would be the same in the mouth as on the cast. Chair time can be reduced by carefully finishing the contacts on the cast.

The occlusion of the castings is checked by marking the occlusal contacts with articulating paper. Any premature contacts are corrected, and their locations are refined by selective grinding. Often prematurities occur where the sprue was attached and insufficient sprue metal was removed. The operator applies a smaller rubber, knife-edge wheel, which should reach some of the remaining areas not accessible to the larger disk (Fig. 18.55A and B). The grooves, fossae, and other most inaccessible regions are smoothed by rubber, abrasive points (Browne and Greenie rubber points; Shofu Dental Corp., San Marcos, CA) (see Fig. 18.55C). Care should be exercised when using the rubber disks and points so that the die surface is not touched and anatomic contours are not destroyed by overpolishing. When finished with the rubber abrasives, the surface of the casting should have a smooth, satin finish. It should be ensured that the contact relationships with the adjacent and opposing teeth have the correct size, position, and intensity.

The technician brushes the occlusal surface of the casting with a soft bristle disk and tripoli (or buffing bar compound [BBC]) (Buffing Bar Compound; Heraeus Kulzer Inc., Armonk, NY) polishing compound, running the disk parallel with the grooves (see Fig. 18.55D). A small felt wheel with polishing compound should be used on the proximal and other accessible surfaces (see Fig. 18.55E). The metal should be so smooth before this application of polishing compound that a beautiful luster should develop in a few seconds. A high sheen may be imparted, if desired, with a felt or chamois wheel and rouge (see Fig. 18.55F and G). As in the application of tripoli/BBC, only a few seconds of rouge application should be required. If more time were expended in the application of these polishing compounds, overpolishing (polishing away) of the margins and die would result. Also such overuse of polishing compounds is often an unsuccessful attempt to mask the fact that the preliminary stages of polishing were not thoroughly completed.

The technician cleans the polished casting of polishing compounds by immersing the die with its inlay in a suitable solvent for 1 or 2 minutes or by scrubbing with a soft brush and soap and water. The technician rinses and removes the casting from the die. No polishing compounds should be found on the preparation side of the casting or on the preparation walls of the die. The presence of such materials on these surfaces indicates that marginal adaptation on the die is not as good as it should be.

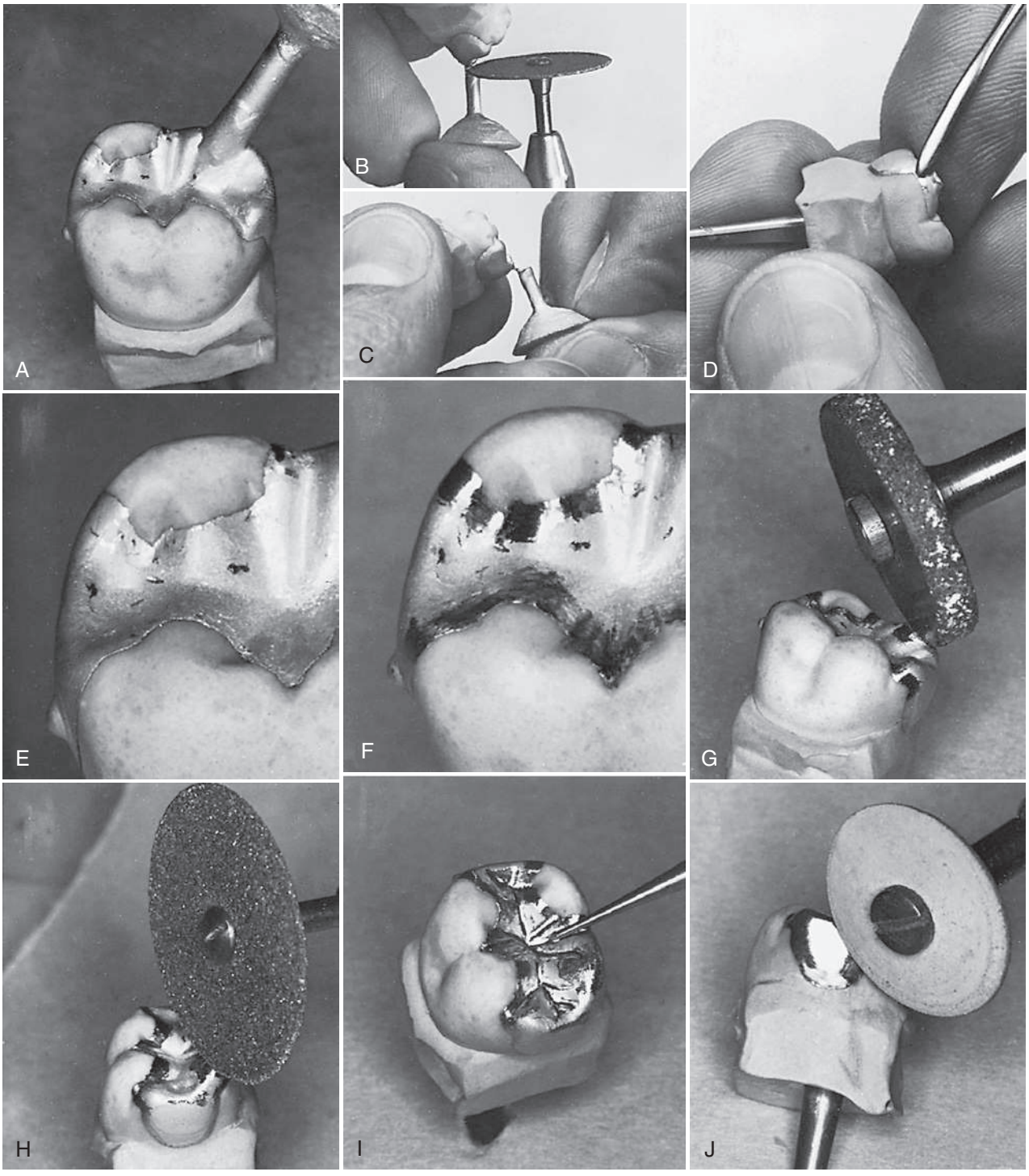
Trying-in the Casting

Preparing the Mouth

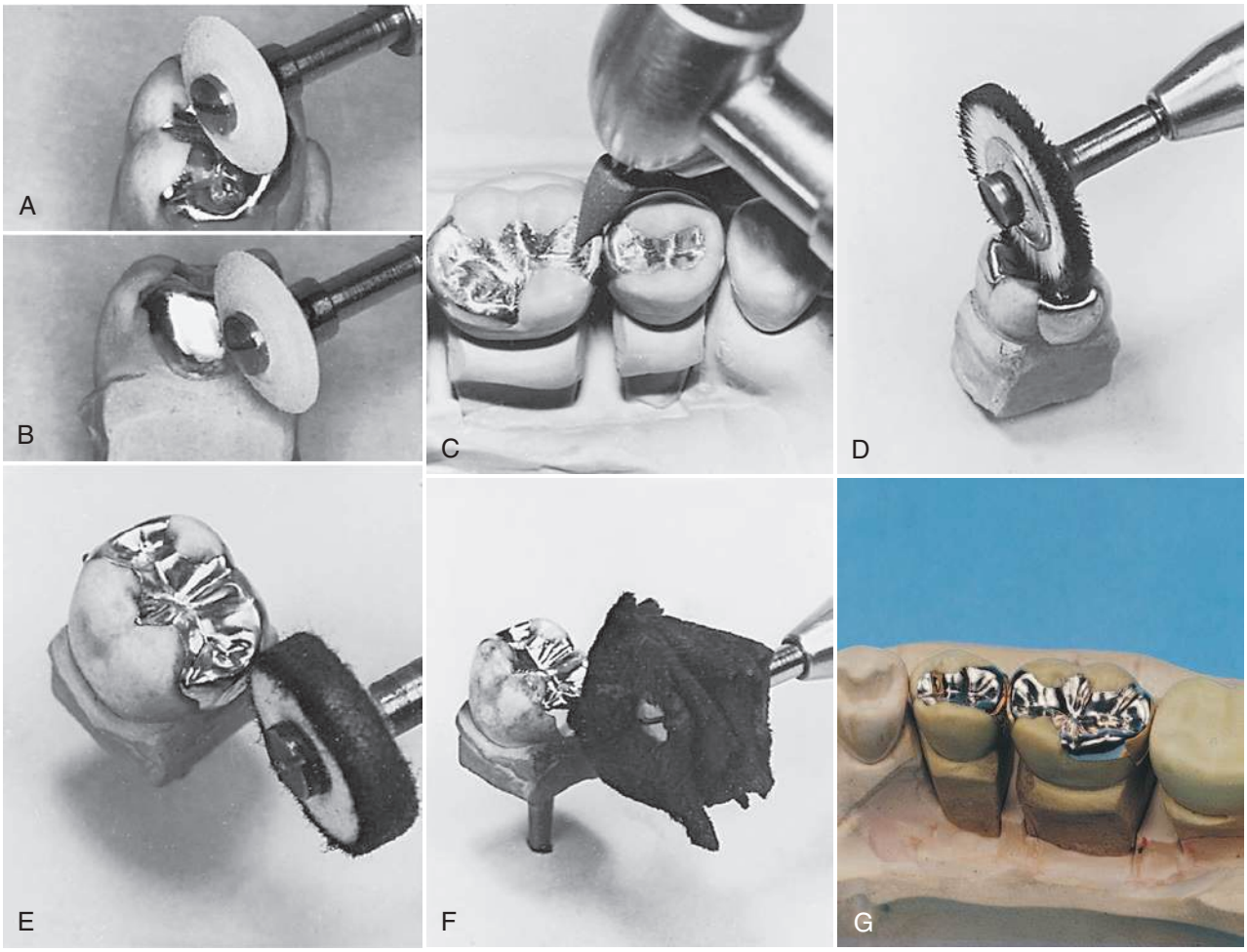
Local anesthesia of the tooth may be necessary before removal of the temporary restoration and the try-in of the casting on the tooth. Anesthesia blocks stimuli from inducing pain and salivation, neither of which is conducive to the best results, particularly in cementation. When teeth are not particularly sensitive, however, an option is to delay or eliminate administering the anesthetic because the patient can tell better if the proximal contacts are tight or if the occlusion is high. The temporary restoration is removed, ensuring that all the temporary cement has been dislodged from the preparation walls and cleared away. To improve visualization, the region is isolated with cotton rolls. Saliva is removed from the tooth operated on and from the adjacent teeth with the air syringe.

Seating the Casting and Adjusting the Proximal Contacts

The operator confirms the fit of the casting on the tooth. A 7.5 × 7.5 cm gauge sponge should be placed as a "throat screen" to catch the casting if it is accidentally dropped (see Fig. 18.60A later in the chapter). The dentist tries the casting on the tooth, using light pressure. *Do not force the casting on the tooth*. If the casting does not seat completely, the most likely cause is an overcontoured proximal surface. Using the mouth mirror, where needed, one views into the embrasures from the facial, lingual, and occlusal aspects. The dentist judges where the proximal contour needs adjustment to allow final seating of the casting, producing at the same time the correct position and form to the contact. Passing dental floss through the contact indicates tightness and position, helping the trained operator identify the degree of excess contact and its location. The dentist applies the floss at an angle and with secure finger-bracing to pass it gently through the contact and not with a snap that is likely to injure interproximal soft tissue. If the floss cannot enter or if it tears on entering, the contact is excessive.



• **Fig. 18.54** A, Cleaned casting should be tried on the die to determine if it has a satisfactory fit. B and C, To remove sprue, a cut that is not quite complete and twice the width of the disk is first made (B), and then the slim, uncut portion is bent and broken (C). D, The inlay is burnished with a No. 2 burnisher along a 1-mm path that is parallel with and adjacent to the margin. E, Magnified view of the casting before burnishing. F, Magnified view of the same marginal region shown in E after burnishing. G and H, Removing the remaining sprue metal with heatless stone (G) or with a carborundum disk (H). I, Accentuating the grooves with a dull No. 1 round bur. J, Smoothing the surfaces accessible to the rubber polishing wheel.



• **Fig. 18.55** A and B, Using a small knife-edge rubber disk on the areas of the occlusal surface that are accessible to this wheel (A) and on proximal surfaces (B). C, Polishing the grooves and other relatively inaccessible areas with a rubber point. D, Applying tripoli/BBC to the occlusal surface using a bristle disk. E, Applying tripoli/BBC to the proximal surfaces using a felt wheel. F, Imparting luster by using a chamois wheel and rouge. G, Polished castings.

Caution: When adjusting a mesioocclusodistal restoration, only one excess contact should be adjusted at a time (the stronger one) before trying again on the tooth and evaluating, unless both contacts feel equally strong. This is done because one excessively strong contact can cause the other to feel strong, when in actuality the latter contact may be correct or even found to be weak (short of contact) after the excessively strong contact is adjusted properly.

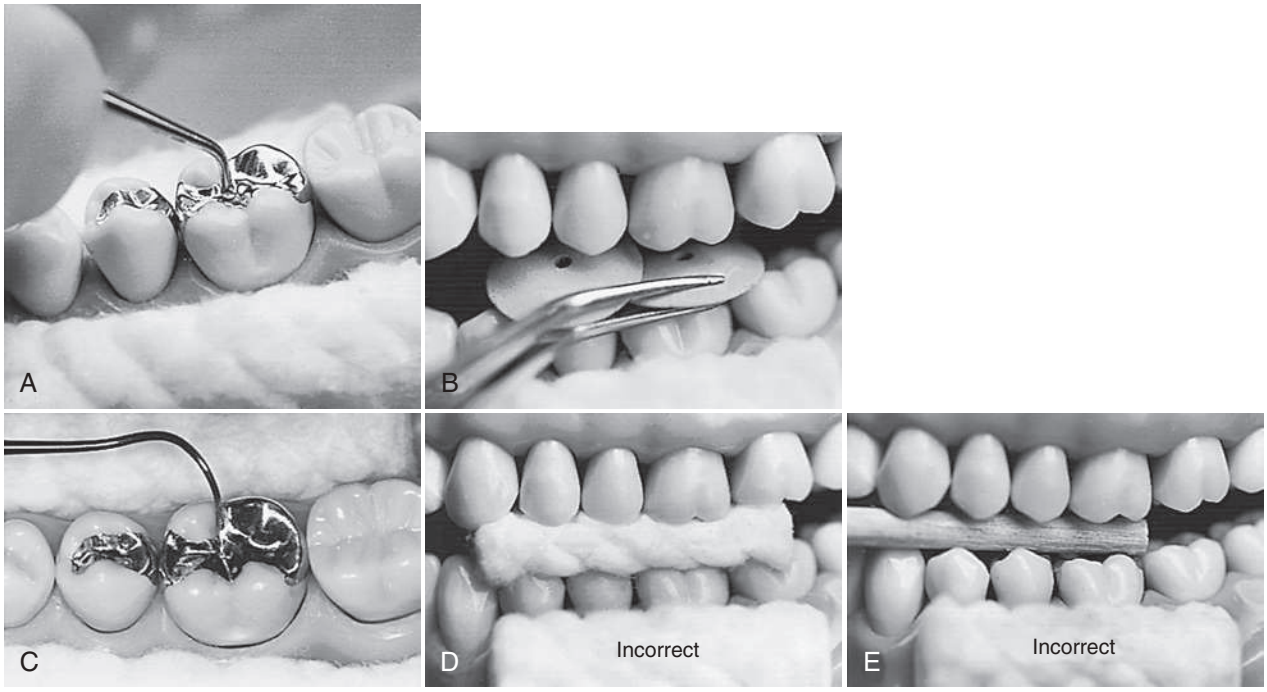
A rubber wheel abrasive is used to adjust the proximal contour and to correct the contact relationship; this often requires several trials on the tooth, but it is best not to remove too much at a time. After each trial and removal, the position of contact is visible in the form of a bright spot on the satiny surface left on the casting from previous surfacing by the rubber wheel. By noting the position of this bright spot in conjunction with observation in the mouth of the contact relationship, the contact position and form can be judged, and the operator can determine whether additional adjustment should be made to alter this position and form. (For retrieval of the casting after each trial on the tooth, see the section on Removing the Casting.)

Often the patient is able to indicate whether the contact is strong, particularly when an anesthetic has not been given. The patient should not be aware of any pressure between teeth after the final adjustment of contacts.

Proper proximal contact occurs when a visual inspection confirms that the adjacent proximal surfaces are touching and that the position and form of the contact relationship are correct. The correct "tightness" of the contacts is best judged with dental floss. This contact should be passive because any pressure between teeth would resolve soon and disappear in unwanted tooth movement.

If the contact is open (short of touching the adjacent tooth), a new contact area must be soldered to the casting. An open contact is best detected by visual inspection with the aid of the mouth mirror. The region must be isolated with cotton rolls and dried with the air syringe. Selection of the proper horizontal viewing angle usually discloses the spaces between teeth. Such an open contact permits the passage of food, which affects and irritates the interproximal gingiva.

When satisfied that the proximal contacts are correct when hand pressure first positions the casting to within 0.2 mm of seating (Fig. 18.56A), the dentist removes the 7.5 × 7.5 cm gauze sponge and ensures that the casting completely seats on the tooth by the



• **Fig. 18.56** A, Hand pressure is used initially to seat the casting on the tooth by applying a ball burnisher in the pit anatomy. B, If the casting fits to within 0.2 mm of the seating, complete seating is ensured by using masticatory pressure when the patient closes on the rubber polishing wheel interposed between the casting and the opposing tooth. C, The marginal fit of the tried-in inlay is inspected. D and E, A cotton roll (D) or piece of wood (E) should not be used in lieu of the rubber polishing wheel method (B).



• **Fig. 18.57** A–C, Castings tried on teeth. These photographs were taken immediately after the restorations were first seated on teeth before any dressing down or burnishing of margins. Neither occlusal adjustment nor contact adjustment was required. Extension of the mesiofacial margin of the second premolar was necessary because of extension of a previous amalgam restoration; extension of the distofacial margins of premolars is caused by skirting (or bracing), which provides maximal resistance form to these weak teeth. Note the area on the mesiofacial margin of the first molar that is to have a composite insert placed after cementation.

application of masticatory pressure. This use of masticatory pressure should be a routine procedure. It is accomplished by positioning a small rubber polishing disk (unmounted) on the occlusal of the restoration and requesting the patient to bite firmly; the patient also is asked to move the jaw slightly from side to side while maintaining this firm pressure (see Fig. 18.56B). At this time, the operator must judge whether the restoration is satisfactory or should be rejected and another casting made. When evaluating the fit (seating) of the casting, the operator should view particularly the margins that are horizontally directed (i.e., margins that are perpendicular to the line of draw). Along at least half the marginal outline, the tip of the explorer tine should move from tooth onto the metal, and vice versa, with barely a catch or a bump (Fig. 18.57; see also Fig. 18.56C). Some operators recommend the use of a cotton roll or a piece of wood for the patient to bite on for seating pressure (see Fig. 18.56D and E). The cotton roll may be too large and too soft to be effective for seating inlays, however, and the piece of wood may not distribute the pressure properly, resulting in less effective seating or tooth fracture. Fig. 18.57 shows the castings tried on the teeth that were first shown in Fig. 18.41H.

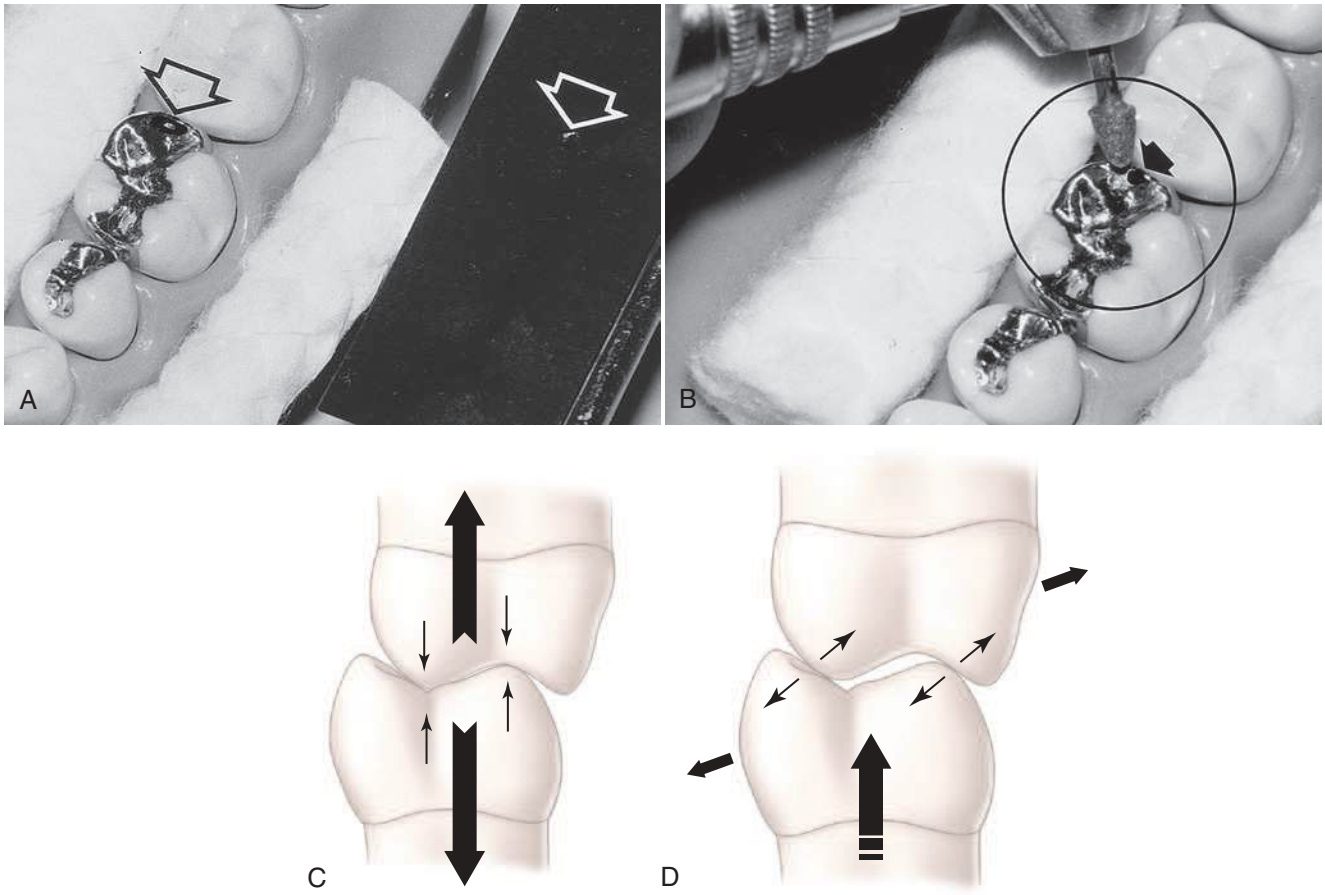
Occluding the Casting

When the proximal contacts have been adjusted and the casting is satisfactorily seated on the tooth, the patient is asked to close

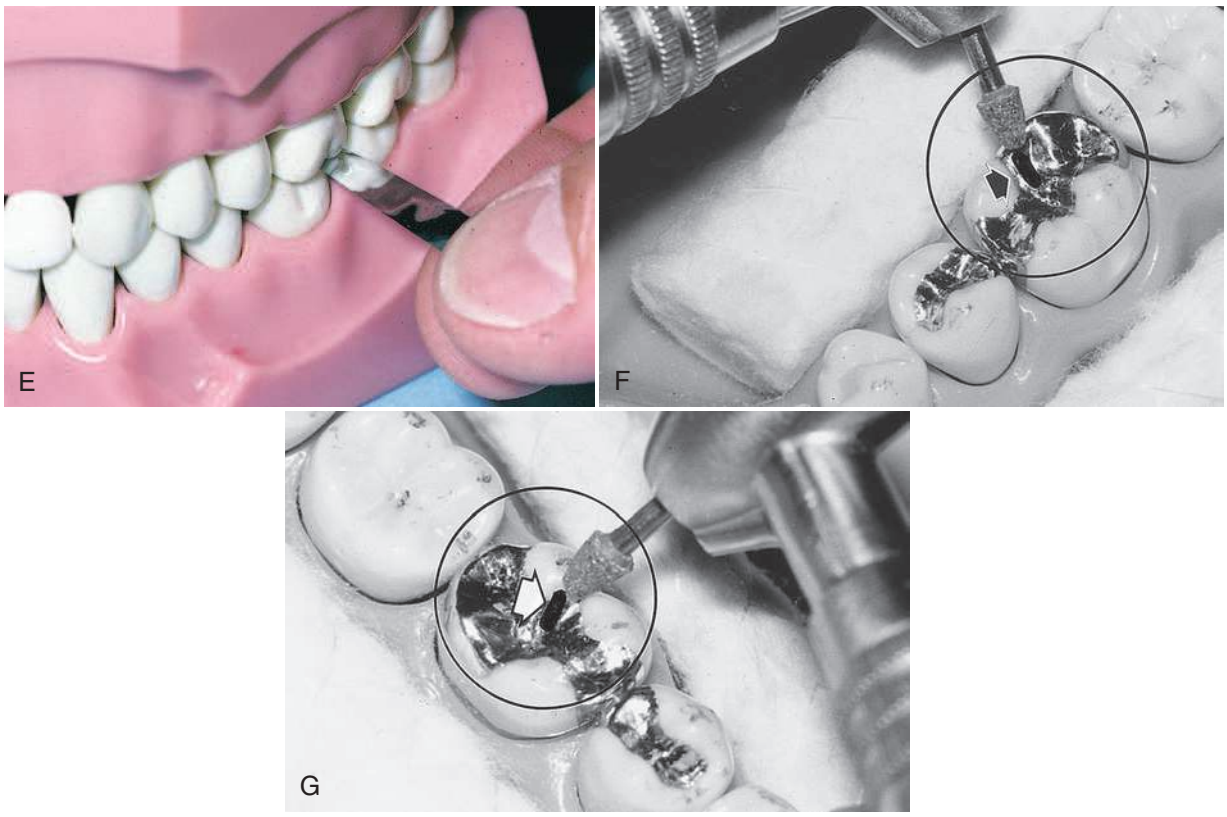
into maximum intercuspation, and the dentist inspects the unprepared adjacent teeth to see if any space exists between the opposing wear facets. Usually the patient can indicate correctly if the casting needs occlusal adjustment; however, the dentist should verify the occlusal relationship objectively. After drying the teeth of saliva, the dentist inserts a strip of articulating paper and requests the patient to close and tap the teeth together (in maximum intercuspation) several times. The dentist removes the paper and examines it by holding it up toward the light for evidence of any areas of penetration caused by the restoration. Any holes can be matched with heavy markings on the casting, and shiny, metal-colored spots may be present in the center of the marks (Fig. 18.58A). Such heavy contacts should be reduced with suitable abrasive stones, while carefully observing the following fundamental concepts for equilibration of occlusion. The space observed between the opposing wear facets of the adjacent unprepared teeth (when the teeth are "closed") is an indication of the maximal amount of vertical reduction of the casting required. Often the "high" occlusal contacts are very broad and extend onto the cusp or ridge slopes. When this occurs, the dentist should grind away the most incorrect portion of the incline contact (a deflective contact), leaving the most correct portion intact (see Fig. 18.58B). Occlusal contacts in maximum intercuspation should be composed of supporting cusp tips placed against flat or smoothly concave surfaces (or into fossae) for stability. The force vector of occlusal contacts should be one that parallels the long axis of the tooth (see Fig. 18.58C). Contacts on inclines tend to deflect the tooth and are less stable (see Fig. 18.58D). The use of articulating paper and the stone is continued until (1) the heavy markings are no longer produced, (2) the contacts on the restoration have optimal position and form, and (3) an even distribution of contacts exists on the casting and the adjacent teeth. Visual inspection should verify that the adjacent unprepared teeth are absolutely touching.

Care must be exercised not to overreduce the occlusal contacts. In the final phase of equilibration, the strength of the occlusal contacts can be tested by using thin plastic shim stock (0.013 mm thick) (Artus Corp., Englewood, NJ) as a "feeler gauge." The dentist tests the intensity of the occlusal contacts of the casting and the adjacent unprepared teeth to see if they hold the shim stock equally (see Fig. 18.58E). It may be helpful to test the occlusal contacts of the adjacent unprepared teeth with the casting out of the mouth for comparison.

When the occlusal contacts have been adjusted in maximum intercuspation, the casting is checked for contacts that occur during



• **Fig. 18.58** Occluding the casting. A, The initial occlusal contact is high and produces a heavy mark with a metal-colored center. Note the corresponding perforation in the articulating paper. B, When adjusting occlusal contacts, the most incorrect portion of the contact is removed, leaving the most correct portion intact. C, The proper occlusal contacts in maximum intercuspation are composed of cusp tips placed against flat or smoothly concave surfaces (or fossae) for stability. D, Incline contacts are less stable and tend to deflect the tooth.



• **Fig. 18.58, cont'd** E, Testing the intensity of the occlusal contacts with a thin (0.013-mm thick) shim stock used as a feeler gauge. F, Removing the undesirable contact (Zone 1, see Chapter 1) that may occur on the working side during lateral mandibular movement. G, Removing the undesirable contact (Zone 3) that may occur on the nonworking side during lateral mandibular movement.

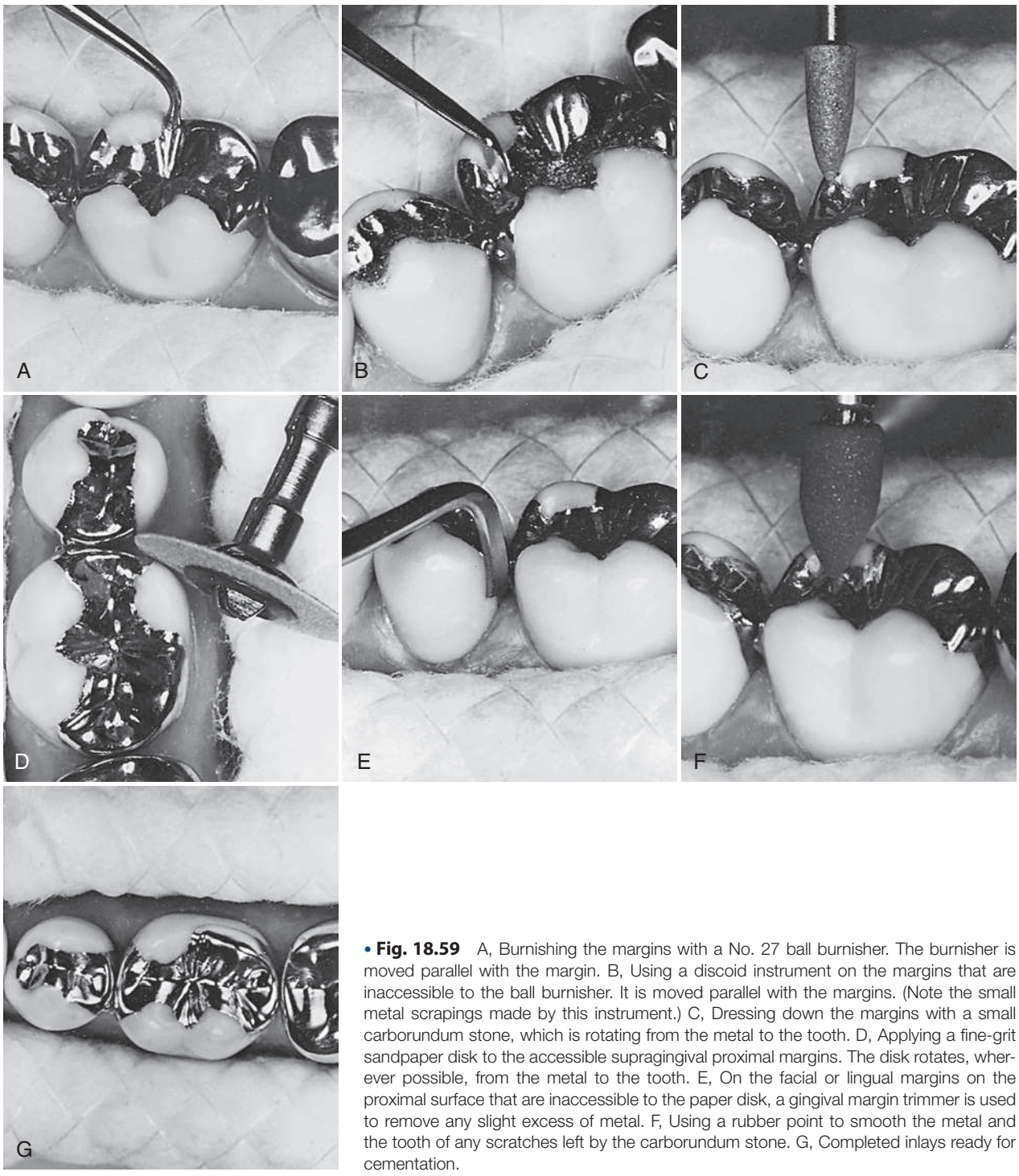
lateral mandibular movements. Lateral working (functional) contacts on the casting are marked by (1) inserting a strip of articulator paper over the quadrant with the casting, (2) having the patient close into maximum intercuspation, and (3) "sliding" the teeth toward the side of the mouth where the casting is located. Contacts between the lingual inclines of the maxillary lingual cusps and facial inclines of the mandibular lingual cusps are considered unusually stressful and should be eliminated (see Fig. 18.58F). Contacts between the lingual inclines of the maxillary facial cusps and the facial inclines of the mandibular facial cusps should remain only if they are passive and a group function pattern of occlusion is desired (see Chapter 1 for further discussion on the implications of various occlusal relationships).

The dentist inserts a strip of articulating paper over the teeth with the castings and has the patient close into maximum intercuspation and slide the teeth laterally toward the opposite side. This action marks any lateral nonworking (nonfunctional) contacts on the restoration. In a normal arrangement of teeth, contacts that might occur during the nonworking pathway are positioned on the facial inclines of the maxillary lingual cusps and the lingual inclines of the mandibular facial cusps. These nonworking contacts must be removed with a suitable stone (see Fig. 18.58G). Complete elimination of nonworking contacts can be verified by using the plastic shim stock. A strip of shim stock is inserted over the casting, and the patient bites together firmly. As soon as the patient begins sliding the mandible toward the opposite side, the shim stock should slip out from between teeth. The dentist examines the casting for interferences in protrusive mandibular movements using the shim stock and articulating paper. The areas that may have to be adjusted to prevent contact are the distal inclines of maxillary teeth and the mesial inclines of mandibular teeth.

Finally, interferences that occur on the casting between centric occlusion and maximum intercuspation are identified and removed. Most patients have a small discrepancy between centric occlusion and maximum intercuspation. Such a "skid" is considered normal for most patients, but the operator should ensure that the casting does not have premature contact at any point between centric occlusion and maximum intercuspation. The preferred technique for manipulating the mandible into centric relation and making teeth touch in centric occlusion is credited to Dawson.[13] When teeth have been marked in centric occlusion, the dentist observes them to ensure that the casting does not have premature contacts in centric occlusion and that it does not exacerbate any centric occlusion–maximum intercuspation skid. If it does, the mesial inclines of maxillary restorations and the distal inclines of mandibular restorations are the areas that may need adjustment.

Improving Marginal Adaptation

The next step is to "dress down" the margins, that is, to adapt the metal as closely as possible to the margins of the tooth. Regardless of how accurately a casting may seat in the preparation, the fit usually can be improved by using the following procedures. With a ball or beaver-tail burnisher, the operator improves marginal adaptation by burnishing the marginal metal with strokes that parallel the margin except for the gingival margin (Fig. 18.59A).



• **Fig. 18.59** A, Burnishing the margins with a No. 27 ball burnisher. The burnisher is moved parallel with the margin. B, Using a discoid instrument on the margins that are inaccessible to the ball burnisher. It is moved parallel with the margins. (Note the small metal scrapings made by this instrument.) C, Dressing down the margins with a small carborundum stone, which is rotating from the metal to the tooth. D, Applying a fine-grit sandpaper disk to the accessible supragingival proximal margins. The disk rotates, wherever possible, from the metal to the tooth. E, On the facial or lingual margins on the proximal surface that are inaccessible to the paper disk, a gingival margin trimmer is used to remove any slight excess of metal. F, Using a rubber point to smooth the metal and the tooth of any scratches left by the carborundum stone. G, Completed inlays ready for cementation.

If the margin is inaccessible to the ball or beaver-tail burnisher (as sometimes occurs at the termination of the casting in groove regions where possibly more enameloplasty or extension could have been employed), the edge of the discoid-type hand instrument serves well as a burnisher. The discoid instrument is held perpendicular to the margin and is moved parallel with the margin (see Fig. 18.59B). The sharp edges of the instrument also trim away any slight excess of metal at the margin. The operator continues on other portions of accessible margins where a slight excess of metal is present. When burnishing the casting on the tooth, the dentist should ensure that the casting is fully seated. Otherwise burnishing may bend the marginal metal, keep the casting from seating, and result in the rejection of the casting.

If necessary, the marginal adaptation and continuity can be improved further by the application of a pointed, fine-grit carborundum stone, especially where the marginal enamel is slightly

"high" and should be reduced or where more than just a slight amount of excess metal should be removed (see Fig. 18.59C). This stone should be used at low speed with light pressure and should rotate either parallel with the margin or from metal to tooth across the margin (never from tooth to metal). After this procedure, the margins are burnished again to enhance marginal adaptation and to smooth the marginal metal.

Another instrument that can be used to improve marginal fit in accessible areas (e.g., the occlusal two thirds of the proximal margins) is a fine-grit paper disk. Wherever possible, the disk should be revolved in a direction from the metal toward the tooth (see Fig. 18.59D). Sometimes these margins are inaccessible to the disk, and a gingival margin trimmer, a gold file, or a cleoid instrument may be helpful to remove a slight excess of metal (see Fig. 18.59E). It is moved in a scraping motion parallel to the margin and burnishes and trims the metal.

The experienced operator, with proper use of the elastic impression material, can produce restoration margins that require little or no burnishing or dressing down. One of the significant advantages of the indirect procedure, when correctly applied, is the high degree of accuracy of the gingival margin adaptation.

The margins should now be such that the explorer tip can pass across the margins smoothly without jumping or catching. The operator should use rubber polishing points of increasing fineness at low speed to smooth and polish the accessible areas of roughness left from adjusting procedures (see Fig. 18.59F and G). An attempt should be made to preserve the anatomic contour and detail. The operator should take care to use light, intermittent pressure when using rubber points to avoid overheating the tooth. The casting surface should be cleaned and dried to verify that it is smooth and free of scratches.

Removing the Casting

When preparing to remove a casting from a tooth, the dentist first places a 7.5 cm × 7.5 cm gauze sponge throat screen to prevent the patient from swallowing or aspirating the casting in the event that it is accidentally mishandled (Fig. 18.60A). If the casting is highly retentive, the dentist first initiates removal with the aid of a sharp Black spoon (15-8-14). The tip of the spoon is inserted as deep as possible in the occlusal embrasure with the back of the spoon resting against the marginal ridge of the adjacent tooth (see Fig. 18.60B). With the tip of the spoon firmly seated against the metal casting, the spoon is pivoted using the adjacent tooth as a fulcrum (see Fig. 18.60C). This procedure is repeated on the other occlusal embrasure if the casting is a mesiooccclusodistal restoration. This should initiate the displacement of the casting, making complete removal thereafter easy.

Cementation

Cement Selection

The selection of cement for permanent cementation is crucial to the success of the final restoration. A discussion of the advantages and disadvantages of each cement can be found in Chapter 13. No cement is without shortcomings. Each product has specific requirements with regard to tooth surface conditioning, casting surface conditioning, and manipulation techniques. To obtain optimal performance from the cement, the dentist should carefully follow the manufacturer's instructions for dispensing, mixing, and application.

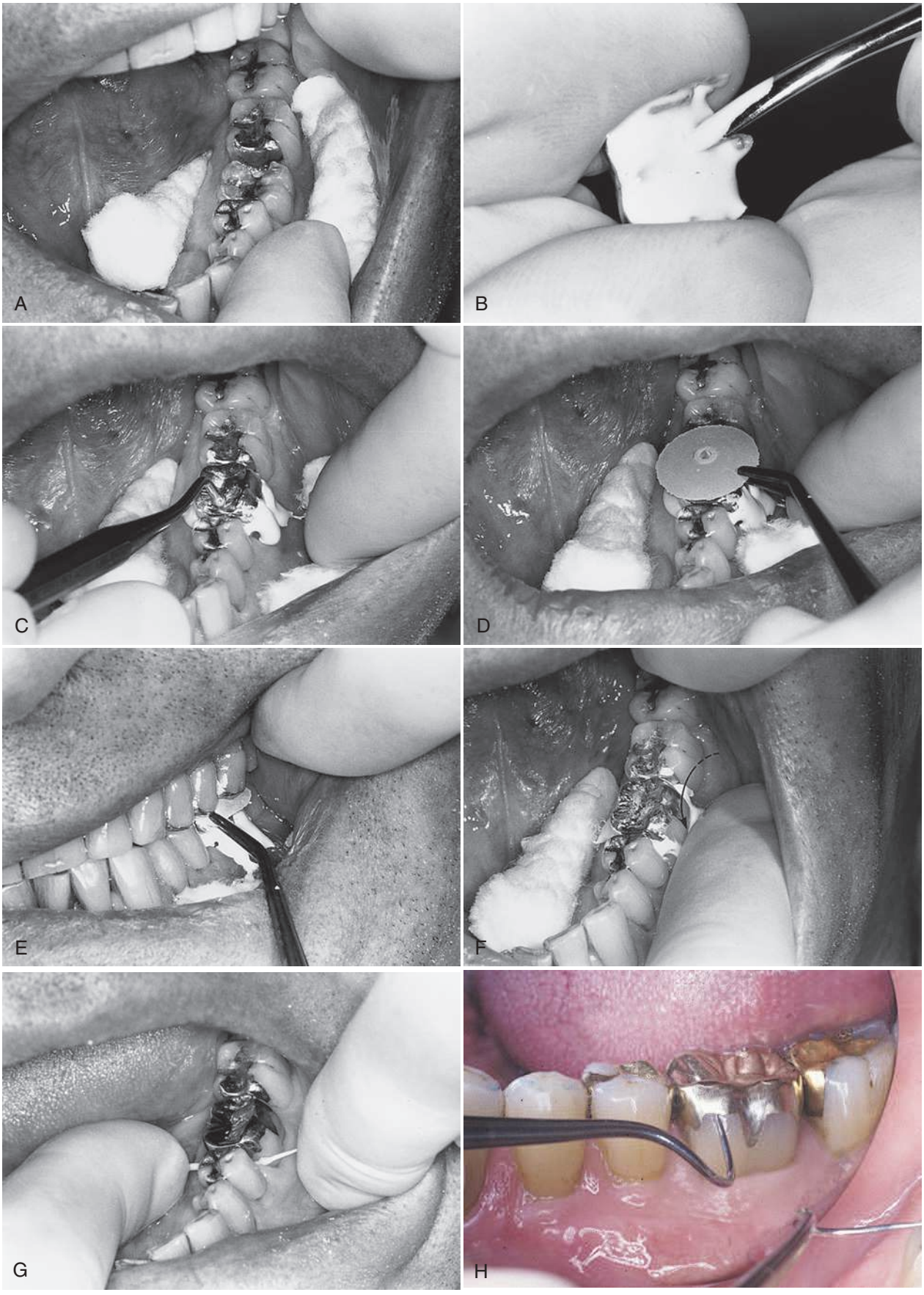
Cementation Technique

Before cementing the casting, the tooth is isolated from saliva with the aid of cotton rolls (and saliva ejector, if necessary) (Fig. 18.61A). With the air syringe, the dentist dries the preparation but does not desiccate it. The cement is mixed according to the manufacturer's instructions. With the cement mix applied generously to the preparation side of the casting (see Fig. 18.61B), the dentist starts to place the casting with the fingers or with operative pliers. Next the dentist places the ball burnisher in the pit areas (first one and then another), exerting firm pressure to seat the casting (see Fig. 18.61C). The dentist places a small flexible rubber polishing disk over the casting, removes the saliva ejector, and requests the patient to close and exert biting force (see Fig. 18.61D and E). The patient also is asked to move the mandible slightly from side to side, while continuing to exert pressure. A few seconds of this pressure is sufficient. When the disk is removed, much of the occlusal area should be clean of the cement mix and easier to inspect and to verify complete seating of the casting. When the cusps are capped, complete seating of the casting is verified by inspection of the facial and lingual margins after wiping the excess cement away (see Fig. 18.61F). While the cement is still soft, all accessible margins are burnished. The saliva ejector is replaced in the mouth and the region kept dry during the setting of the cement. Excess moisture during this setting reaction can weaken many types of cement.

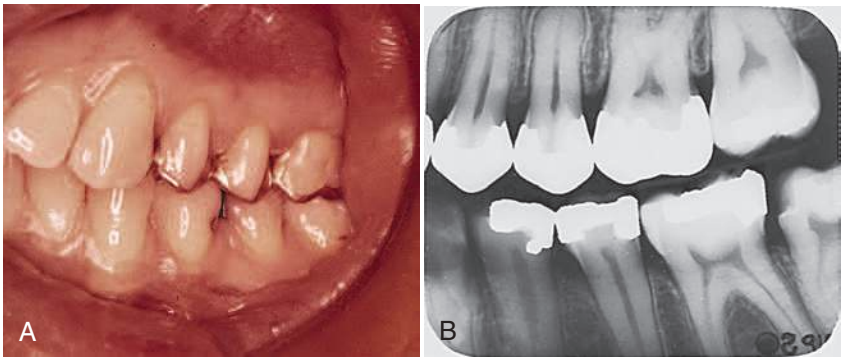
After the cement has hardened, any excess is cleaned off with an explorer and air-water spray. Dental floss should be passed



• **Fig. 18.60** Initiating the removal of the inlay before cementation. A, Place 7.5 × 7.5 cm gauze throat screen to prevent swallowing or aspiration of casting should it be accidentally mishandled. B, The tip of a sharp Black spoon (15-8-14) is inserted first as deep as possible in the occlusal embrasure with the back of the spoon against the adjacent marginal ridge. C, The spoon is pivoted in the direction of the curved arrow by using the adjacent tooth as a fulcrum. The casting has lifted from its seating. After only slight unseating, a similar procedure is applied to the distal aspect.



• **Fig. 18.61** Cementing the cast-metal onlay on the preparation initially shown in Fig. 18.39B. A, Isolating the tooth from saliva with cotton rolls. B, Applying cement with No. 2 beaver-tail burnisher to preparation side of onlay. C, Seating the onlay by using a ball burnisher and hand pressure. D, Placing a rubber polishing disk over the onlay and cementing the cast-metal onlay on the preparation initially shown in Fig. 18.39B. E, The patient is instructed to apply masticatory pressure while slightly moving the jaw from side to side. F, When the disk is lifted from the casting, much of the occlusal aspect is free of cement. With a sweeping, rolling motion of the forefinger, any accessible facial surface margin is cleaned of excess cement to permit visual inspection for verification of proper seating of the onlay. Similarly, any accessible lingual margin is cleaned of excess cement. Full seating also should be verified tactilely with the explorer tine. G, Excess set cement is removed by using the explorer and air-water spray. Dental tape with a small knot is used to dislodge small pieces of interproximal cement. H, Onlay after cementation.



• **Fig. 18.62** A, Cemented castings on teeth first shown in Fig. 18.41E. This photo was taken immediately after cementation and insertion of the composite insert on the molar. B, Bitewing radiograph of the restored quadrant shown in *A*. Note the fit of inlays at the gingival margins and the contour of the proximal surfaces.

through the contact, carried into the interproximal gingival embrasures and sulci, and pulled facially and lingually to help in the removal of cement in this region (see Fig. 18.61G). Tying a small knot in the floss helps dislodge small bits of interproximal cement. Finally, directing a stream of air into the gingival sulcus opens it and reveals any remaining small pieces of cement, which should be removed. When cementing has been properly accomplished, a cement line should not be visible at the margins (see Fig. 18.61H). A quadrant of inlays after cementation is illustrated in Fig. 18.62.

Repair

The weak link of most cast-metal inlays and onlays is the cement seal. At times, the operator may find discrepancies at margins that require replacement or repair. If the restoration is intact and retentive and if the defective margin area is small and accessible, small repairs can be attempted with amalgam or composite. If cement loss is found in one area of the restoration, however, other areas are usually suspect. When defects are found, the most common procedure is to remove the defective restoration and replace it.

Summary

Cast-metal inlays and onlays offer excellent restorations that may be underused in dentistry. The technique requires multiple patient visits and excellent laboratory support, but the resulting restorations have the potential to last for decades. High noble alloys are desirable for patients concerned with allergy or sensitivity to other restorative materials. Cast-metal onlays, in particular, can be designed to strengthen the restored tooth while conserving more tooth structure than does a full crown. Disadvantages such as high cost and esthetics limit their use, but when indicated, cast-metal inlays and onlays provide a restorative option that is less damaging to pulpal and periodontal tissues compared with a full crown.

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